

Labour Scarcity and Productivity: Insights from the Last Nordic Plague

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May 14, 2026

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Abstract

I study the causal impact of labour scarcity on productivity growth, a key driver of long-run economic growth. Exploiting a natural experiment from a 1710s plague outbreak in Northern Europe, I show that plagued regions shift into capital-intensive exports and see an export expansion. Using rich port-level trade data and a Ricardian model, I trace this shift to productivity growth driven by capital deepening. While population levels recover within four decades, the productivity and trade effects persist for almost a century, suggesting long-run changes in comparative advantage. My findings imply that labour scarcity can induce productivity-enhancing reallocation.

Keywords: Labour Scarcity, Productivity, Growth

JEL Classification Codes: F16, O47

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1 Introduction

What determines economies' long-run growth rate? One of the main drivers of economic growth are sustained increases in total factor productivity. A long-standing hypothesis holds that labour scarcity, by raising wages and altering relative factor prices, may spur productivity-enhancing innovation and reallocation (Voth, Caprettini, and Trew, 2025, Habakkuk, 1962). However, the opposing view holds that labour scarcity hampers growth as firms are unable to scale production and competitiveness is harmed by higher wages (OECD, 2024). This debate has taken on an increased importance as advanced economies face both low productivity growth and a shrinking workforce.

In this paper, I study a context where labour scarcity leads to productivity growth. I combine granular geographical variation in labour scarcity from a natural experiment, the last Nordic plague, with novel port-level trade data. This allows me to document how labour scarcity affects the composition and volume of trade. I find that plagued regions shift their exports into capital-intensive goods and conquer larger market shares abroad. While the prior reflects changes in factor proportions, I argue that a Ricardian productivity component is required to explain the latter. I thus rationalise these findings in a Ricardian model with immobile labour and link capital deepening to productivity growth. Finally, I show that these plague-induced comparative advantage changes are still in place half a century after population recovery.

The historical setting of this natural experiment is uniquely suited to study this question. I exploit the 1708-1712 Great Northern War plague outbreak as a plausibly exogenous shock that generated labour scarcity. While earlier plagues affected all of Europe, this later outbreak left many other Northern European regions unaffected. These nearby and similar regions form an ideal comparison group. Further, I study a period decades before the Industrial Revolution, which allows me to abstract from the diffusion and adoption of novel technologies.

Exceptionally detailed trade data allow me to track plagued regions' trade patterns for decades while rigorously incorporating the gravity structure of trade. Two features make the Danish toll records used in this paper the ideal source. First, origins and destinations were recorded at the level of ports rather than countries. I therefore observe trade at the same granular spatial resolution as the treatment. Second, cargoes were disaggregated into goods, which I assign into labour- and capital-intensive sectors. I can thus differentiate between goods of different factor intensities. In sum, these data permit an unprecedentedly detailed analysis of trade following this labour supply shock.

I establish two novel empirical findings. First, I find that plagued regions' share of capital-intensive exports rises by 10pp, compared to a pre-plague mean of 6.6%. New capital-intensive export goods contribute significantly to this finding. As capital became relatively more abundant, the shift into capital-intensive exports is reminiscent of the Rybczynski theorem, which states that a relative rise in a factor's endowment leads to an expansion in sectors using that factor intensively. However, I argue that a factor proportions framework cannot explain this paper's findings.

Second, plagued regions expand their market shares abroad. Their destination market shares rise by 1pp, compared to a pre-plague mean of 0.9%. This finding is surprising as labour scarcity should increase wages and reduce regions' competitiveness. Instead, I find evidence for a pattern of forced experimentation and innovation as plagued regions also add novel goods to their export baskets. While the export expansion is strongest in capital-intensive sectors, it holds across sectors and factor intensities, along both the

intensive and extensive margins. This suggests an important role for productivity growth in the post-plague market share expansion. To eliminate alternative explanations, I show a battery of robustness checks.

Finally, these trade results display path dependence beyond population recovery. By 1750, about four decades after the plague, populations had returned to their pre-plague trends. However, trade and comparative advantage patterns reflect past labour scarcity for almost a century. Thus, the plague's effects on trade patterns outlive population recovery by half a century. This suggests that adjustments to labour scarcity produce persistent changes in comparative advantages.

Next, I build a Ricardian model following Eaton and Kortum, 2002 to rationalise my reduced-form findings. The model's Ricardian nature is motivated by the fact that exports expand across sectors and factor intensities, which a pure Heckscher-Ohlin model cannot explain. I also find direct evidence of post-plague TFP growth in production data. The proposed Ricardian model with immobile labour can reproduce both the shift into capital-intensive exports and the observed export expansion. As labour is immobile across regions, Ricardian forces are married to factor-content forces. The proposed model features two competing channels. First, a Ricardian channel that permits Hicks-neutral changes in productivity (Davis and Weinstein, 2001). I follow Costinot, Donaldson, and Komunjer, 2011 in modelling Ricardian forces within sectors. Second, the model nests non-homothetic preferences as an alternative demand-side channel (Fieler, 2011). I use this model to quantify the relative importance of these channels.

I argue that productivity growth is best-suited to explain my findings. I propose a three-step mechanism that matches my empirical results. First, labour scarcity leads to higher wages. Second, producers employ relatively more capital and exports shift into capital-intensive goods. Third, sectoral productivity growth accelerates as the capital-to-labour ratio rises. I argue that productivity growth as a Ricardian component is necessary to explain the export expansion across sectors and factor intensities. Thus, I suggest that capital deepening increases productivity, and I allow the elasticity between capital deepening and productivity to vary by sector. This reduced-form dependence of productivities on factor proportions allows the model to match the empirical findings.

In support of this mechanism, I employ granular agricultural data from Sweden to show capital deepening *within* agriculture - more horses per capita were used in labour-scarce parishes after the plague - and *across* sectors - agricultural production in labour-intensive arable farming drops and shifts into capital-intensive animal husbandry. Drawing on Swedish parish-level agricultural data, I additionally estimate TFP before and after the plague, and find that TFP growth is significantly larger in plagued parishes. I also construct a proxy for the depths of ports' capital markets and show that financially more developed plagued ports undergo stronger capital deepening. Further, I show that plagued regions add significantly more novel goods, not exported by them before, to their export baskets. This finding is significantly stronger in capital-intensive sectors, reflecting that investments in new production lines are associated with gains in productivity and competitiveness, and that this association is stronger in capital-intensive sectors.

While I treat productivities as fundamentals of the model, there are several micro-foundations for capital deepening increasing productivity. Leaning on Bustos, 2011, I show that technology choice driven by factor prices produces the suggested relationship. As wages rise, more producers switch to capital-intensive technology which confers a productivity boost on them yet requires higher fixed costs. This increases both the capital-to-labour ratio and sectoral productivity. I show that in this framework sectoral

productivity increases in capital deepening. Other possible micro-foundations include learning by doing (Krugman, 1987), cost discovery (Hausmann and Rodrik, 2003), and learning by exporting (Loecker, 2013). All these micro-foundations can produce an ascending relationship between productivity and capital deepening. From the model’s gravity equation, I recover productivity growth as fundamentals (Costinot, Donaldson, and Komunjer, 2011) and project these recovered values onto changes in factor proportions, where institutions and geography are absorbed by fixed effects (Chor, 2010).

Using the model, I first show that wages rise as labour scarcity bites and validate this prediction with historical evidence. The shift into capital-intensive goods, the second step, is attributed to three channels by the model: differences in labour intensity between sectors; relative price effects passed on to destination markets; and sectors’ different elasticities of productivity with respect to the capital-to-labour ratio. While all sectors face the same regional factor proportions, they may differ in their labour shares and in their productivities’ responsiveness to capital. In this historical setting, the latter mainly reflects the availability of productivity-enhancing technologies requiring capital. Such technologies were very limited in labour-intensive sectors (Atack, Margo, and Rhode, 2019, Gallardo and Sauer, 2018, Coleman, 1956). Additional capital relative to labour should produce larger productivity increases in capital-intensive sectors, where such technologies were more readily available; thus, their productivities’ elasticity with respect to capital deepening should be higher.

In support of the third step, I show that productivity growth accelerates after the plague. Capital-intensive agriculture and manufacturing see faster productivity growth than their labour-intensive counterparts. These productivity changes, too, last for almost a century and illustrate path dependence in comparative advantages. I thus argue that labour scarcity leads to productivity growth. However, there is a caveat: institutions shielding producers from these changes stand in the way of the necessary adjustments. I consider one such institution, serfdom, and show that my findings do not apply to regions with serfdom. Labour scarcity was most severe in cities, but serfs could not move there, keeping the hinterland artificially abundant in labour. The lack of productivity effects under serfdom suggests that producers did not face the right set of incentives.

I find no support for a competing pure Heckscher-Ohlin explanation and show that Ricardian forces account for more than 90% of sectoral differences on the supply side in a hybrid framework. Similarly, I find no evidence for a demand-side channel of non-homotheticity, which is frequently discussed in the plague literature (Voigtländer and Voth, 2012). My model nests non-homothetic preferences (Fieler, 2011), under which demand may shift into income-elastic goods as wages rise. However, I find no evidence for this in a direct test and in implied price effects. The absence of changes in import composition speaks against sector-specific embodied technology linked to increasing imports; it also provides another argument in favour of Ricardian and against pure Heckscher-Ohlin forces. Similarly, I argue against venting out and market power as explanations. Further, I find no evidence for human or social capital deepening after the plague: even when controlling for goods’ skill and contract intensities, the only shift in exports after the plague remains into physical capital intensity. However, I acknowledge that factor-biased technological progress and a hybrid model containing both factor content and Ricardian forces could rationalise my findings. Such models would nonetheless require productivity growth after the plague. I thus conclude that the mechanism I propose is well-suited to explain my empirical findings. Finally, I establish that in the counterfactual absence of a productivity channel exports would have contracted. The productivity growth neces-

sary to bridge the counterfactual export contraction and the observed export expansion closely matches the estimated TFP growth figures. Therefore, I propose that productivity growth is necessary to reconcile theory with empirics.

This work builds on several strands of literature. First, I provide novel evidence that labour scarcity leads to productivity growth. I use an exogenous mortality shock, instead of migration or fertility that present endogeneity challenges, to establish these findings. I also go beyond existing work to link labour scarcity not only to the adoption of machines and innovation (as in Voth, Caprettini, and Trew, 2025, Franck, 2024, and Andersson, Karadja, and Prawitz, 2020) but also to productivity growth. This is the ultimate outcome of interest in this debate to which my paper makes a crucial contribution.

Second, this paper makes a key contribution to the literature on path dependence in the spatial economy. My paper demonstrates that temporary labour scarcity is reflected in comparative advantages for almost a century. Thus, I argue that a shock to factor proportions explains persistent changes in comparative advantages. I bring Ricardian theory to unprecedentedly granular trade data, which also allows me to go beyond existing work (Lane, 2025, Juhász, 2018) in establishing path dependence in comparative advantages in a general equilibrium framework. Further, the historical setting of this natural experiment allows me to avoid the pitfalls of technology diffusion and policy endogeneity.

Third, I provide granular evidence and plausible micro-foundations for the mechanisms behind regions' recovery after shocks (Jedwab, Johnson, and Koyama, 2024, Voigtländer and Voth, 2012, Davis and Weinstein, 2002). I show that this recovery entailed active adjustments reflecting economic decisions rather than mechanical forces.

This paper proceeds as follows. The following section summarises the related literature. Section 2 introduces the historical context and the data. Section 3 holds the trade results and section 4 the model. In section 5, I discuss the mechanisms behind my results. Section 7 shows a counterfactual analysis of trade in the absence of a productivity channel and section 8 concludes.

Related Literature

Labour Scarcity & Productivity

The view that labour scarcity promotes productivity growth is well-established in the literature. Habakkuk's seminal work postulates that labour scarcity in the United States led to labour-saving technological progress (Habakkuk, 1962). Acemoglu, 2010 generalises the relationship between factor proportions and the factor bias of technological progress. He shows that labour scarcity encourages technological progress if the latter reduces the marginal product of labour. Allen, 2009b considers Britain a high wage, cheap energy economy and argues that British innovations resulted from these factor proportions.¹

Recent work has added causal identification to this debate. Voth, Caprettini, and Trew, 2025 use variation in warships' coastal access to identify the positive effect of labour scarcity on the adoption of labour-saving machines. Franck, 2024 exploits temperature variation to analyse how labour scarcity led to technology adoption and innovation. Andersson, Karadja, and Prawitz, 2020 use an IV approach to identify the causal effects of emigration on technological change, linking higher capital intensity to innovation.

In current-day debates, labour scarcity is considered an obstacle to economic growth

¹This assertion is subject to criticism by Humphries and Schneider, 2020, Humphries and Weisdorf, 2019, and Stephenson, 2018.

(OECD, 2024, Caldara, Iacoviello, and Yu, 2024). An emerging literature shows that firms respond to labour scarcity by competing for workers, adjusting their input mix, and innovating in management techniques (Börschlein, Bossler, and Popp, 2024, Groiss and Sondermann, 2023).

My paper’s key empirical contribution is to test existing theories. In particular, my paper forms the first test of the Habakkuk thesis in an open economy setting. Further, the commodity- and port-level trade data set is an empirical contribution in itself. On the theoretical side, I link labour scarcity and capital deepening-driven productivity growth in a general equilibrium framework. This addresses the criticism of Temin, 1966 towards Habakkuk, 1962 regarding factor proportions in general equilibrium.

Path Dependence in Comparative Advantage

An established literature proposes micro-foundations for path dependence in comparative advantage. Krugman, 1987 shows how learning-by-doing feeds into lasting productivity changes. A related explanation lies in learning by exporting. Clerides, Lach, and Tybout, 1998 find no general evidence that firms become more productive after entering export markets. Loecker, 2013 argues to overcome previous identification challenges and finds that entering export markets leads to productivity growth. More generally, Allen and Donaldson, 2022 build a model to explain persistent effects of temporary shocks.

A growing body of literature tests this empirically. Juhász, 2018 interprets the British blockade of France as temporary trade protection. She finds that more protected regions increased their spinning capacity and had higher value added per capita in industry for several decades. Lane, 2025 studies industrial policies in South Korea and argues that they successfully changed comparative advantages.²

I contribute to this literature by showing that an exogenous increase in capital intensities is followed by a lasting shift in comparative advantages. Juhász, Lane, and Rodrik, 2024 argue that assessing policies is complicated by identification challenges. Instead, I study an exogenous shock, which largely exempts my findings from this criticism.

Economic Effects of Plagues

My paper also relates to the literature on plagues. Population recovery after the Black Death took about two centuries (Jedwab, Johnson, and Koyama, 2024). Wages, however, did not recover in all parts of Europe (Jedwab, Johnson, and Koyama, 2022, Álvarez-Nogal, Prados de la Escosura, and Santiago-Caballero, 2020).³ Living standards rose: Broadberry et al., 2014 document that the caloric composition in England shifted towards dairy and meat. Gelman, 1982 finds that English agriculture shifted from arable into pastoral farming. Bosshart and Dittmar, 2025 and Dittmar and Meisenzahl, 2019 stress the political economy of plague outbreaks. Voigtländer and Voth, 2012 argue that sustained wage increases were maintained by directly creating downward pressure on urban populations. Jedwab, Johnson, and Koyama, 2022 document the increased bargaining power of workers beyond higher wages.

My paper adds to this literature by focusing on granular trade in a general equilibrium framework and establishing novel empirical facts. My findings reject the predic-

²More broadly on the time path of changes in trade patterns, Hanson, Lind, and Muendler, 2018 stress the empirical relevance of dynamic comparative advantage.

³In the short-run, Jedwab, Johnson, and Koyama, 2022 find that wages actually dropped. They explain this through Smithian growth going into reverse.

tions of Malthusian models (Cantoni, 2015) and of models of non-homothetic demand (Voigtländer and Voth, 2012). Instead, I find productivity growth effects that have been implicit to findings in existing work. A theoretical contribution lies in my model’s ability to explicitly parse out non-homotheticity (Fieler, 2011). This paper thus helps to establish a firmer foundation of our understanding of the economic effects of plagues.

2 Context and Data

This paper studies the last plague outbreak in Northern Europe, which coincided with large-scale warfare. The war that broke out in 1700 between Sweden, Russia, and their allies affected most countries on the Baltic Sea. Early campaigns led the Swedish army into the Baltics and later deep into the Polish-Lithuanian Commonwealth. As the war turned in Russia’s favour, Swedish troops receded to Northern Germany, and Russian troops marched far into Sweden. After the decisive Swedish defeat in Poltava in 1709, the Danish navy blocked ships between Sweden and its Northern German possessions, and the Russian navy raided the Baltic coast. As a result, ship passages at the Sound collapsed. This so-called Great Northern War ended in 1721 and marked the end of Sweden as a great power. Swedish territories in the Baltic were lost to Russia, and about half of Swedish Pomerania to Prussia.

The war was accompanied by a severe outbreak of the plague. The plague mostly followed army routes, reaching East Prussia in 1708, most of the Baltic Sea by 1711 and Hamburg by 1712. At this point, the Swedish army was on her way back from what is today Ukraine and sought refuge and fresh supplies in Swedish Pomerania. Appendix Figure 8 shows digitised army marching routes based on Spruner and Menke, 1880 and Barraclough, 1997 to illustrate how armies spread the plague. This historical evidence speaks towards war, not trade, spreading the plague. Figure 1 shows plagued regions in my sample.⁴ 22 regions contain cities that were besieged during the war, while seven regions were both plagued and besieged.⁵ Most plagued cities caught the plague from their hinterlands through which armies marched.

2.1 Plague & Mortality Data

Appendix Table 12 presents the plague data I collected. Virtually all plagued regions should be accounted for: the literature covers all areas around the Baltic, and it is reasonable to assume that a plague incident will have been recorded. Even Bremen’s mortality rate of 0.7% is recorded. I restrict myself to coding plagued regions. While Raster, 2023 presents hand-collected rural mortality data for Northern Estonia, this remarkable effort can hardly be extended to the entire region. For towns covered in both the trade data and in Raster, 2023, the plague status overlaps precisely.

Mortality estimates are available for half of plagued cities, with the median mortality rate at 36%. For the remaining plagued cities, I predicted mortality with the timing of the plague, geographical controls, and the proximity to army routes. The proximity to armies, displayed in Appendix Figure 8, is motivated by the finding that armies spread the plague. Appendix Table 14 shows the regression results.

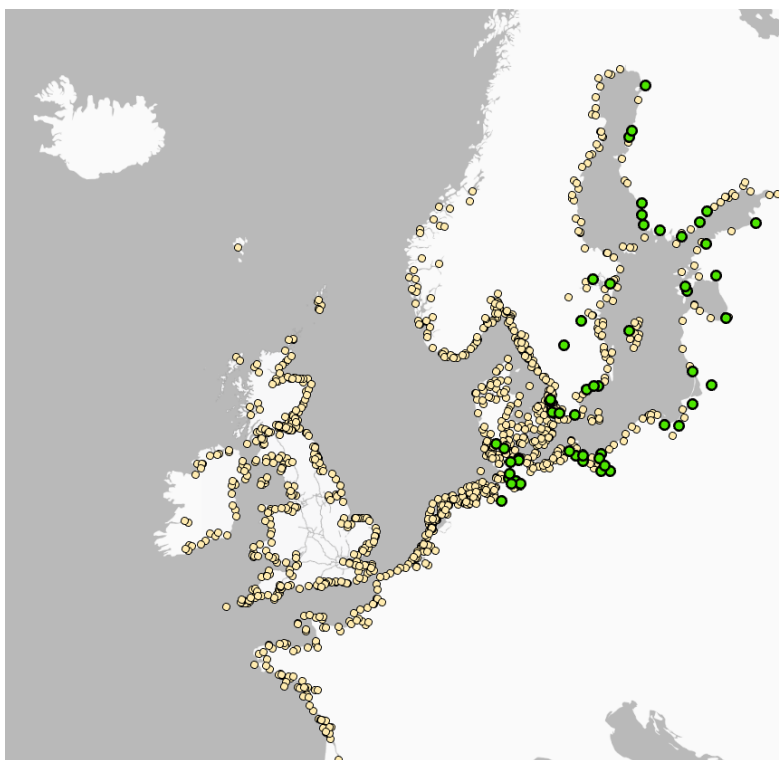
⁴Appendix Table 32 lists plagued region across all of Europe, even outside my sample.

⁵See Appendix Table 13.

While both mortality estimates and a plague dummy are used, there is doubt about the quality and comparability of mortality figures (Roosen and Curtis, 2018, Dittmar and Meisenzahl, 2019), pointing to the plague dummy as being more robust. There is no evidence for demographic heterogeneity in mortality (Raster, 2023, Alfani and Murphy, 2017) Moreover, there is no evidence that the plague was spread by trading specific commodities.⁶

Figure 1 shows the 55 out of 594 regions that were plagued. As only seven plagued regions were additionally directly besieged, I expect the destruction of physical capital to play a minor role. Appendix Table 15 shows that plagued cities were not significantly larger before the plague. I also show that plagued cities were not significantly smaller by 1750.⁷ To test regional and not only urban population recovery, I show in Appendix Table 16 that plagued Swedish parishes had recovered their populations by 1750. In Appendix B.3, I discuss the contributions of increased fertility, lowered mortality, and migration to this population recovery.

Figure 1: Plagued regions



Notes: **Green circles:** plague. **Beige circles:** no plague.

The main analysis identifies treatment as a plague outbreak in a region. As a robustness check, I incorporate plague outbreaks across all of Europe (Appendix Table 32 and Appendix Figure 30). I construct an indirect plague treatment variable that assigns higher weights to closer by and larger plagued regions. All details can be found in Appendix C.5. In Appendix Table 33, I show that the indirect plague treatment variable is

⁶Frandsen, 2009 mentions that in some places, textiles of those who perished were burnt as they were believed to be contagious. However, the finding in Appendix Figure 44 that relative demand did not shift into textiles after the plague speaks against the salience of this anecdote.

⁷Given low urbanisation rates, I consider urban population recovery indicative of regional population recovery.

never significant, and that the main results established for the direct plague treatment go through as before. Therefore, I continue to focus on the direct effects of the plague.

2.2 The Little Ice Age

The period after the plague coincides with warming at the end of the Little Ice Age (Waldinger, 2022). I control for temperature change when recovering agricultural productivity: warming in a cold region with long winters raises agricultural productivity and lengthens growing seasons.⁸ I use temperature data (Xoplaki, 2005, Luterbacher et al., 2004) and match regions to the four closest observations, weighting by the inverse distance to these grid points. I define growing seasons as spring and summer.

2.3 Serfdom

Appendix Figure 42 shows regions by their serfdom status. While Denmark is the only country in this area that re-introduced serfdom in 1733 (Gary et al., 2022), Raster, 2023 argues that labour coercion increased in Northern Estonia as a result of post-plague labour scarcity. My paper does not seek to disentangle the margins of labour coercion. Instead, I conceptualise serfdom as a mobility restriction and collect data on the presence of serfdom. While serfs accounted for less than 10% of the population in Estonia and 10-30% in Lithuania (Baten, Szoltysek, and Campestrini, 2016), labour coercion will have restricted the mobility of a significantly higher share of the rural population. My main sources are Raster, 2023 and Peters, 2022 (see Appendix E.3.1).

2.4 The Soundtoll Data

A novel granular trade panel is constructed based on the Soundtoll data (see also Marcinek, Maurer, and Rauch, 2025). I report key descriptives in Appendix Table 5. Virtually all trade between the Baltic and North Seas is recorded.

Classification of Goods

I map 143,855 cargo descriptions in Old Danish to 227 goods and to five sectors: labour-intensive agriculture, capital-intensive agriculture, labour-intensive manufacturing, capital-intensive manufacturing, and remaining unclassified goods. As a robustness exercise, I use Claude’s Opus 4.6 model to classify cargoes. This simple exercise produces the same goods classification as my manual efforts for 84% of traded value, and for 96% of traded value after some inconsequential differences⁹ are accounted for (Appendix A.1).

In Section 4, I argue that the differentiation between labour- and capital-intensive goods is crucial, and that land intensities are not a driving factor.¹⁰ I classify goods by recognising that capital-intensive goods require a high baseline amount of capital, contained in buildings, furnaces, machinery, tools, carts, and animals. Labour-intensive

⁸The documented shift into capital-intensive agriculture actually runs counter to climate warming. Under warmer temperatures, regions were likely to reverse their shift into pastoral farming that had occurred during the Little Ice Age (Degroot et al., 2022, Degroot, 2018). This reversal should go in the opposite direction of what I document below.

⁹These are iron works/iron, planks/wood, goods/mercery, cloth/sail, meat/beef.

¹⁰I find in Appendix Figure 14 that there is no relative shift into land-intensive commodities (arable crops, livestock and animal products, plantation goods, raw forestry) after the plague.

goods, on the other hand, are dominated by labour-intensive tasks (Heblich, Redding, and Zylberberg, 2025) and limited in the degree to which labour can be saved.

Labour-intensive agriculture is arable farming. Such farming certainly uses some capital in the form of implements such as carts and ploughs. However, arable farming is dominated by the numerous labour-intensive tasks it entails, such as ploughing, sowing, weeding, harvesting and threshing (Heblich, Redding, and Zylberberg, 2025). Pastoral farming, mining, and processed foods are capital-intensive agriculture. The infrastructure and tools required to mine and process foods make them capital-intensive. For pastoral farming, the importance of capital is well-established, with livestock itself classified as capital (Broadberry et al., 2014, Allen, 2005). Livestock require capital-investments for grazing (fencing, drainage, and roads) and for milking and slaughtering (buildings and tools) and otherwise graze freely in fields, requiring little labour input.

In manufacturing, I classify textiles as labour-intensive. Broadberry and Gupta, 2009 and Pomeranz, 2006 discuss how traditional production methods in textiles were very labour-intensive, with tasks such as spinning, weaving, and dyeing requiring high labour inputs. Textile production in the early 18th century Baltic region will have employed traditional production methods as these were agrarian societies before the Industrial Revolution. Iron and metal works and ship building, on the other hand, are naturally more capital-intensive in that they use furnaces, buildings, docks, and tools.

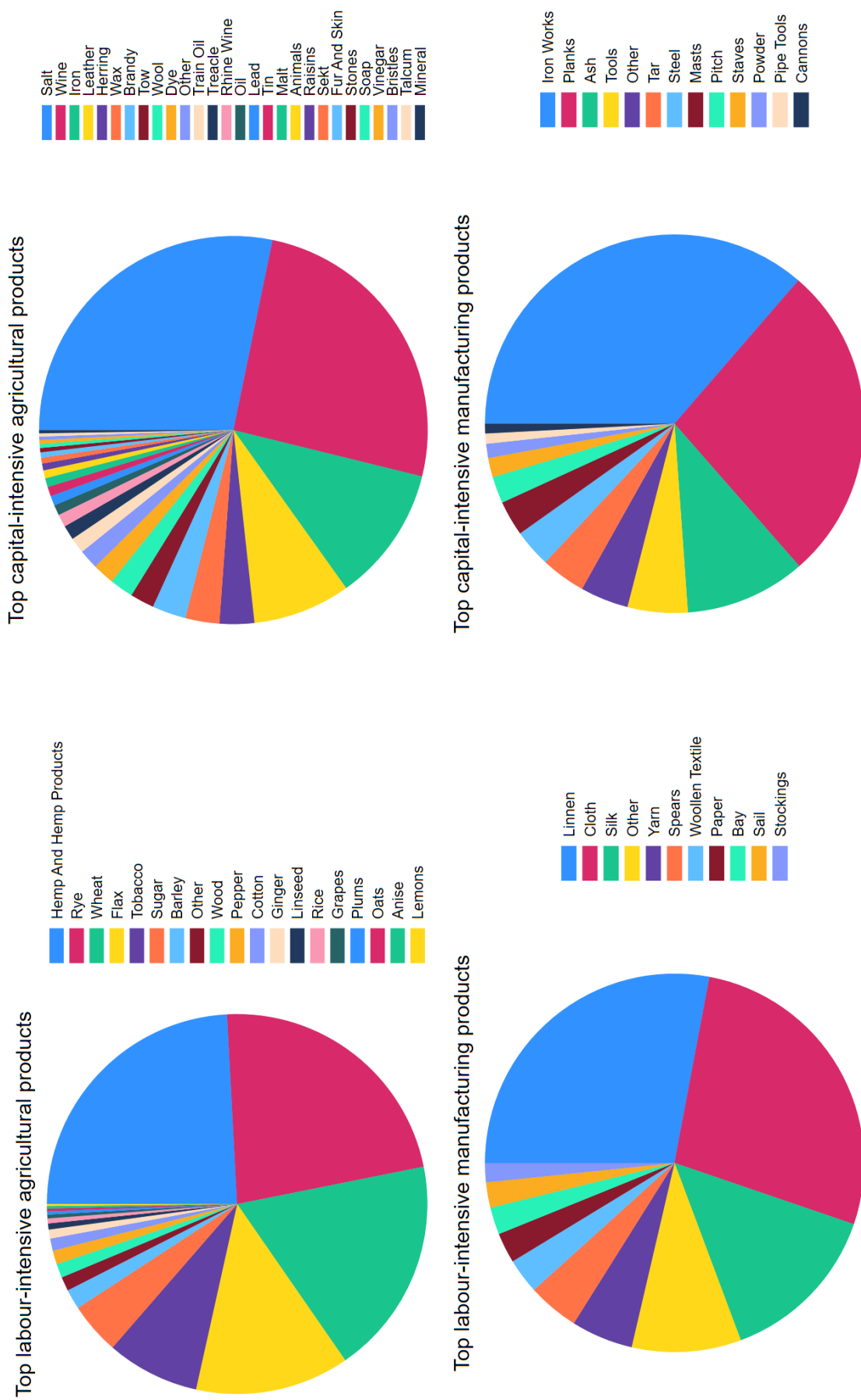
Figure 2 shows the most common goods for each sector by value between 1668 and 1750. Appendix Table 6 ranks the top 30 goods, accounting for 90% of traded value during this period, and shows their assignments. I run a robustness exercise in which I reassign four debatable goods: wine and tow to labour-intensive agriculture and planks and ash to labour-intensive manufacturing. This does not change my results.

Classification of cities, duties, currencies, and units

The Soundtoll data record origins and destinations at the level of ports. I map 90,737 original port names to 3,085 unique port identifiers and assign them into one of 56 areas. Areas are mostly equivalent to NUTS 2 regions; for the United Kingdom, France, Germany, and Spain, they are NUTS 1 regions to ensure similarly sized areas and to avoid areas with a single port (Appendix B). However, my estimates are robust to grouping ports into NUTS 1 or NUTS 2 regions. Only ships passing the Sound are recorded. As such, I only observe trade between the Baltic Sea and the North Sea, but none within either. I confirm my main empirical findings in trade data for Hamburg (Appendix Table 31) for which I see trade with all partners. I also present simulation results in Appendix Figure 31 on a simulated productivity increase. Both in the hypothetical full and in the actually observed sample the productivity increase is recovered. In both cases the estimates are centred around the simulated effect, suggesting no bias.

Details on tolls and duties are given in Appendix A.3. I follow the literature in interpreting the proportional duty as a proxy for value (Waldinger, 2022). Additionally, I verify that duty rate variation does not affect my findings (Appendix A.4). Currency concerns, as elaborated in Appendix A.2, are minor as 99.76% of trade transactions are carried out in Danish coins. I also use shipments' weights to construct unit prices (Appendix Table 9). While there are reasons to doubt the accuracy of reported tolls and to suspect corruption, a truth-telling mechanism was in place to mitigate incorrect reporting: the Crown could purchase goods at the stated value to discourage underreporting.

Figure 2: Goods by sector



Notes: This figure shows the most common goods per sector by overall traded value. The 2% least common goods have been summarised as 'other' for legibility.

3 Trade

This Section presents novel descriptives on trade following a plague outbreak. Labour scarcity¹¹ increases wages (Jedwab, Johnson, and Koyama, 2022), making capital relatively cheaper.¹² One needs to classify goods as labour- or capital-intensive. For agriculture, I classify arable farming as labour- and pastoral farming as capital-intensive (Heblich, Redding, and Zylberberg, 2025, Broadberry et al., 2014, Allen, 2005). For manufacturing, I distinguish labour-intensive production of textiles from capital-intensive metal works, ship building, and machinery-intensive production (Broadberry and Gupta, 2009).¹³ Details on goods classifications can be found in Section 2.4.

Three adjustments could appear. First, an adjustment *within* sectors. As long as factor prices are not equalised across regions, a higher capital-to-labour ratio will be employed in the production of goods. In the case of arable farming, this might take the form of metal ploughs or drainage. In manufacturing, machinery and tools reflect a common way of increasing the capital-to-labour ratio. This adjustment *within* sectors is difficult to observe in trade data, so I turn to production data below.

The second adjustment is a compositional effect *across* sectors and as such observable in trade data. Labour scarcity should induce regions to increase their production of capital-intensive goods relative to labour-intensive goods. This adjustment *across sectors* is predicted by the Rybczynski theorem as part of the Heckscher-Ohlin theory of trade. At constant relative goods prices, it states that the rise in the endowment of capital, compared to labour, leads to a more than proportional expansion of output in capital-intensive sectors. This change in regional relative factor endowments does not lead to changes in regional factor prices as long as the region is sufficiently small. This prediction of factor price equalisation is rejected by the data, which is one of the reasons why I propose a Ricardian model.¹⁴ The Ricardian model also predicts an adjustment *across sectors*, as labour is assumed immobile across regions, and can further rationalise an overall export expansion through productivity growth.

Third, labour scarcity may lead to productivity growth. If it does not, export would decline overall as wages rise; the first two adjustments would lead to changes in the capital intensity and composition of exports. If labour scarcity does lead to productivity growth, one should observe both a compositional change and an expansion of exports.

I expect the compositional adjustment across sectors to become visible in trade data almost immediately as producers respond to higher wages as soon as mortality reduced labour supply. It may be useful to consider the scope of this shock as the median mortality rate was 36%. Productivity growth, however, should take several years to materialise as producers experiment with new production methods and best practices take time to spread throughout the economy (Juhász, Squicciarini, and Voigtländer, 2024). Furthermore, trade volumes during the second half of the war were severely depressed: after 1709, the Danish navy blocked ships; the Russian navy raided many Baltic coastal areas and intercepted vessels. This reduced the number of passages at the Sound (Appendix Figure 10). I collect detailed data on Russian naval raids and blocked ports and exploit

¹¹The terms scarcity and shortage are used interchangeably (Caldara, Iacoviello, and Yu, 2024). I speak of scarcity throughout, as labour became scarcer compared to capital.

¹²In Appendix Table 34, I follow Keller, Shiue, and Wang, 2021 in inferring interest rate from monthly grain price variation, and find that interest rates do not rise after the plague.

¹³I assume unclassified goods as labour-intensive; my results are unchanged when dropping them.

¹⁴Trefler, 1993 reconciles the same empirical rejection of FPE with factor content theory by introducing factor-augmenting productivity differences. I will return to this point in Section 6.

the fact that the Danish navy was operating its blockade out of the island of Christiansø. In Appendix Table 19, I show that halving a port's distance to Christiansø is associated with 27-34% less ships sailing from that port during the blockade, controlling for other direct blockades, Russian raids, and a host of fixed effects. Crucially, this adverse effect on trade does not vary by a port's plague status. This increased risk of shipping thus affected both plagued and unplagued ports and limits our ability to distinguish their export composition and levels after the plague. Therefore, trade data from the first decade after the plague are relatively uninformative, and most effects become visible only after the peace treaty of 1721 restored shipping routes. This is confirmed by studying the export volumes and composition of the main plagued port on the North Sea, where shipping was not disrupted: Hamburg's trade (see Appendix Table 31) shows an effect on impact.

Identification

I provide evidence to rule out that the plague was endogenous to trade. In the literature on the Black Death, trade has been proposed to have propagated the plague; for this outbreak, however, armies have been argued to have spread the plague and rarely directly targeted cities. The main reason for this is that during the period when trade could have spread the plague, the Danish blockade severely reduced trade volumes, which supports my claim that this particular plague was exogenous to trade. There is no evidence of the plague as a biological weapon. In a balance test, to-be-plagued cities do not differ in population size before the plague (Appendix Table 15). For trade, Table 1 shows that a future plague dummy does not predict a different volume, composition, growth, or number of exports. Balancedness in this regard is confirmed in that I tend to not find any significant pre-trends in event studies.

Table 1: Pre-plague balance checks.

	Log Exports	Export Growth	Log # of Exported Goods
	(1)	(2)	(3)
To be Plagued	0.588 (0.427)	-0.055 (0.064)	0.221* (0.120)
To be Plagued x Capital-Int.	-0.437 (0.367)	0.080 (0.063)	0.043 (0.107)
<i>Fixed Effects:</i>			
– Destination x Year	✓	✓	✓
Years	1668-1708	1689-1708	1668-1708
Observations	15,826	47,160	31,604

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are log export volumes (column 1), annual export volumes divided by total exports between 1668 and 1688 (column 2), and the log number of exported goods (column 3). The independent variable is a plague dummy, equal to one for to-be-plagued regions, and this dummy's interaction with a dummy for a capital-intensive sector. Years are restricted to before 1708 to exclude any plague years. Additional controls are annual growing season temperature, latitude, and longitude.

Fact #1: Capital-intensive exports increase more than labour-intensive exports.

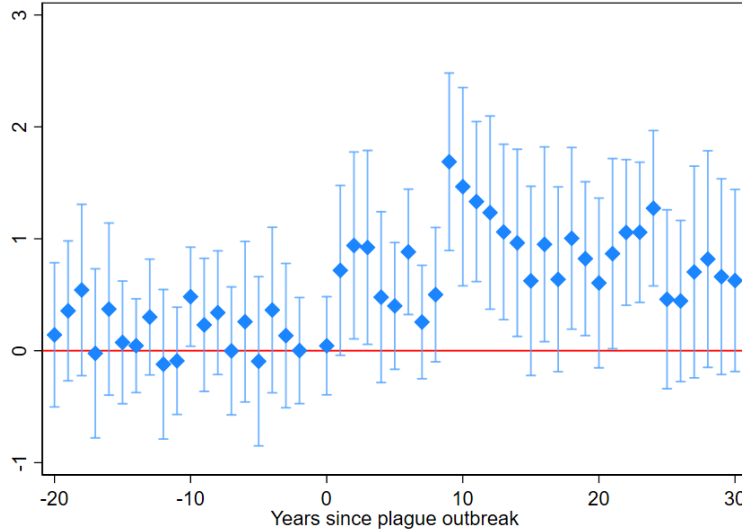
The plague induced labour scarcity, which will have translated into higher wages and an increased capital intensity of production. The plague's effects should thus differ by sectors' factor intensities. To test this, I estimate an event study using OLS on log exports, weighted by exports in levels.¹⁵ In particular, I estimate:

$$T_{ikt} = \sum_{l=-20 \setminus -1}^{30} \beta_l^f \cdot \mathbb{1}(K_{it} = l) + \alpha_{ik} + \alpha_{kt} + \gamma_f x_{it} + \epsilon_{ikt}, \quad (1)$$

where i are origin ports, k one of 5 sectors, f is factor intensity (capital- or labour-intensive), and t are years. T_{ikt} are (log) exports.¹⁶ K_{it} is the time difference to the plague, α_{ik} are origin-sector fixed effects, α_{kt} sector-time fixed effects, and x_{it} are annual growing season temperatures. I am particularly interested in the difference between β_l^C and β_l^L . Throughout, standard errors are clustered at the origin-level.

Figure 3 shows the difference between β_l^C and β_l^L : after the plague, capital-intensive exports expand relative to labour-intensive exports. The estimates imply a 10pp increase in the share of capital-intensive exports, compared to a pre-plague mean of 6.6%.¹⁷

Figure 3: Capital-intensive exports expand relative to labour-intensive exports



Notes: Estimation of equation 1. Standard errors clustered at the origin level.

In the Appendix, I present further results. Appendix Figure 11 shows that an alternative assignment of four debatable goods produces almost identical results. Appendix Table 20 shows PPML and OLS results in a DiDiD set up. Appendix Table 21 shows that

¹⁵This recovers OLS estimates similar to PPML estimates. Mayer, Vicard, and Zignago, 2019 and Head and Mayer, 2013 show that PPML places more weight on high trade flows; weighting OLS on log trade by levels thus brings OLS close to PPML results (Mayer, Vicard, and Zignago, 2019).

¹⁶Throughout, these are the value of exports. 99.76% of transactions were carried out in Danish coins.

¹⁷As discussed above, the shock becomes particularly visible after a decade, when war-induced shipping disruptions unrelated to the plague had been resolved (Appendix Figure 10 and Appendix Table 19).

army proximity and sieges are not associated with a shift into capital-intensive exports. This also serves as a placebo test: for unplagued regions, army proximity and sieges do not predict changes in export composition. Appendix Figure 12 confirms the above finding using the imputation estimator in Borusyak, Jaravel, and Spiess, 2021. Appendix Table 22 shows that the relative expansion of capital-intensive exports occurred both in manufacturing and in agriculture. Further, Appendix Figure 14 shows that no shift into land-intensive exports takes place after the plague. I analyse the plague’s effects on the composition of exports at the goods level in Appendix Figure 13.

Directly on production, I show that after the plague Swedish farms switched out of labour-intensive rye and barley production and into raising calves and foals. These effects are stronger the closer a farm is located to a plagued city (Section

This shift into capital-intensive exports outlasts the shock to factor proportions by at least half a century. In Appendix Tables 15 and 16, I show that by 1750 plagued cities and parishes were no smaller than unplagued ones. Given population recovery, factor proportion theories of trade would suggest that the shift across sectors should be reversed. To the contrary, Appendix Figure 15 shows that the shift into capital-intensive exports persists for almost 90 years after the plague. Capital intensity thus remains significantly elevated despite population recovery.

Fact #2: Plagued regions capture larger shares of destination markets.

I estimate a second event study incorporating the gravity structure of trade:

$$T_{ijt} = \sum_{l=-20 \setminus -1}^{30} \beta_l \cdot \mathbb{1}(K_{it} = l) + \alpha_{ij} + \alpha_{jt} + \alpha_{at} + \gamma x_{it} + \epsilon_{ijt}, \quad (2)$$

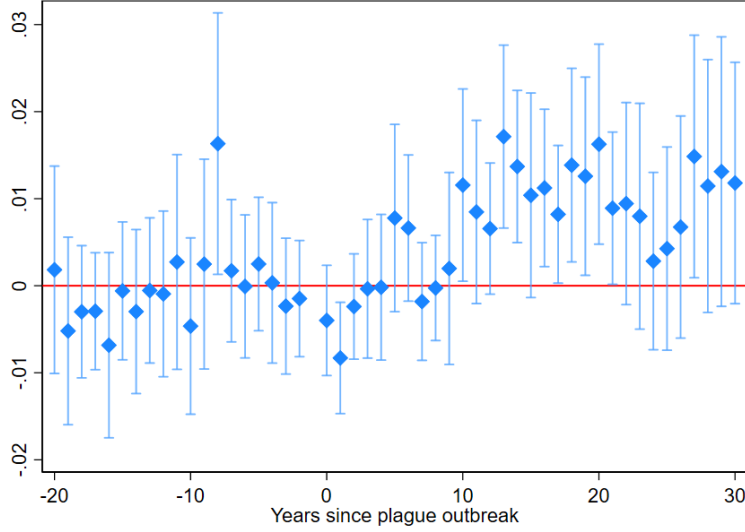
where T_{ijt} is one of several outcomes for trade from origin i to destination j in year t . K_{it} is the time difference between the current year and the plague, α_{ij} is a bilateral fixed effect, and α_{jt} are destination-time fixed effects absorbing the demand side. ϵ_{ijt} is the error term. α_{at} are area-time fixed effects.¹⁸ These permit for areas’ different time trends and thus permit heterogeneity by area. In particular, they absorb area-wide changes in market access due to territorial changes, such as Estonia and Latvia becoming a part of Russia, and area-wide shocks such as famines. x_{it} are time-varying origin controls, in particular growing season temperature, longitude $\times$ year, and latitude $\times$ year. β_l are the outcomes of interest.

For Fact #2, T_{ijt} is either the volume of exports from i to j in t , or the export share that origin i captures in destination j : $T_{ijt} = \frac{x_{ijt}}{\sum_{i \in I} x_{ijt}}$.¹⁹ This formulation of T_{ijt} as the share of exports from origin i in destination j in year t motivates the Ricardian model below, which predicts trade shares. For results on the extensive margin and on new exports, T_{ijt} is the number of goods exported from i to j in t .

¹⁸Areas are displayed in Appendix Figure 7. For most of Europe, these are equivalent to NUTS2; for the UK, Germany, France, and Spain, they are NUTS1. This is to ensure similarly sized areas and to avoid areas with a single port (Appendix B).

¹⁹I set market shares of i in j to 0 whenever j shows no imports in my data. As the data capture only trade between the North and Baltic Seas, I thus assume that every region has some imports, even if unobserved to me.

Figure 4: Market shares expand



Notes: Estimation of equation 2 on i 's share of j 's imports, $T_{ijt} = \frac{x_{ijt}}{\sum_{i \in I} x_{ijt}}$. Standard errors clustered at the origin level.

Contrary to concerns of an export contraction, Figure 4 shows that plagued cities increase their export shares by about 1pp, which is close to the pre-plague mean export share of 0.9%. Appendix Figure 16a also shows an expansion of export volumes.

The export expansion begins shortly after the plague, with the first significant point estimates for export volumes three years later. Ten years after the plague, these regions capture significantly larger market shares in destination markets. This time difference reflects the above-discussed war-related disruptions to shipping in the Baltic Sea, which were unrelated to the plague (Appendix Figure 10 and Appendix Table 19).

Notably, these results reject the hypothesis that trade simply bounced back. There is a decade of peace during which to-be-plagued ports do not display different export patterns. Appendix Table 23 presents results from a difference-in-differences set up using OLS and PPML, excluding the war years of 1700 to 1721. Appendix Figure 17a extends the pre-period to 30 years and again finds no pre-trends. To more flexibly control for temperature changes, Appendix Figure 20a additionally controls for temperatures with lags of 10 and 20 years. Appendix Figure 17b constructs market shares from cleaned underlying value correcting for duty rates (Section 2.4 and Appendix A.4). Appendix Figure 18 finds very similar results with NUTS 1-time and NUTS 2-time fixed effects instead of the proposed area-time fixed effects.

Appendix Figure 20b implements the estimator by Borusyak, Jaravel, and Spiess, 2021. As the summation of ATEs may involve negative weights, I check and find that all weights are positive and conclude that the standard estimator is appropriate (Chaisemartin and D'Haultfoeuille, 2022). Similarly, I address staggered treatment timing by implementing the decomposition in Goodman-Bacon, 2021. I find that more than 99.1% of the identifying variation comes from comparisons between never-treated and eventually-treated ports, with less than 0.9% coming from comparisons across treatment timings. My estimates are therefore virtually unaffected by potential bias of this kind.

Appendix Figure 19 decomposes the intensive margin results into sectors. Two decades after the plague, I find that capital-intensive manufacturing and agriculture con-

tribute each a quarter to the overall export expansion, with the remaining half accruing to labour-intensive agriculture. Labour-intensive manufacturing plays virtually no role. The outsized role of labour-intensive agriculture is not surprising as it makes up 55% of traded value between 1668 and 1750. Aggregating sectors by factor intensity instead, Appendix Figure 21 shows that both saw export expansions. These results choose T_{ijft} to be the volume of exports by factor intensity. At a more granular level, Appendix Figure 22 presents results at the goods level.

The increase in destination market shares may be indicative of productivity growth as plagued regions became more competitive. This export expansion outlives population recovery which had been achieved by 1750. Appendix Figures 23a and 23b show that up to 80 years after the plague, destination market shares and export volumes are still larger in previously plagued regions.

Extensive Margin: Plagued regions export a larger variety of goods.

The extensive margin plays an important role in the export expansion: Appendix Figure 16b shows that plagued regions increased the number of goods they export. Through the lens of a Ricardian model (Eaton and Kortum, 2002), the extensive margin expansion drives the results on market shares. Plagued regions becoming the most competitive suppliers of more goods to destination markets is therefore an indication of productivity growth and observationally consistent with an export expansion along the intensive margin. Appendix Figure 24 decomposes the extensive margin expansion by sector. Appendix Figure 25 shows these results overall for capital- and labour-intensive sectors. Appendix Figure 26 shows results at the level of goods.

3.1 Other Robustness Results

I provide results from a PPML panel regression on trade levels in Appendix Table 26. The results show that trade increased after one side suffered a plague outbreak. Considering the set of fixed effects, however, this result draws on different variation from my main result. Further, Appendix Table 27 shows results for the probability of a region exporting in a given year. After a plague outbreak, regions' probability to be active exporters increases significantly. I present heterogeneity results in Appendix Table 28. While the effects are larger for regions that exported more before the plague, they are significant even for regions that had no exports before the plague. Appendix Table 30 shows that the presence of a besieged city does not affect the post-plague trade findings. Finally, I check if the specification of treatment is sufficient in focusing on plague outbreaks in regions themselves. In Appendix C.5, I describe how I construct a continuous plague treatment variable that captures plague outbreaks across all of Europe. I show in Appendix Table 33 that the indirect treatment variable is never significant.²⁰ Further, my findings on the direct plague treatment are robust to including the indirect plague treatment.

3.2 Mechanical Explanations

I argue that my findings are not driven by mechanical forces. First, the observed export expansion might reflect a reallocation of exports from the Baltic Sea to the North Sea. Heterogeneity across sectors speaks against a simplistic reallocation. Further, this raises

²⁰This also holds for imports. Results available on request.

the question of how plagued exporters became competitive in new markets where they competed against existing exporters. The combination of new markets and longer distances implies that regions in the Baltics would have had to overcome the cost of distance and higher wages through higher productivity to be competitive.

To more rigorously address this concern, I show simulation results in Appendix Figure 31 for a simulated productivity increase and its effect on trade flows. Both in the hypothetical full and in the actually observed sample, where only trade passing the toll station is registered, the productivity increase is reflected in significantly higher exports. For both samples, the point estimates are centred around the simulated effect, suggesting no bias.

Territorial changes may also affect the results. Sweden lost control of modern-day Estonia, Latvia, and parts of Northern Germany. This affected these regions' market access to Sweden, but also Swedish ports' market access to the lost territories. This concern is addressed by including area-time (e.g. Estonia-year) fixed effects. The variation I draw on is between ports within Estonia, all of which underwent changes in market access. These fixed effects thus account for territorial changes. Relatedly, the late 17th and early 18th century saw several particularly harsh winters and led to famines. As little localised information is available on famines, I use area-time fixed effects to absorb area-wide famines. In Appendix Figure 20a, I additionally control for growing season temperatures and for temperatures with a lag of 10 and 20 years to control for harsh winters.

A second mechanical explanation concerns the role of harbours as export hubs. If a harbour exports hinterland production, a share of these products will be consumed in the port itself. Following a plague outbreak, local consumption may drop, and, assuming constant hinterland production, exports should rise.²¹ To test whether local demand contraction in export hubs drives my results²², I show in Appendix Figure 29 that exports shipped to plagued regions increase. The mechanical explanation predicts a decline in imports, however. Further, Appendix Table 28 shows that also the smallest harbours that are unlikely to have served as export hubs see an export expansion.

Relatedly, one could argue that my findings are driven by exporter-specific declines in trade costs related to the plague or to a higher capital intensity of exports. After the plague, harbour operations may have become more efficient, and capital-intensive goods may be faster to load onto ships. To address this issue, I count the number of days between two appearances at the Sound of the same captain (identified by full name and home town) on return trips, where the first leg's destination overlaps with the second leg's origin. This is an exporter-specific measure of both travel days and handling time at harbours, where captain-specific and origin-time determinants of days at sea are absorbed by fixed effects. In Appendix Table 29, I show that travel and loading times are unaffected by either the plague or by capital intensity. Thus, trade costs are unrelated to the shock.

Finally, I provide evidence in Appendix Table 31 against both concerns - selection and mechanical reallocations - by drawing on data by Baasch, 1929 that cover all of Hamburg's trade by commodity between 1702 and 1713 (Pfister, n.d.). Exports are recorded with all of Hamburg's trading partners. This provides more detail than the Soundtoll data, which are restricted to trade between the Baltic and North Seas. I find evidence that

²¹I challenge the notion that the hinterland's production would not have been affected. The plague was spread by moving armies, which plundered the hinterland and disrupted production.

²²In national accounting terms, exports rise if domestic absorption falls by more than income, such that production decreases less than consumption.

exports expand after the plague and shift into capital-intensive commodities. This dispels concerns that Baltic exporters simply entered into markets that plagued Hamburg had left. Similarly, this finding confirms the export expansion and shift into capital-intensive exports for one port for which I have export data for all destinations. As Hamburg was unaffected by naval warfare and sees effects on impact, this suggests that the time lag discussed above is indeed driven by war. In sum, mechanical forces are unlikely to drive my findings.

4 A Ricardian Model

I propose a Ricardian model to explain my empirical findings. In Malthusian models with fixed productivity, an adverse labour supply shock would be followed by an export decline as wages increase. If only Heckscher-Ohlin forces were at play, one would observe a reallocation to capital-intensive production but no overall increase in exports. I argue more extensively against pure Heckscher-Ohlin forces in Section 6. Finally, factor theories predict factor price equalisation.²³ As all three are rejected by my findings, I build a Ricardian model with intra-industry variation in productivity (Donaldson and Hornbeck, 2016, Costinot, Donaldson, and Komunjer, 2011, Eaton and Kortum, 2002).

There are many regions, $i=1,\dots,I$, and two factors of production: labour L_{ik} and capital K_{ik} . There are five sectors indexed by k : labour-intensive agriculture (LA), capital-intensive agriculture (CA), labour-intensive manufacturing (LM), capital-intensive manufacturing (CM), and unclassified goods (U). Throughout, I am dropping the t subscript indicating time for simplicity.

Unique Spatial Equilibrium: Population recovery is an indicator of a unique spatial equilibrium (Davis and Weinstein, 2002). I use city population data by Buringh, 2021 to show that by 1750, plagued cities and parishes had returned to their pre-plague growth paths (Appendix Table 15 and 16). While for a sufficiently small shock population may also recover in a NEG setting such as Krugman, 1991, I argue that the median mortality rate of 36% is anything but small. Thus, the model has a unique spatial equilibrium.

Factors and Sectors: The model features labour and capital. Each sector has a labour share γ_k . Capital is assumed to be freely mobile across regions and sectors. For each sector k the production function takes the Cobb-Douglas form:

$$P(L_{ik}, K_{ik}) = A_{ik} L_{ik}^{\gamma_k} K_{ik}^{1-\gamma_k}. \quad (3)$$

One might wonder whether land ought to be included. I argue that land was abundant in the Baltics: Raster, 2023 shows that population densities were so low that after the plague entire manors were abandoned; Allen, 2003 proposes that serfdom in the Russian Empire was an institution to preserve the power of the owners of abundant land over scarce peasants. Therefore, I argue that labour was the constraining factor. Had land

²³With more than two countries, Helpman and Krugman, 1985 show that FPE continues to hold if all countries fall within the FPE set, which comes down to endowments not being too dissimilar. Schott, 2003 shows that FPE is achieved for many countries if they are in the same diversification cone: as countries become more similar in endowments, FPE holds approximately also for $n>2$, and precisely when endowments are identical.

Table 2: Spatial Separation & Capital Intensity

Region		Labour-intensive		Capital-intensive	
		Labour-int. manufacturing		Capital-int. manufacturing	
	City	(LM)	$\gamma_{LM} > \gamma_{CM}$	(CM)	
	Hinterland	Labour-int. agriculture	$\gamma_{LA} > \gamma_{CA}$	Capital-int. agriculture	(CA)

Notes: Assumptions on spatial separation, sectors, and factor intensities. γ_k denotes the labour share in each sector k .

been a constraining factor, the increased relative supply of land should benefit the production of land-intensive goods; Appendix Figure 36 finds no evidence in production data and Appendix Figure 14 find no shift into land-intensive exports. Further, my findings are also present within manufacturing, where land intensities should play a role (Appendix Table 22 and Appendix Figure 21). Therefore, I argue that the crucial comparison is between labour- and capital-intensive goods and do not model land as a factor.

I assume cities to produce manufacturing and hinterlands to produce agricultural goods. This assumption finds historical support in policies that restricted manufacturing to cities. In Sweden, the textile industry was restricted to cities, with rural producers supplying only the input goods; similarly, urban blast furnaces were to be supplied by rural pig iron (Magnusson, 2007). To ensure this separation, a ‘town economic policy’ was introduced in the 17th century. In a similar vein, Klein and Ogilvie, 2015 describe how urban institutions hindered rural industrial production.

I define a region as a city and its hinterland. Larger cities on navigable rivers may boast larger hinterlands; all I assume is that this does not change over the study period. Swedish cities, for example, enjoyed trade monopolies over their surrounding area, which only changed in 1765 when the Commodity Act was withdrawn (Magnusson, 2007). All 1,425 cities in the data are coastal, and the average diameter of the hinterland is 25 kilometres. Table 2 summarises these assumptions.

Labour Mobility: With labour mobility across regions, wage differentials are eliminated through migration. One would have to observe an immediate population recovery under this assumption; however, I find that population recovery was achieved only by 1750. The literature on plague outbreaks finds significant wage increases (Jedwab, Johnson, and Koyama, 2022), implying local labour scarcity and the absence of factor price equalisation. The latter is supported by Davis and Weinstein, 2001. Accordingly, I model labour as immobile across regions but mobile across sectors within a region. Plagued cities, where mortality rates were higher than in the hinterland, can recover via rural-to-urban migration. This is true in areas without serfdom. Under serfdom, however, peasants cannot move to cities and a wedge between urban and rural wages persists.

Workers are mobile within agriculture and within manufacturing in a given region, implying wage equalisation.²⁴ Let $w_i^{CM} = w_i^{LM} \equiv w_i^M$ and $w_i^{CA} = w_i^{LA} \equiv w_i^A$. Without serfdom, all four sectors pay equal wages in a region. With serfdom, there are equalised wages within agriculture and within manufacturing. The wedge between urban and rural wages, ϕ_i , will be region-specific and depends on mortality differences between the two parts of a region. ϕ_i also captures language and skill frictions. Equation 4 summarises

²⁴While serfs are not paid wages and cannot move between different agricultural employments, the assumption here is that their lords efficiently allocate them to equalise the marginal products of labour between labour-intensive and capital-intensive agriculture.

the assumptions on labour mobility. Appendix Figure 43 shows that a wedge ϕ_i appeared in Denmark after the re-introduction of serfdom.

$$w_{iM} = \begin{cases} w_{iA}, & \text{without serfdom,} \\ (1 + \phi_i)w_{iA}, & \text{with serfdom.} \end{cases} \quad (4)$$

Productivity: Each sector's output features in an infinite number of varieties, $\omega \in \Omega$. Productivity A_{ik} is a random variable, drawn independently for each (i, k, ω, t) from a Fréchet distribution as in Donaldson and Hornbeck, 2016:

$$F_{ik}(z) = 1 - \exp(-A_{ik}z^{-\theta}), \quad (5)$$

where $A_{ik} > 0$ and $\theta > 1$. A_{ik} captures fundamental productivity of region i in sector k , and encompasses productivity-affecting fundamentals that pertain to all producers in that region and sector. θ reflects intra-industry heterogeneity.²⁵ In this model, productivity is therefore total factor productivity and not factor biased (Acemoglu, 2002).²⁶

Capital Intensity: Producers take the variety-level draw $A_{ik}(\omega)$ as given when allocating labour and capital. Factor prices equal marginal products, so the production of each variety ω becomes more intensive in capital after a wage increase:

$$\frac{K_{ik}(\omega)}{L_{ik}(\omega)} = \frac{1 - \gamma_k}{\gamma_k r} w_{ik}.$$

The capital-to-labour ratios rise also overall, with $K_{ik} = \int_{\omega \in \Omega} K_{ik}(\omega)$. The model therefore predicts the shift *within* sectors into capital-intensive production. The strength of this shift depends on the elasticity of substitution between factors: this equation is an artefact of the Cobb-Douglas production function, where the elasticity of substitution between capital and labour is 1. Alternatively, production functions could be linear (with an infinite elasticity of substitution) or Leontieff (with an elasticity of substitution of 0). However, a linear production function has empirically unsupported corner predictions for factor demand; further, the rising post-plague capital-to-labour ratio (see Section 5) speaks against a Leontieff function.

Marginal Costs: The marginal cost of production is given by:

$$MC_{ik}(\omega) = \frac{(w_{ik})^{\gamma_k} r^{1-\gamma_k}}{A_{ik}(\omega)}. \quad (6)$$

This is a model of constant marginal costs. My finding relates to the theory of venting out (Almunia et al., 2021), in which a domestic demand slump leading to an export boom is rationalised by non-constant marginal costs of production. In Section 6, I argue that one can better rationalise my findings without assuming non-constant marginal costs.

Trade Costs: I assume the standard iceberg form. Trade costs are symmetric, $d_{ijk} = d_{ji,k}$. The no-arbitrage condition $d_{il,k} \leq d_{ijk}d_{jl,k}$ is also imposed. Trade costs may vary by sector and time, thus subsuming any potential tariffs.

²⁵Costinot, Donaldson, and Komunjer, 2011 elaborate on the implications of assuming $\theta_k = \theta \forall k$.

²⁶Davis and Weinstein, 2001 argue that a model of Hicks-neutral technical differences is best suited to explain global production structures.

Market Structure & Prices: Markets are perfectly competitive and consumers in j purchase a variety ω from its cheapest supplier location i :

$p_{jk}(\omega) = \min_{1 \leq i \leq I} c_{ijk}(\omega)$, where $c_{ijk}(\omega)$ is assumed to be strictly positive. Following Redding and Venables, 2004 and Donaldson and Hornbeck, 2016, consumer market access and firm market access are defined as the price indices:²⁷

$$CMA_{jk} = (P_{jk})^{-\theta} = \chi_k \sum_{i=1}^I A_{ik} (w_{ik})^{-\gamma_k \theta} d_{ijk}^{-\theta} \quad (7)$$

$$FMA_{ik} = \chi_k \alpha_k \sum_{j=1}^J d_{ijk}^{-\theta} (CMA_{jk})^{-1} Y_j. \quad (8)$$

Wage Rate & Labour Force: Wages are determined sectorally. Combining market clearing conditions and plugging in firm market access yields:

$$w_{ik} = \zeta_k \left(\frac{A_{ik} FMA_{ik}}{L_{ik}} \right)^{\frac{1}{1+\gamma_k \theta}}. \quad (9)$$

Preferences: The representative consumer has a two-level utility function, where the upper tier is Cobb-Douglas and the lower tier is CES. α_k are the Cobb-Douglas weights in the upper tier and $\sigma_k > 1$ is the elasticity of substitution between differentiated varieties in the lower tier. I assume $\sigma_k < 1 + \theta$. A continuum of these differentiated varieties ω are consumed. I will later follow Fieler, 2011 in introducing non-homotheticity.

Trade Flows: Trade flows take the gravity form:

$$X_{ijk} = \frac{\chi_k A_{ik} (w_{ik})^{-\gamma_k \theta} d_{ijk}^{-\theta}}{CMA_{jk}} \alpha_k Y_j. \quad (10)$$

Region i exports more to region j in sector k if it has a higher productivity, A_{ik} , lower trade costs, d_{ijk} , or lower wages, w_i . Region i further exports larger volumes to j if region j has low consumer market access in a sector, which means low competition for i .

Equation 10 can be used to recover productivity growth. Adding a time dimension to the procedure in Costinot, Donaldson, and Komunjer, 2011, I estimate using PPML:

$$X_{ijkt} = \exp(\alpha + \delta_{ijt} + \delta_{jkt} + \delta_{ikt}) \times \epsilon_{ijkt}, \quad (11)$$

where all δ parameters are fixed effects and ϵ_{ijkt} is the error term. The denominator in equation 10 is absorbed by δ_{jkt} fixed effects, which also absorb inward multilateral resistance (Fally, 2015, Anderson and Wincoop, 2003). There is no structural outward multilateral resistance term. However, wages absorb the general equilibrium feedback that would show up in a structural outward multilateral resistance term. The critical assumption is that wages are common across sectors and differenced out (see Appendix Section D.1). In this case, there is no outward multilateral resistance contamination in the structural sense, and δ_{ikt} recovers relative productivity growth.

²⁷Note that the existence of these CES price indices requires the assumption $\sigma_k < 1 + \theta$.

²⁸ $\chi_k = \left(\Gamma\left(\frac{\theta+1-\sigma_k}{\theta}\right) \right)^{\frac{-\theta}{1-\sigma_k}} r^{-(1-\gamma_k)\theta}$; $\zeta_k = \left(r^{1-\gamma_k} \right)^{\frac{-\theta}{1+\gamma_k \theta}}$

5 Mechanism

I propose capital deepening-driven productivity growth to explain plagued regions' export success. In response to labour scarcity (Step #1), producers increase the amount of capital they use relative to labour (Step #2). While not internalised by producers, a sector-wide increase in the capital-to-labour ratio increases productivity (Step #3).

Importantly, I do not endogenise productivity. The literature recovering productivities requires the assumption of variety-level productivity draws from a given distribution (Costinot, Donaldson, and Komunjer, 2011) and views the distribution's scale parameters as fundamentals. Origin-sector variation in trade data has been projected on observables to assess the determinants of productivity (Chor, 2010). Following this literature, I relate recovered productivity growth to the plague as a shock to factor proportions while absorbing other determinants with fixed effects.

I propose a reduced-form relationship to lend economic meaning to sectoral productivities. While producers take the variety-level draw $A_{ik}(\omega)$ as given, I suggest that the Fréchet distribution's scale parameters A_{ik} are a function of the capital-to-labour ratio $\frac{K_{ik}}{L_{ik}}$, a vector permitting permanent effects of shocks, H_{ik} , and sectoral, regional, and region-sector determinants x_i, x_k, x_{ik} :

$$A_{ik} = g\left(\left(\frac{K_{ik}}{L_{ik}}\right)^{\beta_k}, H_{ik}, x_i, x_k, x_{ik}\right). \quad (12)$$

The function $g(\cdot)$ is increasing in the capital-to-labour ratio, whose impact on productivity is scaled by β_k . Sectors differ in the degree to which increased capital can move productivities and technological barriers present the main hurdle. Appendix Table 35 lists examples of capital deepening available in this period. In the production of wheat, e.g., capital deepening in the early 18th century had limited effects on productivity, as no modern farm machinery was available beyond limited tools. The scope for capital to increase agricultural productivity was severely limited in this period, with most productivity-increasing devices centuries old (Gallardo and Sauer, 2018 and Coleman, 1956). Attack, Margo, and Rhode, 2019 concludes that few technologies were available before the 19th century to increase productivities in labour-intensive manufacturing. In capital-intensive iron and steel making, on the other hand, capital deepening and investments into more efficient furnaces more readily translate into productivity improvements.

H_{ik} accounts for lasting effects of shocks. While my model is static, I allow for permanent productivity effects of shocks. This formulation generalises the arguments in Krugman, 1987, Juhász, 2018, and Lane, 2025. As there are no granular data for this period to test the elements of x_i and x_k , I absorb this variation with fixed effects. Origin-destination-time and destination-sector-time variation are absorbed when decomposing gravity equation 11. Below, I include origin-sector fixed effects to absorb the time-invariant part of vector x_{ik} . I also absorb area-sector-time variation with fixed effects, leaving only origin-sector-time variation within areas.

I then relate the remaining variation to the plague as a proxy for the increased capital-to-labour ratio. This simplifies the reduced-form equation above:

$$A_{ik} = \left(\frac{K_{ik}}{L_{ik}}\right)^{\beta_k}. \quad (13)$$

Given the variation which I propose to be driven by capital deepening, one alternative are human and social capital. If these were sectorally varying - in levels, they are absorbed

by fixed effects - and respond to labour scarcity, the numerator in equation 13 may not capture only physical capital K_{ik} , but rather the sum of physical, human, and social capital, which plausibly all shift in the same direction after the plague. Goods that are intensive in physical capital may also require higher social or human capital. I thus need to check whether, holding factor intensity constant, there is a relative export shift into goods that intensively use human and social capital. In Section 6, I find no evidence for human or social capital deepening. Even when controlling for how intensively human and social capital are used in the production of goods, I document a strong relative shift into capital-intensive exports after the plague. Thus, I propose that equation 13 captures the most relevant reduced-form relationship between factor proportions and productivity.

Micro-Foundation

I propose a micro-foundation that produces equation 13 in Appendix E. I introduce technology choice as an endogenous component to productivity (Bustos, 2011). Producers may use a labour-intensive technology that requires low fixed costs. Alternatively, they may incur a higher fixed cost to produce under a capital-intensive technology. Examples of capital-intensive technology in this period are provided in Appendix Table 35. This provides an endogenous boost to the exogenous Fréchet productivity draw, which I expect to be larger in capital-intensive sectors. The trade-off defines a threshold productivity draw beyond which the capital-intensive technology is employed. As wages rise, this threshold productivity drops; productivity and the capital-to-labour ratio rise. This defines an equation mirroring equation 13. The parameter β_k now reflects sectoral differences in the productivity boost conferred on users of the capital-intensive technology.

Further, learning by doing (Krugman, 1987), learning by exporting (Loecker, 2013), and cost discovery (Hausmann and Rodrik, 2003) can micro-found my proposed mechanism. Faced with a labour supply shock, plagued regions have to experiment with new factor proportions and new products. In doing so, they may reap the benefits of learning by doing or exporting and may discover their true comparative advantage. These all involve a higher capital-to-labour ratio and predict productivity growth. While I am unable to differentiate between these competing micro-foundations, all of them would correctly be picked up as productivity growth below. Further, they all imply that comparative advantage patterns are not fully determined by contemporaneous fundamentals. In both senses, these micro-foundations are consistent with my proposed mechanism.

Step #1: The Plague Induces Labour Scarcity

The plague reduced urban (regional) populations by 36% (20%)²⁹ on average, which increases wages. According to equation 9, supposing $\gamma_k = 0.5$ and $\theta = 5$, a 20% decrease in employment increases sectoral wages by 6.6%. The same is true for regional wages, as sectoral employment adjusts so as to arbitrage away sectoral wage differences. Appendix Figure 32 shows model-implied wage growth for the whole observed range of mortality rates and different labour shares.

²⁹Mortality estimates in Appendix Table 12 pertain to cities. 20% is in the middle between the upper and lower bound for regions. Mortality rates were lower in hinterlands (Voigtländer and Voth, 2012). Assuming 36% to hold for regional population is thus an upper bound. Bairoch, Batou, and Chèvre, 1988 finds a European urbanisation rate of 11.4% in 1700, similar to the 11.9% in Vries, 2013. As a lower bound, if there had been no mortality in the hinterland, regional mortality would lie at 4.3%.

Evidence from Population, Wages, and Interest Rates

In data by Buringh, 2021, I show that plagued cities' populations had recovered by 1750. Additionally, I propose the number of captains reporting to live in a region as a proxy providing annual data.³⁰ Appendix Figure 33 shows an event study on the number of captains, where the average point estimate corresponds to a 50% population drop. The number of captains climbs back to pre-plague levels after 15 years. This likely overstates both the population shock and recovery, as captains were highly skilled and mobile. Thus, labour scarcity is present in plagued regions for 15-50 years.³¹

Model-based wage growth is low compared to the Black Death. Jedwab, Johnson, and Koyama, 2022 argue that high mortality disrupted trade and increased transaction costs, decreasing real wages in England and Spain for two years. Following Alfani and Murphy, 2017, I suggest that markets did not unravel after the Great Northern War plague outbreak as institutions were better prepared for a mortality shock. Thus, wage increases after this plague outbreak are likely much closer to theoretical predictions.

There is evidence from the Black Death that wage increases were real. Broadberry et al., 2014 show that dairy and meat provided a larger share of calories in England after the plague. Galofré-Vilà, Hinde, and Meera Guntupalli, 2018 document an average height increase of 6.59 cm between 1348 and 1400. Pfister, 2017 provides evidence closer to my setting: real wages of unskilled urban male labourers in Germany doubled as the Thirty Years' War induced labour scarcity through war and plagues.

Importantly, wages rose relative to return rates on capital. Data on city-level interest rates in the Baltic and North Sea region are scarce in the early 18th century. Even the most extensive collections of sources (Huang, Chilosì, and Sapoznik, 2019, Schmelzing, 2019) cover only three of the plagued cities: Hamburg and Danzig, yet only before the plague, and Bremen, which had a mortality rate of 0.7%, whose factor proportions were thus barely affected, and which accordingly saw virtually no variation in interest rates. Thus, I instead follow Keller, Shiue, and Wang, 2021 who infer interest rates from monthly variation in grain prices. I use price data by Allen and Unger, 2018 for Danzig in 1704-1793. Appendix Table 34 regresses interest rates recovered from wheat, rye, and barley prices on a post-plague dummy and finds that interest rates do not increase after the plague. I conclude that both wages rise absolutely and relatively.

Step #2: Production Becomes More Capital-Intensive

Labour scarcity is followed by an increase in capital intensity along two margins. First, a higher capital-to-labour ratio is used:

$$\frac{K_{ik}}{L_{ik}} = \frac{1 - \gamma_k}{\gamma_k r} w_{ik}.$$

Second, a shift into capital-intensive sectors occurs. Taking ratios of gravity equation 10 for manufacturing and plugging in equations 13 and factor market clearing yields:

³⁰This variable is the home port as in Marczinek, Maurer, and Rauch, 2025. To count captains once, I omit observations in which first name, last name, and region are duplicates in a given year.

³¹I explored the Hyde data. However, these are either imputed or at the national level, using linear interpolations between centuries, without providing genuine localised annual population data. As I need regionally granular and ideally annual population data, the Hyde data are not the ideal source either.

$$\frac{X_{ijCM}}{X_{ijLM}} = \xi w_{iM}^{-\theta(\gamma_{CM}-\gamma_{LM})} w_{iM}^{\beta_{CM}-\beta_{LM}} \frac{CMA_{jLM}}{CMA_{jCM}}. \quad (14)$$

Three channels operate. First, capital-intensive sectors expand as they have smaller labour shares. This is the **labour cost channel**. Second, capital-intensive sectors expand as capital deepening increases productivities. For $\beta_{CM} > \beta_{LM}$, the increased capital-to-labour ratio increases productivity in capital-intensive manufacturing more than in labour-intensive manufacturing. This results in the second term in equation 14. This **productivity channel** further increases the expansion of capital-intensive sectors.

In general equilibrium, $\frac{CMA_{jLM}}{CMA_{jCM}}$ decreases, counteracting some of the shift into capital-intensive exports. In a labour-intensive sector, wage growth weighs more heavily on an origin's contribution to a destination's market access. Beyond that, differences in β may lead to slower productivity growth in labour-intensive sectors. Both work to decrease $\frac{CMA_{jLM}}{CMA_{jCM}}$. In other words, plagued regions' labour scarcity and productivity advances increase the relative (export) prices of labour-intensive compared to capital-intensive goods. This relative price channel reduces the value of capital-intensive exports compared to labour-intensive exports. I control for this channel by estimating equation 2 with destination-time fixed effects.

Evidence from Capital Stocks

Combining market clearing conditions and sectoral wages 9 reveals:

$$K_{ik} = \frac{(1 - \gamma_k)\zeta_k}{\gamma_k r} \left(A_{ik} F M A_{ik} \right)^{\frac{1}{1+\gamma_k\theta}} L_{ik}^{\frac{\gamma_k\theta}{1+\gamma_k\theta}}. \quad (15)$$

Equation 15 shows the elasticity of capital stocks with respect to employment, which is less than 1. Thus, the capital-to-labour ratio increases. While no granular data are available, I employ a proxy for capital stocks: the number of ships owned in a region.³³ Ships are an expensive infrastructure investment and highly mobile.³⁴ The plague leads to a larger number of ships (Appendix Table 36). For two decades after the plague, significantly less ships are registered in plagued regions (Appendix Figure 34). A comparison with the number of captains (Appendix Figure 33) fits an elasticity of less than 1. 50 years after the plague, there were significantly more ships registered in previously plagued regions; this higher capital stock remained in place until the 1850s. The drop and recovery in population and lasting productivity gains, following equation 15, can explain these dynamics. Beyond ships, I find that plagued ports import larger volumes of capital goods (metals, tools, nails, building materials), also relative to an overall growth in imports (Appendix Figure 35). This suggests that capital stocks are being accumulated.

Evidence from Production Data

In trade data, the shift *across* sectors is observable (Figure 3). I find further evidence in farm-level data from Sweden: farms shift out of labour-intensive agriculture and into

³² $\xi = \frac{\alpha_{CM}}{\alpha_{LM}} \frac{\chi_{CM}}{\chi_{LM}} \left(\frac{d_{ijCM}}{d_{ijLM}} \right)^{-\theta}$

³³ The labour cost and productivity channels determine the allocation of capital stocks across sectors.

³⁴ The export expansion is not mechanically linked to an accumulation of ships. Merchants from other regions could sail to productive locations.

capital-intensive agriculture.³⁵ I draw on data by Olsson and Svensson, 2017. After the plague, Scania’s farms reduce their production of labour-intensive rye and barley and shift into capital-intensive calves (Appendix Figure 36).³⁶

Thus, insights from trade data carry over to production data. To test the link with labour scarcity, I construct for each farm i a measure of distance to Ystad and Malmö:

$$plaguedist_{it} = \sum_j Distance_{ij} * Plague_{jt}. \quad (16)$$

Farms located closer to plagued Ystad or Malmö have lower values of $plaguedist_{it}$. I expect greater proximity to plagued cities to be associated with higher mortality. Accordingly, the shift into capital-intensive agriculture should be increasing in proximity to plagued cities. In Appendix Table 37, I find that proximity to plagued cities is associated with higher growth in calves production and lower growth in rye and barley production.³⁷

As for the shift *within* sectors, I use data for 2,344 Swedish parishes in 1690 and 1750 on land usage, animals, and population (Andersson Palm, 2025). Plagued parishes use significantly more horses, controlling for population, after the plague (Appendix Table 38). This holds both in 1750 and in 1810.³⁸ By the 1700s, horses had become the main draught animal in much of Europe (Langdon, 1986), though oxen may still have played a role in Sweden. Thus, more horses per capita reflect capital deepening *within* agriculture, as also argued by Olsson and Svensson, 2008.

In Appendix Table 39, I introduce interactions with Scanian farms’ soil quality. Despite the labour scarcity-induced shift out of rye, I find that farms with bad soil close to plagued cities shift into rye - the hardest grain. Faced with both scarce labour and poor soil, farms shift into what is easiest to grow. Labour scarcity hits barley production uniformly independent of soil quality. The overall shift out of rye and into foals is driven by farms with good soil close to plagued cities. This is capital deepening in arable agriculture: horsepower is being substituted for lost labour, and particularly so with rich soil. Soil quality does not interact with other animals, supporting that this is about draught animals specifically.³⁹ By 1750, population had recovered in plagued parishes (Appendix Table 16). Thus, both capital levels and the capital-to-labour ratio were significantly higher in 1750 and in 1810 in plagued parishes despite population recovery.

Evidence from Exports of Novel and Innovative Goods

Investment and innovation are difficult to study in this period. As a proxy, I investigate novel export goods that indicate innovation and investment into new production lines. An export expansion may indicate both more intensive usage of existing production knowledge and a creation of new production knowledge, which one may refer to as innovation. A higher capital-to-labour ratio may have allowed plagued regions to make the investments necessary to acquire new production knowledge. This link between innovation and labour scarcity finds empirical support in Andersson, Karadja, and Prawitz,

³⁵The same pattern was observed in English agriculture after the Black Death (Clark, 2016).

³⁶While calves grow up to be cows and will produce meat and milk, some regard raising foals and calves as a metric of capital deepening in agriculture (Olsson and Svensson, 2008).

³⁷The overall regional trend out of rye and barley and into calves is accounted for by time trends.

³⁸The number of ships as a proxy for ports’ capital stocks is also significantly higher at and beyond this time horizon (Appendix Figure 34).

³⁹Beyond draught animals, there is evidence of capital deepening through metal ploughs, early threshing machines and modern farm buildings, where plagued Scania stands out as an early adopter (Magnusson, 2007).

2020. I partial out goods that regions first exported between 1689, 20 years before the first year of the plague, and 1732, 20 years after the end of the plague.

Appendix Figure 37 shows that the number of new export goods increases significantly faster in plagued regions. Novel export goods contribute a quarter to the overall extensive margin expansion. Appendix Figure 27c shows that novel export goods are significantly more frequent in capital-intensive sectors. Results at the goods level are shown in Appendix Figure 28. Appendix Tables 24 and 25 show PPML results. Even after 20 years, plagued regions continue to innovate more than non-plagued ones. This fits with the findings of Juhász, Squicciarini, and Voigtländer, 2024.

Durability of Capital

I argue that my finding of capital deepening after the plague is not driven by different durabilities of capital. As blast furnaces last longer than a plough, capital deepening in capital-intensive sectors may be longer-lived than in labour-intensive sectors. I provide three arguments against this. First, the time horizon over which I find capital deepening suggests continued reinvestment rather than depreciation of a one-off increase in capital stocks. Following Allen, 2009a, suppose the aggregate depreciation rate sits at 2.4%. This implies a half-life of 29 years for the capital stock. However, the capital intensity of exports and the number of horses per capita were higher almost a century after the plague (Appendix Figure 15 and Appendix Table 38). Second, suppose a conservative depreciation rate of 2% in capital-intensive and a higher rate of 7% in labour-intensive sectors. The half-lives are then 35 and 10 years, respectively. Thus, long-lasting capital deepening cannot be explained by durability alone and requires continued reinvestment. Third, comparing durabilities in agriculture illustrates the limited explanatory power of depreciation. While horses and oxen may live longer than a plough lasts, their lifespans are unlikely to exceed 20 years; crucially, their working life spans are estimated at 5-7 years by Langdon, 1982. Thus, the relevant depreciation rate for draught animals should be very similar to that of tools in arable agriculture. Despite similar depreciation rates, I find strong evidence for long-lasting capital deepening in agriculture. Thus, I suggest that differences in the durability of capital do not drive my results.

Step #3: Productivity Grows; More So In High β_k Sectors

I make two assumptions to recover productivity growth (Appendix D.1). First, all sectors have the same labour share.⁴⁰ Second, the labour mobility friction does not vary within but only across areas, such that $\phi_{ikt} = \phi_{jkt} \forall (k, t)$ and i, j within the same area.⁴¹

Sectors may have different elasticities between capital deepening and productivity growth. As discussed above, I do not endogenise productivity but propose a reduced-form relationship that determines productivity as a function of factor proportions, where other determinants are absorbed by fixed effects:

$$A_{ik} = \left(\frac{K_{ik}}{L_{ik}} \right)^{\beta_k}.$$

⁴⁰Capital-intensity then comes down to different efficiencies β_k with which sectors turn capital deepening into productivity growth. Below, I put bounds on the bias induced by a violation of this assumption.

⁴¹This explicitly permits the reintroduction of serfdom in Denmark in 1733 and worsening serfdom in Estonia (Raster, 2023). Alternative assumptions are discussed in Appendix D.1.

If all sectors operate the same production function, capital deepening would still produce different productivity gains if $\beta_k \neq \beta_{k'}$. I expect this efficiency to be higher in capital-intensive than in labour-intensive sectors.⁴² At the time, few machines and technologies existed in labour-intensive sectors that could have led to capital deepening increasing productivity growth (Atack, Margo, and Rhode, 2019, Gallardo and Sauer, 2018, Coleman, 1956). To test the impact of the plague on sectoral productivity growth, I estimate equation 17 on data between 1668 and 1749:

$$\text{productivity growth}_{ikt} = \delta_k \text{plague}_{it} + \alpha_{ik} + \alpha_{akt} + \epsilon_{ikt}, \quad (17)$$

where productivity growth $_{ikt} = \delta_{ikt}$ are the fixed effects recovered from equation 11.⁴³ α_{ik} are region-sector fixed effects, absorbing time-invariant region-sector-specific determinants of productivity, such as availability of resources or time-invariant institutions. α_{akt} are area-sector-time fixed effects, absorbing area-wide time variation in sectoral productivity. These absorb sectoral productivity growth across an entire area, such that I compare the plague's effects of productivity growth within an area.⁴⁴ Examples of this are area- and country-wide economic and trade policies. Crucially, if there is time variation in the serfdom-induced mobility friction, these are absorbed by α_{akt} as long as they do not vary within areas. The fixed effects α_{ik} and α_{akt} thus absorb the time-invariant and area-level time-varying parts of vector x_{ik} in the general functional form for productivities in equation 12. ϵ_{ikt} are the error terms. δ_k is the coefficient on the plague.

Table 3 shows significant productivity growth in capital-intensive sectors relative to labour-intensive sectors after the plague. This difference of productivity growth by factor intensity holds both within agriculture and manufacturing.

Table 3: Impact of plague on sectoral productivity growth, by sector and factor intensity

	Labour-intensive ag.		Capital-intensive ag.		Labour-intensive man.		Capital-intensive man.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Plague Dummy	0.047 (0.178)		0.624*** (0.208)		-1.239*** (0.164)		0.073 (0.169)	
Mortality Rate		0.215 (0.464)		1.811*** (0.544)		-3.006*** (0.428)		0.635 (0.441)
<i>Fixed Effects:</i>								
– Region	✓	✓	✓	✓	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓	✓	✓	✓	✓
Observations	50,010	50,010	50,020	50,020	50,020	50,020	50,020	50,020

Notes: Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is log sectoral productivity growth. The independent variable is first a plague dummy that equals 1 for plagued regions after the plague hit. The second independent variable is the mortality rate, which for half of regions is imputed as the predicted value from regression results presented in Appendix Table 14.

Sectorally differing productivity effects come down only to differences in β_k under the assumption that all sectors have the same labour share. A violation of this assumption

⁴²If the elasticity of substitution between capital and labour were to differ sectorally, this would also show up in β_k . In the proposed model, these elasticities are 1 for all sectors under Cobb-Douglas; given the available data, I do not think that I can convincingly estimate these elasticities.

⁴³See Appendix D.1 for details.

⁴⁴Areas are displayed in Appendix Figure 7. As before, these are equivalent to NUTS2 for most of Europe; for the UK, Germany, France, and Spain, they are NUTS1, to have similarly sized areas and avoid areas with a single port (Appendix B).

does not affect the level effect for productivities, which is absorbed here and for which I find significant estimates below. Wages can only rise after the plague: as exports go up, there needs to be productivity growth in a Ricardian framework. However, unaccounted for differences in labour shares lead to an overestimation in sectoral differences in β_k . In Appendix Figure 38, however, I show that for model-implied wage growth and even extreme differences in labour shares, this scaling factor leads to an overestimation in point estimates of only 17.3-37.6%. Further, I find similar results when accounting for spatial correlation (Conley, 1999) in Appendix Figure 39.

How did productivities respond once the adverse labour supply shock had been overcome (Appendix Tables 15 and 16)? Extending the productivity analysis to the long-run, I find immediate productivity growth for both agriculture and manufacturing in the first post-plague decade, with capital-intensive manufacturing (agriculture) seeing faster productivity growth than labour-intensive manufacturing (agriculture). Plague-induced productivity changes persist for half a century longer than the underlying changes in factor proportions. This suggests that temporary shocks to factor proportions can produce persistent productivity changes.

Evidence from Production Data

To complement the gravity-based recovery of productivity, I estimate a Cobb-Douglas production function and recover logarithmic productivities (TFP) as residuals. I use agricultural data by Andersson Palm, 2025 for Swedish parishes in 1690 and 1750. As revenue for labour-(capital-)intensive agriculture, I use the value of harvests (animal output⁴⁵). I then regress logarithmic revenue on logarithmic population (labour), sowed acreage (land), and draught animals (capital), and recover the residuals. Table 4 shows that TFP levels are significantly higher in plagued parishes. The effect is larger in capital-intensive than in labour-intensive agriculture. All this is in keeping with the previously-established productivity findings that relied on recovery from the gravity equation.

Recovering productivity as a residual from regressing revenues on inputs has well-documented issues: beyond simultaneity bias, the residual mixes TFP with demand-side price variation (Klette and Griliches, 1996). In this period, however, there is no information on intermediate inputs (Levinsohn and Petrin, 2003) or investments (Olley and Pakes, 1996). As the data are recorded every 60 years, the scarcity of time periods thwarts me from employing estimators such as Akerberg, Caves, and Frazer, 2015. Thus, recovering TFP as residuals may be the most suitable approach. I follow a literature estimating firm-level production functions (Griliches and Mairesse, 1984, Griliches and Mairesse, 1998, Hall and Mairesse, 1995). These papers use revenue, capital, and employment as these are the only variables in typical firm accounts.

Similarly, the residual approach faces the objection of input mismeasurement: after the plague, the quality of labour may have risen, as survivors may be systematically healthier. Capital utilisation may have changed, too, and land quality per worker may have improved. Such factor quality upgrading contaminates the residual which I claim to capture TFP. However, I argue that the larger TFP growth in capital-intensive sectors is hard to explain by factor quality upgrading, which should be uniform across sectors. Further, the post-plague observation is in 1750, four decades after the plague; selection effects in surviving workers should have disappeared by then. Below, I also rule out

⁴⁵Specifically the number of calves, cows, sheep, goats, pigs, and heifers; the output of e.g. milk and meat should be proportional to this.

Table 4: Parish-level TFP growth after the plague

	Agricultural TFP by Factor Intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
Post Plague	0.716*** (0.0463)	0.749*** (0.0520)	0.693*** (0.0467)	0.736*** (0.0559)	0.729*** (0.0485)	0.748*** (0.0528)
Post Plague x Capital Intensive	0.399*** (0.0828)	0.322*** (0.0841)	0.304*** (0.0842)	0.142 (0.0886)	0.274*** (0.0906)	0.715*** (0.116)
Outcome	Log TFP	TFP	Log TFP	TFP	Log TFP	TFP
Draught Animals	Horses and Oxen	Horses and Oxen	Horses	Horses	Oxen	Oxen
<i>Fixed Effects:</i>						
– Parish x Factor Intensity	✓	✓	✓	✓	✓	✓
– Factor Intensity x Year	✓	✓	✓	✓	✓	✓
Observations	3,914	3,914	3,748	3,748	2,632	2,632

Notes: Standard errors clustered at the parish level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are parish-level (logarithmic) TFP by factor intensity estimated from regressing logarithmic harvests or stock of animals (for labour- and capital-intensive agriculture, respectively) on logarithmic population, sowed acreage, and horses and oxen (columns 1-2, or only horses in columns 3-4, or only oxen in columns 5-6) as draught animals. The independent variable is a post-plague dummy and its interaction with a dummy for capital-intensive agriculture.

differential skill acquisition that may have improved human capital. This suggests that, indeed, the plague induces TFP growth.

There is also historical evidence for productivity growth after the plague. Clark, 2016 and Broadberry et al., 2014 show that the Black Death was followed by productivity gains. For this outbreak, Appendix Figure 40 shows a brief drop in output per farm, adjusting for size, based on Scanian data (Olsson and Svensson, 2017). Production surpasses pre-plague levels after a few years, consistent with a productivity effect.

Beyond agriculture, one Swedish manufacturing industry that saw important changes was ship building. Using data on ships from the Swedish East India company, I show in Appendix Figure 41 that the vast majority of ship yards built in Sweden after the plague was built in previously plagued regions.⁴⁶ These also built the largest and most productive ships, as measured by the number of successful trips to East Asia.

5.1 Evidence from Factor Frictions

The mechanism can be validated by studying settings where it is constrained by factor market frictions. I study two such frictions: for labour, serfdom which limits the mobility between cities and their hinterlands. For capital, I investigate pre-war variation in the depth of capital markets which should affect the degree of capital deepening.

5.1.1 Labour Frictions

Serfdom severely limited labour mobility between cities and their hinterlands. I analyse how capital deepening and productivity growth differ for regions with serfdom (Section 2.3 and Appendix Figure 42). Serfdom hindered rural-to-urban labour mobility by binding serfs to their lords' lands. Further, serfs in the Eastern Baltic were not allowed to learn

⁴⁶Construction of the Polhemsdockan in Karlskrona began in 1717, five years after the plague hit the city. This dry dock used pumps rather than tides and is considered by some the Eighth Wonder of the World.

a trade and typically spoke Estonian or Latvian. Skill and language barriers thus further prevented serfs from moving to (mostly German-speaking) cities.

Under full labour mobility, wages are equalised between a city and her hinterland. However, under mobility restrictions, higher urban wages are no longer equalised to those in the hinterland as peasants cannot move. Denmark presents a testing ground for this hypothesis as serfdom was re-introduced in 1733 (Gary et al., 2022). There was no wage gap between urban and rural workers before 1733; however, a gap appears just after serfdom was brought back (Appendix Figure 43). I formalise this wedge of ϕ_i between urban and rural wages: $w_{iM} = (1 + \phi_i)w_{iA}$.

I consider two forms of labour scarcity. Under labour mobility, labour scarcity from higher urban mortality rates (Voigtländer and Voth, 2012) is equalised across sectors. I refer to this as *absolute* labour scarcity, as for all sectors, labour has become scarce. With serfdom, however, labour mobility does not equalise labour scarcity across sectors. Beyond absolute labour scarcity, cities suffer from *relative* labour scarcity as workers from the hinterland cannot move to benefit from higher urban wages. Thus, labour is artificially scarce in cities and artificially abundant in rural areas. This *relative* labour scarcity manifests itself in a wage wedge.

In Appendix E.3.1, I show how this logic affects production and exports in my model. Urban manufacturing wages increase more under serfdom; thus, the shift into a higher capital intensity of exports should be larger in plagued regions with serfdom. In agriculture, the opposite is true: under serfdom, rural agricultural wages rise less after the plague, leading to a smaller shift into capital-intensive agricultural exports under serfdom. Appendix Table 41 confirms these expectations.

5.1.2 Capital Frictions

In imperfect capital markets, capital deepening is constrained by financial development: ports with deeper pre-war capital markets should undergo stronger capital deepening. Data on loans and interest rates are very limited for this period. I summarise three tiers of financial development for large ports in the area: Hamburg, Amsterdam, and Bremen were the most developed (Huang, Chilos, and Sapoznik, 2019, Carlos and Neal, 2011, Denzel, 2010). Danzig may be grouped in a second tier (Wee, 1977). Stockholm, Königsberg, Riga, Reval, and Lübeck were the least financially developed major ports.⁴⁷ Going deeper, empirical evidence for financial variables is lacking for virtually all ports.

Given these constraints, I borrow an insight from Böhm-Bawerk, 1889 and Antràs and Kulesza, 2026: financially developed regions have a comparative advantage in sectors with a high average period of production (APP). In sectors where more time passes between spending on inputs and revenue raised from selling final goods, financing costs are higher. Thus, ports' pre-plague APP in exports reflects their financial development.

I construct each port's export-weighted pre-war APP.⁴⁸ I ask whether the shift into a higher capital intensity and the export expansion are stronger for ports with a higher pre-war APP. Throughout, I control for all interactions with pre-war export levels by sector, as

⁴⁷For example, shadow interest rates in Stockholm were twice as high as in Bremen (Huang, Chilos, and Sapoznik, 2019, Edvinsson, 2011), indicating poor access to credit. Königsberg, Riga, and Reval do not even appear in financial series before 1700 (Denzel, 2010), suggesting that credit may have depended entirely on the financial markets of Hamburg and Danzig.

⁴⁸All details in Appendix E.3.2. Ranking ports by their average APP, I find that the major ports are ranked as follows: Hamburg, Bremen, Amsterdam, Reval, Riga, Danzig, Stockholm, Königsberg, Lübeck. This matches the above-described historical evidence on financial development very well.

APP is higher in manufacturing and export-weighted APP is mechanically higher in larger ports. In Appendix Table 43, I find that, even when controlling for export volumes and sectoral structure, plagued ports with a higher pre-war APP see a significantly larger shift into capital-intensive exports and a significantly larger export expansion. Appendix Table 44 shows that manufacturing productivity growth is stronger in regions with a higher pre-war APP. This suggest that the observed effects are explained by capital deepening: my findings are significantly stronger for ports that were less financially constrained and could more easily engage in capital deepening.

6 Alternative Mechanisms

As an alternative mechanism, I can rule out pure Heckscher-Ohlin forces and show that Ricardian forces account for more than 90% of variation in a hybrid framework. I present historical evidence against the presence of a factor bias in technical change. However, I cannot rule out that the true model is a hybrid containing Ricardian and Heckscher-Ohlin forces; similarly, I cannot rule out that technology is factor-biased. I nonetheless argue that productivity growth after the plague is required to explain my findings. Further, I find no evidence for human and social capital deepening after the plague. I also conclude that “venting out” and market power are ill-suited to explain my findings. Finally, I test for the presence of non-homotheticity (Fieler, 2011).

Heckscher-Ohlin Forces

I relate my model to Romalis, 2004, Morrow, 2010, and Fadinger and Fleiss, 2011 and argue that Heckscher-Ohlin forces may be present, but are neither required nor sufficient to explain the findings of this paper; in a hybrid model, more than 90% of the difference in sectors’ supply-side responses are driven by Ricardian forces.

Fadinger and Fleiss, 2011 propose a version of their model in the spirit of Eaton and Kortum, 2002 that does not require production data. In the resulting gravity equation⁴⁹ exporter-industry fixed effects capture TFP but are contaminated with the interaction of factor abundance and factor intensity:

$$\hat{\delta}_{ikt} \propto \frac{\exp(\lambda_{ikt}/\beta_k)}{\prod_f (w_{it}^f)^{a_{fk}/\beta_k}} = \frac{\exp(\lambda_{ikt}/\theta_k)}{(w_{it})^{\gamma_k/\theta_k} r^{(1-\gamma_k)/\theta_k}}, \quad (18)$$

where $\exp(\lambda_{ikt}/\beta_k)$ is productivity and w_{it}^f are factor prices and a_{fk} are factor shares. The identity introduces the notation used in my paper, where $\beta_k = \theta_k$, $w_{it}^L = w_{it}$, $w_{it}^K = r$, $\alpha_{Lk} = \gamma_k$, and $\alpha_{Kk} = 1 - \gamma_k$. While I cannot precisely separate Ricardian forces from factor terms without data on factors’ prices and shares in the production function, I calculate the scaling factor for different wage growth and labour share figures. In Figure 5, I show the extent to which my productivity growth estimates need to be scaled down. For model-implied wage growth of 6.6% and a difference in labour shares of 0.5, the Ricardian term needs to be scaled down by 0.64%⁵⁰; even for large wage growth and labour share differences, Ricardian forces continue to account for more than 90%.⁵¹

⁴⁹Equation (A.5) of the Online Appendix in levels: $M_{ijk} = \frac{\exp(\lambda_{ik}/\beta_k)}{\prod_f (w_i^f)^{a_{fk}/\beta_k}} \cdot \frac{1}{\tau_{ijk}^{1/\beta_k}}$

⁵⁰As to the (here unidentified) level effect of productivity growth, the estimates of δ_{ikt} would actually need to be scaled *up* by 0.6% in a hybrid framework.

⁵¹For $\theta = 2$ as a very low estimate of the trade elasticity, this figure is still above 80% throughout.

Figure 5: Scaling factor required to turn δ_{ikt} into pure Ricardian term



Notes: This figure shows the scaling factor by which recovered δ_{ikt} would need to be divided to clean for Heckscher-Ohlin forces in the hybrid model of Fadinger and Fleiss, 2011 and to arrive at the remaining Ricardian forces. When taking ratios of equation 18 with a reference period and sector, the scaling factor is $\left(\frac{w_{it}}{w_{it'}}\right)^{\frac{\gamma_{k'} - \gamma_k}{\theta}}$.

Outside of a hybrid framework, can pure Heckscher-Ohlin forces rationalise my results? Labour-scarce exporters see their exporter-industry fixed effect drop relatively in labour-intensive sectors (Table 3). This is consistent with two mechanisms: first, Heckscher-Ohlin forces operate and TFP changes are absent or uniform across sectors. In the latter case, symmetric TFP growth is absorbed. The observed rotation into capital-intensive exports is then truly the Rybczynski effect. Second, there is sector-biased TFP growth favouring capital-intensive sectors and Heckscher-Ohlin forces are not necessary.

I argue against the first explanation. The export share expansion in Figure 4 cannot be purely reflective of Heckscher-Ohlin forces. Combining Romalis, 2004 with a national accounting logic, reduced labour supply increases wages, and under standard assumptions, GDP falls. Imports fall and domestic absorption likely falls, too. Under balanced trade, overall exports must then also fall. For exports to, instead, increase, domestic absorption would have to fall by more than income. Contradicting this point, I find in Appendix Figure 29 that imports rise after the plague. Additionally, Appendix Figures 19 and 21 show that the export expansion is concentrated in capital-intensive sectors. The absorption story cannot account for this pattern. In the absence of falling domestic absorption, wage increases reduce competitiveness across sectors, just less strongly in capital-intensive ones. Exports, then, cannot increase in the absence of TFP growth that shifts up competitiveness. Further, the finding of productivity growth in levels that is stronger in capital-intensive sectors (Table 4) directly contradicts the prediction of a Heckscher-Ohlin framework that output patterns are shifted, yet that productivity residuals are unaltered. This shift in TFP is direct evidence of Ricardian forces at play.

Thus, there is no support for the pure Heckscher-Ohlin interpretation. I propose that TFP growth favouring capital-intensive sectors is in the driver's seat. For exports to rise in Costinot, Donaldson, and Komunjer, 2011, controlling for bilateral and demand-side determinants, wages need to drop or productivity needs to rise. The export expansion is

thus indicative of productivity growth that is large enough to dominate plague-induced wage growth. Further, I argue below that a pure Heckscher-Ohlin framework mechanically implies changes in relative imports. Instead, I find that relative imports are unaffected by the plague, which a Ricardian framework can rationalise.

However, both Ricardian and Heckscher-Ohlin forces may be operating. Under the proposed reduced-form dependence of productivities on capital deepening (equation 13), factor proportions enter through the backdoor, and are proposed to determine fundamental productivities. The difference is that under Heckscher-Ohlin, the production frontier is held fixed and factor are reallocated across sectors. My story is that the production frontier itself shifts as a result of capital deepening, which explains the rising capital intensity of exports, growing market shares in destination ports, and TFP growth. The above decomposition in the framework of Fadinger and Fleiss, 2011 suggests that indeed Ricardian forces account for the vast majority of my findings.

Factor-biased technological progress

Labour scarcity could lead to directed (e.g. labour-saving) technical change and productivity growth (Alfaro et al., 2024, Acemoglu, 2010, Acemoglu, 2002). This is consistent with the proposed mechanism. However, history does not support the presence of a labour-saving factor bias in this era. Labour-saving mechanisation in crop agriculture began succeeding only in the 19th century (Gallardo and Sauer, 2018). An early example of a crop harvesting machine that explicitly substituted labour for capital was the cotton gin by American inventor Eli Whitney, patented only in 1794. Focusing on the 17th century in England, Coleman, 1956 argues that labour-saving technologies in agriculture were practically non-existent. Those that existed, such as mills, spinning wheels, and water-powered blast furnaces, were centuries old. In manufacturing, Atack, Margo, and Rhode, 2019 document how innovative machinery consolidated multiple hand tasks into a single machine task. They consider this the most labour-saving type of technological transition, which, however, only came to bear during the 19th century. Coleman, 1956 argues for a political will to absorb rather than substitute labour in this period, resulting in opposition to the development of labour-saving devices in agriculture and manufacturing alike. This points to unbiased technical change.

On factor bias in international trade, Trefler, 1993 reconciles factor content theories with the empirical rejection of factor price equalisation by introducing factor-biased technological progress. The threat to separating wages from productivities is that the factor bias correlates with labour shares and that I am comparing sectors with different labour shares. In that case, wage growth contaminates the productivity comparison. While I cannot test for such a correlation directly, the findings of Table 3 permit a role for factor-biased rather than Hicks-neutral productivity growth. In this case, an interpretation as pure TFP overstates the Ricardian story. Above, I study the bias introduced by a violation of my assumption on labour shares: I find that productivity growth remains largely in place as the upper limit on the inflation of productivity estimates is 32%. Thus, factor-biased technological progress is consistent with my findings, yet leaves the productivity growth story largely in place.

Human and Social Capital Deepening

Equation 13 may contain human and social capital if they have sectorally varying components. After the plague, survivors may choose to invest more in their and their offsprings'

human capital; similarly, there may be selection in survivors in that better connected individuals have higher survival chances. If these adjustments have a sectoral component, human capital deepening and social capital deepening may enter in equation 13. I do not expect this to be a major concern: First, human and social capital are typically region-time variation, which is absorbed in my recovery of productivities. An increase in human or social capital across sectors is inconsequential for my productivity estimates. Second, this paper’s setting in an early modern period implies that sector-specific human capital was very limited, suggesting that the absorbed level effect of any potential human capital deepening would be the main concern.

To more rigorously address these concerns, I test whether there is any evidence for human or social capital deepening. If these additional types of capital were sectorally-varying and responding to labour scarcity, I would expect to see a relative export shift into goods that use these factor intensely, applying the logic of equation 14. I construct measures of goods’ skill intensity and contract intensity (see Appendix E.4). In Appendix Table 45, I show that across specifications the only consistently positive and significant estimates are for a shift into a higher physical capital intensity. There is no shift into a higher contract or skill intensity of exports. This analysis has to be caveated by the fact that both measures of human and social capital intensity are based on modern data for the U.S. I assume that there is a sufficiently high correlation over time in goods’ relative skill and relative contract intensities. The evidence presented here suggests that it is indeed physical capital deepening, and not human or social capital deepening, that explains the post-plague export expansion.

Venting Out & Market Power

Almunia et al., 2021 propose non-constant marginal costs to explain “venting out”, for which this historical setting provides an ideal testing ground: while the Black Death hit all of Europe, the 1708-1714 plague outbreak left most of Europe as unperturbed export markets to counteract the local demand shock of the plague. Under venting out, regions would increase their exports in line with my findings. I suggest the time dimension as a crucial difference, however. Almunia et al., 2021 track exports for four years, while this paper studies trade for decades after the plague. Short-run marginal costs are unlikely to explain this persistent export boom.

Market power may reconcile short-run marginal cost changes with my long-run findings. If plagued regions capture market shares through lower marginal costs, a combination of market power and switching costs could explain persistent market share gains. However, I present evidence in Appendix Table 48 that export prices of plagued origins drop significantly. While in the short-run this may be explained through venting out, export prices of capital-intensive goods keep falling for 30 years. This pattern is incompatible with increased market power, but fits with productivity growth.

Non-Homotheticity

Under non-homothetic demand, demand shifts relatively into more income-elastic goods after labour scarcity leads to real wage gains. Voigtländer and Voth, 2012 place non-homothetic demand at the centre of their argument. The authors study the much earlier Black Death of 1346, which encompassed all of Europe. Therefore, my paper is not a test of Voigtländer and Voth, 2012. However, as non-homothetic demand is frequently

discussed in this context, I gather evidence that it played no role for this outbreak.⁵² I assume that capital-intensive agriculture (manufacturing) is more income-elastic than labour-intensive agriculture (manufacturing).

In Appendix E.5, I show how my model nests non-homothetic demand (Fieler, 2011) and present three pieces of evidence against its relevance here. First, I show that my estimates of productivity growth are robust to non-homothetic demand. There is thus no contamination of productivity growth estimates with demand-side factors. The question is therefore not whether there is one or the other but whether there is both productivity growth and non-homotheticity.

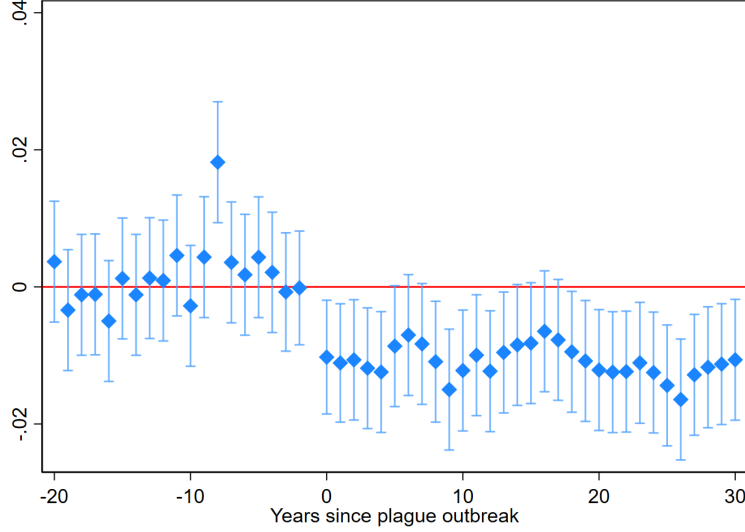
Second, I compare price predictions between my model and one of non-homothetic demand. As demand shifts relatively into more income-elastic capital-intensive goods, these should become relatively more expensive. Under the proposed productivity mechanism, however, capital-intensive goods should become relatively cheaper. In Appendix E.5, I show that prices for capital-intensive goods decline relatively in plagued ports.⁵³ Further, the Soundtoll data allow me to test granular price predictions: comparing import prices to export prices permits disentangling supply-side from demand-side factors. Under non-homotheticity, changes in relative demand translate into relative import price changes. Controlling for the supply side, I find that relative import prices of capital-intensive goods are unaffected by the importer's plague status. This suggests homothetic demand. Export prices, on the other hand, are directly affected by the proposed productivity growth: export prices of capital-intensive goods should decline relatively. This is precisely what I find.

Third, I test directly for a relative demand shift after the plague by constructing a test on relative imports (Fieler, 2011). There is no evidence of any relative demand-side changes, which also rules out that sector-specific embodied technology linked to increasing imports drives my TFP findings. Constant relative imports actually provide further support for a Ricardian rather than a Heckscher-Ohlin framework. In the latter, labour scarcity should shift exports into more capital-intensive sectors, mirrored by imports shifting into more labour-intensive sectors. However, this mirroring need not be present in a Ricardian framework with heterogeneous TFP growth, which can shift the export composition without mechanically shifting the import composition. The relative ordering of labour-intensive goods may be unaffected; further, the composition of imports is subject to the supply side's technology distribution, which was not affected by the plague. The fact that relative imports do not change accordingly supports a Ricardian framework of heterogeneous TFP growth. All three elements of this analysis thus find no support for non-homothetic demand and instead support this paper's productivity growth mechanism.

⁵²The only mention reminiscent of non-homothetic demand in the historical literature for this plague outbreak is discussed in Frandsen, 2009: in some places, it was believed that textiles and bedding spread the disease, which led to survivors burning the textiles of those who died. After the plague, this might imply a relative demand shift into textiles, which are summarised as labour-intensive manufacturing here. I show below that there is no indication for a relative demand shift into textiles.

⁵³These price effects reconcile rising meat and dairy consumption after the plague (Broadberry et al., 2014) with homothetic demand.

Figure 6: Counterfactual market shares without productivity channel



Notes: Counterfactual export shares, shutting down productivity channel. Estimate equation 2 on counterfactual export shares after the plague and observed export shares before the plague.

7 Counterfactual

In a counterfactual analysis, I shut down productivity growth. I find that market shares of plagued regions would have contracted rather than expanded as wages rise and competitiveness falls.⁵⁴ Counterfactual market shares t years after the plague can then be decomposed into a time-invariant term, s_{ijkPRE} , that contains productivities, trade costs, and incomes, and a time-varying component reflecting employment:

$$s_{ijkt}^c = \frac{L_{ikt}^{\frac{\gamma_k \theta}{1+\gamma_k \theta}}}{L_{ikPRE}} s_{ijkPRE}. \quad (19)$$

As no annual population data are available, I suppose an upper bound for population recovery and impose exponential population growth. In Figure 6, I repeat the event study in equation 2 but replace post-plague export shares of plagued regions by counterfactual shares. I find significantly lower export shares as a smaller population led to higher wages and lower competitiveness. This stands in stark contrast to my empirical findings. To get from the average point estimate in counterfactual export shares (a contraction of 1.03 p.p.) to the observed expansion of export shares (by 0.98 p.p.), average productivity needs to rise by 210%. This - admittedly high - estimate mirrors the magnitudes found in the TFP estimation in Table 4, where the point estimate for labour-intensive (capital-intensive) TFP growth (column 1) implies TFP growth of 105% (205%).

⁵⁴I plug in gravity equation 10, equation 9 for wages and equations 7 for firm and consumer market access to arrive at an equation defining market shares as a function of productivities, sectoral employment, trade costs, wages, and total income. A_{ik} is fixed. All details are in Appendix F.

8 Conclusion

This paper studies the relationship between labour scarcity and productivity growth. I establish novel empirical facts: First, capital-intensive exports expand relatively after the plague. Second, plagued regions expand their market shares in destination markets.

To conceptualise my findings, I build a Ricardian model. I suggest that productivity is a function of factor proportions, where other determinants are absorbed by fixed effects. From this model, I back out productivity and show that productivity grows, and more so in capital-intensive sectors. I propose that capital deepening leads to productivity growth and provide additional supporting evidence from farm- and parish-level agricultural data. These productivity changes are in place for almost a century, even though populations had recovered after 40 years.

I also present arguments against alternative mechanisms. Two alternatives can rationalise my findings: a hybrid model of Ricardian and Heckscher-Ohlin forces; and factor-biased technological progress instead of TFP. Nonetheless, to explain the empirical findings of this paper, both require productivity growth, though the point estimated would differ. In counterfactual analysis, I shut down the productivity channel and find that exports decline. The difference between this counterfactual export contraction and the observed export expansion is well-explained by and fits in magnitude with my productivity growth estimates.

The findings of this paper suggest several take aways. Many industrialised economies grapple with persistently low productivity growth rates. At the same time, their societies are aging and preferences on working hours appear to be changing. Further, long-term health conditions are reducing labour force participation rates in several large economies (Di Meo and Eryilmaz, 2025). Thus, labour scarcity in industrialised economies reflects both declines in the working age population and in labour force participation. I argue that this labour scarcity can lead to productivity growth if agents adjust to changes in factor prices and comparative advantage.

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A Data Appendix

Table 5 holds key descriptives for the trade panels constructed from the Soundtoll records.

Table 5: Descriptive Statistics

	Obs.	Mean	Std. Dev.	Median	Min	Max
<i>Panel A: Bilateral Trade, Intensive Margin by Sector (origin $\times$ destination $\times$ sector $\times$ year)</i>						
Exports (levels)	2,514,900	2.685	101.486	0.000	0.000	32110.792
Capital-intensive sector	2,514,900	0.400	0.490	0.000	0.000	1.000
Exports: capital-int. agriculture	502,980	4.101	78.956	0.000	0.000	8592.281
Exports: capital-int. manufacturing	502,980	1.588	45.716	0.000	0.000	7047.719
Exports: labour-int. agriculture	502,980	6.722	206.383	0.000	0.000	32110.792
Exports: labour-int. manufacturing	502,980	0.596	20.881	0.000	0.000	5183.438
Share of zeros (%)		97.7				
600 origins, 501 destinations, years 1668–1750						
<i>Panel B: Bilateral Trade, Intensive Margin Overall (origin $\times$ destination $\times$ year)</i>						
Exports (levels)	502,980	13.424	268.081	0.000	0.000	35175.146
Market share	502,980	0.013	0.099	0.000	0.000	1.000
Share of zeros (%)		95.1				
<i>Panel C: Bilateral Trade, Extensive Margin (origin $\times$ destination $\times$ year)</i>						
Number of exported goods	502,980	0.431	2.622	0.000	0	84
Any exports (dummy)	502,980	0.094	0.292	0.000	0	1
<i>Panel D: Sectoral Productivity (origin $\times$ year)</i>						
Log prod.: labour-int. agriculture	50,174	-1.237	5.128	-1.462	-39.675	34.770
Log prod.: capital-int. agriculture	50,184	0.560	6.425	-0.233	-47.423	31.429
Log prod.: labour-int. manufacturing	50,184	-0.452	4.165	0.000	-26.075	30.098
Log prod.: capital-int. manufacturing	50,184	5.105	8.704	2.020	-24.293	47.999
600 cities, years 1668–1750						
<i>Panel E: City-Level Exports (origin $\times$ year)</i>						
Total exports (levels)	49,800	135.585	1296.781	0.000	0.000	40149.656
Capital-intensive share	49,800	0.096	0.276	0.000	0.000	1.000
Zero total exports (dummy)	49,800	0.861	0.346	1.000	0.000	1.000
600 origins, years 1668–1750						

Notes: Summary statistics for the main datasets. Panel A is the bilateral origin-destination-sector-year panel. Panel B is the bilateral origin-destination-year panel used for market share regressions. Panel C summarises the extensive margin (number of goods exported per bilateral pair and year). Panel D reports log sectoral productivity estimates used in the productivity regressions. Panel E aggregates to the city-year level.

A.1 Commodities

The Soundtoll data do not provide researchers with cleaned information on commodities. Instead, the variable ‘soort’ denotes entries in words, specifically in the words of the custom official at the time. 143,855 separate descriptions are reported in Danish, which changed over the centuries. I use the Soundtoll Project’s list of products translated into English and double-check in Danish etymological dictionaries. I manually clean goods as models trained on modern Danish are not suitable for Old Danish. Further, a lot of similar sounding words introduce complexity and require intuition, as closest neighbour matching can be highly misleading.

In 104,132 of 3.66 million shipments (2.8%, or 2.5% of value)⁵⁵, several goods are reported. I cannot discern which value accrues to either of the goods. Besides the relative uncommonness of this, there are two more reasons not to worry: for one, these two goods are usually within the same category. Also, these goods were regarded as substitutes by toll officials and carried the same duties, and only for this reason were reported together in one line rather than separately.

I code goods as granularly as I can. This granularity reflects the source’s time period: many types of fish and textiles are recorded, whereas some shipments are mentioned to contain generic goods. I code individual types of fish as their own good; the compound category of fish contains all types that are not further specified. These differences in granularity matter little as I almost always aggregate trade up to the sectoral level.

For textiles, I distinguish raw materials (cotton, linen, wool) from processed textiles (band, gaze, ribbon, yarn, tissue, stockings), luxury textiles (flannel, satin, silk, velours, velvet) and naval textiles (sail/canvas, flag cloth). In cases where the data are insufficiently specific, I use compound categories (cloth, cotton textile, woollen textile). The compound categories distinguish, if possible, at least by cotton and wool. The good ‘clothes’ contains all types of fabric, textile, and clothes which are not further distinguishable, either by material or by degree of processing.

I arrive at 227 goods, 8 out of which only exist in the early trade data without bilateral coverage, and which therefore are never part of the analysis. In total, 143,855 goods descriptions are mapped onto 219 goods, so the average commodity appears in 9,000 variations. An example of different spellings is sturgeon, appearing in four variations: Stor, Større, Støre, and Stører. Another example of varieties of goods and spellings is iron, of which there are six variations: jerren, jeern, jern, jeren, osmundt jern, and osmunds jern, where the latter two denote specifically Swedish iron. Many variations in fact contain information on value or amounts: lead oxide appears in 15 variations, six of which specify value, e.g. ‘gleide 3 ort’ and ‘gleide a 6 skilling’.

To complement my manual cleaning efforts with a simple robustness check, I used Claude’s Opus 4.6 model to classify raw cargo descriptions.⁵⁶ For 84% of traded value, the algorithmic assignment is the same as the manual one. When accounting for a few inconsequential disagreements, I arrive at 96% overlap between the two approaches. The disagreements are between very closely related goods in the same sector, usually at a different aggregation, such as iron and iron works, planks and wood, cloth and sail, brandy and wine, goods and mercery, and meat and beef. Thus, the two approaches produce almost identical results.

From the manually cleaned goods, I then construct five sectors: labour-intensive agriculture, capital-intensive agriculture, labour-intensive manufacturing, capital-intensive manufacturing, and the remaining sector of unclassified goods. I distinguish agricultural from manufacturing goods based on their likely location of production, namely the countryside rather than the city. Capital-intensive goods require a high amount of capital that cannot be swapped for labour. This capital may consist of factories, furnaces, machinery, or tools necessary to produce the good. Labour-intensive goods are limited in

⁵⁵Counting as double whenever ‘og’ (also) or ‘etc’ are contained in a goods description.

⁵⁶The prompt was: ”In this folder, you find the file ”list from raw unique.dta”. It contains a list of raw trade records, contained in the variable ”soort”, the value of the transaction (”totaldaler”) and my manually cleaned commodity classificaton (”comm” and ”good id”). Disregarding this manual classification, please make a new classification from scratch. A commodity that you can assign each entry of ”soort” into has to be one of the commodities in ”comm”, however, or you can leave an entry as unclassified.”

Table 6: Top 30 goods by value

Good	Sector	Share of traded value	Cumulative share of traded value	Rank
Hemp	Labour-int. Agriculture	11.25	11.25	1
Rye	Labour-int. Agriculture	10.57	21.82	2
Salt	Capital-int. Agriculture	9.75	31.57	3
Wine	Capital-int. Agriculture	8.84	40.42	4
Wheat	Labour-int. Agriculture	8.63	49.05	5
Flax	Labour-int. Agriculture	6.1	55.15	6
Iron Works	Capital-int. Manufacturing	4.31	59.46	7
Iron	Capital-int. Agriculture	3.9	63.36	8
Tobacco	Labour-int. Agriculture	3.68	67.04	9
Planks	Capital-int. Manufacturing	3.22	70.27	10
Leather	Capital-int. Agriculture	2.81	73.07	11
Sugar	Labour-int. Agriculture	2.05	75.13	12
Unclassified	Unclassified	1.32	76.44	13
Ash	Capital-int. Manufacturing	1.22	77.67	14
Goods	Unclassified	1.22	78.88	15
Linnen	Labour-int. Manufacturing	1.14	80.03	16
Cloth	Labour-int. Manufacturing	1.12	81.14	17
Herring	Capital-int. Agriculture	1	82.14	18
Wax	Capital-int. Agriculture	.97	83.12	19
Brandy	Capital-int. Agriculture	.97	84.09	20
Barley	Labour-int. Agriculture	.8	84.89	21
Tow	Capital-int. Agriculture	.69	85.58	22
Wool	Capital-int. Agriculture	.68	86.26	23
Dye	Capital-int. Agriculture	.63	86.88	24
Tools	Capital-int. Manufacturing	.61	87.49	25
Silk	Labour-int. Manufacturing	.58	88.07	26
Wood	Labour-int. Agriculture	.56	88.62	27
Pepper	Labour-int. Agriculture	.55	89.17	28
Cotton	Labour-int. Agriculture	.48	89.65	29
Train Oil	Capital-int. Agriculture	.45	90.1	30

Notes: This table shows the top 30 goods by value of trade between 1668 and 1750.

the degree to which labour can be saved in their production. Labour-intensive agriculture is all of arable farming, whereas pastoral farming, mining, and processed foods including alcohols are capital-intensive agriculture. Labour-intensive manufacturing are largely textiles. Capital-intensive manufacturing produces goods such as processed metals, tools, and ship building materials. All 227 goods, accompanied by an assignment into the four sectors, are reported in Table 7. 2% of traded value is in unclassified goods, 99% of which are generic goods and mercery. This implies that 98% of traded value is assigned into goods.

Table 6 ranks the 30 most common goods traded between 1668 and 1750, which together make up 90% of the value of trade.

Table 7: Goods Classification

Animals	Capital-int. ag.	Rhine Wine	Capital-int. ag.	Ship Nails	Capital-int. man.	Tea	Labour-int. ag.
Arrack	Capital-int. ag.	Rum	Capital-int. ag.	Soda	Capital-int. man.	Tobacco	Labour-int. ag.
Beef	Capital-int. ag.	Salmon	Capital-int. ag.	Staves	Capital-int. man.	Trees	Labour-int. ag.
Beer	Capital-int. ag.	Salpeter	Capital-int. ag.	Steel	Capital-int. man.	Wheat	Labour-int. ag.
Bones	Capital-int. ag.	Salt	Capital-int. ag.	Swords	Capital-int. man.	Wood	Labour-int. ag.
Brandy	Capital-int. ag.	Seed	Capital-int. ag.	Tar	Capital-int. man.	Baize	Labour-int. man.
Bread	Capital-int. ag.	Sekt	Capital-int. ag.	Tiles	Capital-int. man.	Band	Labour-int. man.
Bristles	Capital-int. ag.	Soap	Capital-int. ag.	Tools	Capital-int. man.	Bay	Labour-int. man.
Buckling	Capital-int. ag.	Sprat	Capital-int. ag.	Turpentine	Capital-int. man.	Bowls	Labour-int. man.
Butter	Capital-int. ag.	Starch	Capital-int. ag.	Almonds	Labour-int. ag.	Chests	Labour-int. man.
Caviare	Capital-int. ag.	Stockfish	Capital-int. ag.	Amber	Labour-int. ag.	Cloth	Labour-int. man.
Chalk	Capital-int. ag.	Stones	Capital-int. ag.	Anise	Labour-int. ag.	Confectionaries Cotton Textile Flag Cloth Flannel Folio Furniture Gaze Gloves Hat Lace Lamps Linnen Mats Paper Ribbon Ropes Sail Satin Silk Spears Spokes Stockings Tissue Trousers Velours Velvet Woollen Textile Yarn Ballast Chinese Goods Comm English Goods Faroeer Goods French Goods Goods Icelandic Goods Luggage Mercery Norwegian Goods Provisions Sailors Ship Sully Soldiers Swedish Goods Waste	Labour-int. man.
Cheese	Capital-int. ag.	Sturgeon	Capital-int. ag.	Apples	Labour-int. ag.		Labour-int. man.
Cidre	Capital-int. ag.	Sulphur	Capital-int. ag.	Barley	Labour-int. ag.		Labour-int. man.
Coal	Capital-int. ag.	Talcum	Capital-int. ag.	Beans	Labour-int. ag.	Labour-int. man.	Labour-int. man.
Codfish	Capital-int. ag.	Tartar	Capital-int. ag.	Bushel	Labour-int. ag.	Folio	Labour-int. man.
Copper	Capital-int. ag.	Tin	Capital-int. ag.	Capers	Labour-int. ag.	Furniture	Labour-int. man.
Downs	Capital-int. ag.	Tow	Capital-int. ag.	Cardamon	Labour-int. ag.	Gaze	Labour-int. man.
Dye	Capital-int. ag.	Train Oil	Capital-int. ag.	Chestnuts	Labour-int. ag.	Gloves	Labour-int. man.
Eal	Capital-int. ag.	Treacle	Capital-int. ag.	Cinnamon	Labour-int. ag.	Hat	Labour-int. man.
Feather	Capital-int. ag.	Vetch	Capital-int. ag.	Cloves	Labour-int. ag.	Lace	Labour-int. man.
Fish	Capital-int. ag.	Vinegar	Capital-int. ag.	Cocoa	Labour-int. ag.	Lamps	Labour-int. man.
Flounders	Capital-int. ag.	Virriol	Capital-int. ag.	Coffee	Labour-int. ag.	Linnen	Labour-int. man.
Flour	Capital-int. ag.	Wax	Capital-int. ag.	Cork	Labour-int. ag.	Mats	Labour-int. man.
Fur And Skin	Capital-int. ag.	Wine	Capital-int. ag.	Cotton	Labour-int. ag.	Paper	Labour-int. man.
Gin	Capital-int. ag.	Wool	Capital-int. ag.	Cucumbers	Labour-int. ag.	Ribbon	Labour-int. man.
Grass	Capital-int. ag.	Zinc	Capital-int. ag.	Dates	Labour-int. ag.	Ropes	Labour-int. man.
Gymsum	Capital-int. ag.	Alkali	Capital-int. man.	Dodder	Labour-int. ag.	Sail	Labour-int. man.
Haddock	Capital-int. ag.	Anchors	Capital-int. man.	Figs	Labour-int. ag.	Satin	Labour-int. man.
Hair	Capital-int. ag.	Ash	Capital-int. man.	Flax	Labour-int. ag.	Silk	Labour-int. man.
Ham	Capital-int. ag.	Barrels	Capital-int. man.	Frankincense	Labour-int. ag.	Spears	Labour-int. man.
Herring	Capital-int. ag.	Bells	Capital-int. man.	Fruit	Labour-int. ag.	Spokes	Labour-int. man.
Honey	Capital-int. ag.	Black Powder	Capital-int. man.	Ginger	Labour-int. ag.	Stockings	Labour-int. man.
Horn	Capital-int. ag.	Bottles	Capital-int. man.	Grapes	Labour-int. ag.	Tissue	Labour-int. man.
Horses And Horse Products	Capital-int. ag.	Bowsprits	Capital-int. man.	Groats	Labour-int. ag.	Trousers	Labour-int. man.
	Capital-int. ag.	Brass	Capital-int. man.	Hops	Labour-int. ag.	Velours	Labour-int. man.
	Capital-int. ag.	Bullets	Capital-int. man.	Ivory	Labour-int. ag.	Velvet	Labour-int. man.
Iron	Capital-int. ag.	Brass Works	Capital-int. man.	Juniper	Labour-int. ag.	Woollen Textile	Labour-int. man.
Lead	Capital-int. ag.	Cannon	Capital-int. man.	Lemons	Labour-int. ag.	Yarn	Labour-int. man.
Leadoxide	Capital-int. ag.	Caps And Helmets	Capital-int. man.	Linseed	Labour-int. ag.	Ballast	Unclassified
Leather	Capital-int. ag.	Cement	Capital-int. man.	Millet	Labour-int. ag.	Chinese Goods	Unclassified
Lings	Capital-int. ag.	China	Capital-int. man.	Muscade	Labour-int. ag.	Comm	Unclassified
Liquorice	Capital-int. ag.	Glass	Capital-int. man.	Nuts	Labour-int. ag.	English Goods	Unclassified
Mackarel	Capital-int. ag.	Glue	Capital-int. man.	Oats	Labour-int. ag.	Faroeer Goods	Unclassified
Malt	Capital-int. ag.	Hoes And Shovles	Capital-int. man.	Olives	Labour-int. ag.	French Goods	Unclassified
Marble	Capital-int. ag.	Iron Works	Capital-int. man.	Peas	Labour-int. ag.	Goods	Unclassified
Mead	Capital-int. ag.		Capital-int. man.	Pepper	Labour-int. ag.	Icelandic Goods	Unclassified
Meat	Capital-int. ag.		Capital-int. man.	Plums	Labour-int. ag.	Luggage	Unclassified
Mineral	Capital-int. ag.	Medicine	Capital-int. man.	Potatoes	Labour-int. ag.	Mercery	Unclassified
Mustard	Capital-int. ag.	Nails	Capital-int. man.	Resin	Labour-int. ag.	Norwegian Goods	Unclassified
Oil	Capital-int. ag.	Oars	Capital-int. man.	Rice	Labour-int. ag.	Provisions	Unclassified
Ox Products	Capital-int. ag.	Pipe Tools	Capital-int. man.	Rubber	Labour-int. ag.	Sailors	Unclassified
Oysters	Capital-int. ag.	Pipes	Capital-int. man.	Rye	Labour-int. ag.	Ship Sully	Unclassified
Paint	Capital-int. ag.	Pistols	Capital-int. man.	Safran	Labour-int. ag.	Soldiers	Unclassified
Pimento	Capital-int. ag.	Pitch	Capital-int. man.	Spices	Labour-int. ag.	Swedish Goods	Unclassified
Pollock	Capital-int. ag.	Planks	Capital-int. man.	Sugar	Labour-int. ag.	Waste	Unclassified
Quicksilver	Capital-int. ag.	Powder	Capital-int. man.				
Raisins	Capital-int. ag.	Sheet Metal	Capital-int. man.				
Rays	Capital-int. ag.						

Notes: List of 227 goods and classification into capital-intensive and labour-intensive agriculture and manufacturing, and unclassified goods.

A.2 Currencies and Units

Foreign currency is a non-issue in this data set as virtually all trade is recorded in Danish coins. In principle, the Soundtoll Data contain a variety of coins in which toll transactions are recorded. Over all recorded passages, there are 40 types of coins, of which 48.88% are Daler and 50.88% Skilling. I use data starting in 1668 that cover 4.38 million coin transactions. From 1668 onwards, 48.8% of coin transactions are recorded in Daler and 51.2% are in Skilling. Out of 4.38 million transactions, 347 use the Ort and one uses the Rosenobel, an English gold coin. I convert it based on its gold content. Thus, only 0.008% of transactions are in non-Danish currency. Table 8 records conversion rates.

Table 8: Currency conversion rates

	Daler	Skilling	Ort	Rosenobel
Daler	1	0.0104	0.25	3.94
Skilling		1	24	378
Ort			1	15.76
Rosenobel				1

Table 9 lists the conversion of historical units of account into kilograms, liters, meters, and counts.

Table 9: Conversion of units of account

Kilogram			Liter / Meter			Unit		
Unit	Commodity	Value	Unit	Commodity	Value	Unit	Commodity	Value
Boeter	Butter	3.96	Ahmer		150	Bale		1
Centner		50	Ancker		38.5	Ladning		1
Skippund		159	Boeter	Wine	585	Ries		0.1
Qtr. Skippund		39.75	Boeter	Oil	1225	Qtr. Bale		0.25
Drompt		35.6	Korff		24.8	Dbl. Bale		2
Faad		950	Drompt		431	Daler		1
Half Faad		475	Faad		930	Gulden		1
Qtr. Faad		237.5	Half Faad		465	Deger		10
Sacker		165	Qtr. Faad		232.5	Dosin		12
Half Sacker		82.5	Flaske		0.7245	Fierdinger		0.25
Tonder	Grain	100	Tonder	Beer	131	Kister	Glass	120
Tonder	Butter	112	Tonder	Salt	170	Kister	Tin	225
Half Tonder	Grain	50	Tonder	Herring	108	Half Kister	Glass	60
Half Tonder	Butter	56	Half Tonder	Beer	65.5	Half Kister	Tin	112.5
Kister	Tea	25	Half Tonder	Salt	85	Qtr. Kister	Glass	30
Half Kister	Tea	12.5	Half Tonder	Herring	54	Qtr. Kister	Tin	56.25
Qtr. Kister	Tea	6.25	Oxehoffde		232	Stocker		1
Pund		0.5	Oxehoffde	Beer Vin.	197	Half Stocker		0.5
Lispund		8	Moyer		876	Hundred		100
Last		1905	Ottinger		16.945	Ort		1
Last	Hemp, Hops	952	Piber		464	Paar		2
Skaeppe	Grain.	12.5	Qtr. Piber		116	Ring		240
Steen		15	Skaeppe		17.39	Skilling		1
Woger		17.856	Allen (m)		0.63	Schock		60
						Thilther		12
						Thommer		40
						Tylvt		12
						Thussend		1000

A.3 Duties

For every passage, the type and amount of duty is recorded, and usually there are several duties payable, some fixed and some proportional. As an example, I picked passage number 304. This ship paid 4 Daler of fire money ('fyrpenge', a fixed fee for the maintenance of the lighthouse) and passed the Sound on April 27, 1790. Its captain, Daniel Wendlandt, citizen of Windau, received a captain's compensation of 1.5 Daler, which was deducted from the total duty payable. This is a proportional tax free allowance. The first sub-total is 37.5 Daler, which I call proportional duty. When subtracting the captain's allowance, the second sub-total comes to 36 Daler, and the total duty payable is 40 Daler, where the fire money of 4 Daler has been added. The ship is sailing with 75 last of rye; the recorded tax amount is 37.5 Daler, which equals the proportional duties and is the basis for my value calculations. The payable duties account for the allowance and the fixed fee. Table 10 shows the summation exercise.

Table 10: An example from the Soundtoll Data to show the different duties and sub-totals

Proportional duty	37.5 Daler
First sub-total	37.5 Daler
Captain's allowance	-1.5 Daler
Second sub-total	36 Daler
Fixed duty (fire money)	+4 Daler
Total payable duty	40 Daler

The Soundtoll team digitised the toll books in setting up three separate files.

1. The first one, called 'belastingen' or duties, records for each passage and line the type and amount of duty.
2. The second file, called 'ladingen' or carriage, records for each passage the origin and destination, the goods transported, and the overall amount of duty paid per good.
3. The third data set is called 'doorvaarten', or passages. For every passage, it records the day of passage through the Sound, information on the captain, discounts (or 'korting') applied to the duty, and two sub-totals and a total. A 'korting' is a discount, 87% of which are 'føring', a compensation for the captain in terms of a tax free allowance at their disposal.

The duty amount I use as a proxy for value is the proportional duty as in Waldinger, 2022. Gøbel, 2010 elaborates on the mechanics of the proportionality of this duty, which is summarised in Waldinger, 2022 as a duty of 1-2% of cargo value. In order to discourage understating cargo value, a truth-telling mechanism was put in place: the toll official could purchase the cargo at the total value indicated in the customs forms. In Section A.4 below, I use the information on duty variation in Gøbel, 2010 to back out underlying values from toll data by specifying a functional form for duty rates. This exercise produces virtually identical results, suggesting that it is appropriate to focus on the proportional duty.

A.4 Recovering underlying value

Underlying value, rather than the amount of duties paid, is a measure of value which I use as a robustness check. Gøbel, 2010 notes that the rate at which proportional duties were applied varied by flag, with Danish ships paying lower tax rates, by good, with e.g. salt facing an additional duty, and also by time, with Swedish cargoes taxed at a lower rate only until 1720. Importantly, the origin and destination were not important for the determination of the proportional duty rate.

Motivated by this finding, I assume that taxed value, $\phi_{ijo,kt}$ is the product of a proportional duty $\psi_{o,kt}$, varying by flag o, sector k and time t, and underlying value $v_{ijo,kt}$:

$$\phi_{ijo,kt} = \psi_{o,kt} v_{ijo,kt} = \psi_{o,kt} p_{ijkt} q_{ijo,kt}, \quad (20)$$

such that the duty rate paid for shipping sector k goods from i to j under the flag of o in time t depend on the value of these goods multiplied by a proportional duty, $\psi_{o,kt}$, which varies only by flag o, sector k, and time t.

Turning equation 20 into logs and absorbing log proportional duties by $d_{o,kt}$, I can recover proportional duties by estimating:

$$\ln \phi_{ijo,kt} = d_{o,kt} + \ln v_{ijo,kt}. \quad (21)$$

I let flag o be the country in which a ship is registered in the Soundtoll records.⁵⁷ Sectors k are the five sectors used throughout. While I estimate the following on annual data, I let the time dimension of fixed effects be decades. I set Denmark as the reference flag o, unclassified goods as the reference sector k, and the 1670s as the reference decade.⁵⁸

Under the assumption that there are no time trends in the proportional duty rate of unclassified goods shipped under Danish flags, this recovers proportional duty rates by country, sector, and time. Choosing Denmark as the reference is motivated by the fact that Danish merchants paid the lowest rates throughout. With these estimated tariff rates, I turn the observed tariff amounts into underlying value by dividing toll amounts by the normalised duty rate, in line with equation 20.

B History and Geography Appendix

Only ships passing the Oresund appear in the data. Thus, trade flows between cities on either side of the Sound (say, between London and Amsterdam) is unobserved. For these flows, I do not impose zero trade, but rather I will not include these. The Soundtoll data, however, cover almost all trade between the Baltic and North Sea areas.

It is highly unlikely that any trade between, say, Danzig and London occurred other than through the Sound. Other straits, such as the Little Belt between Jutland and Funen, were difficult and dangerous to navigate for larger ships such as the ones used for trading with the West. The main alternative to passing the Sound, between Sweden and

⁵⁷The main countries are Sweden, Russia, the German states, Denmark, the Netherlands, France, and Britain. Within these larger states, I also allow Norway, Belgium, Estonia, Lithuania, Latvia and Finland to potentially have different duty rates.

⁵⁸Using $d_{o,kt} = \ln \psi_{o,kt}$ and writing in terms of $\psi_{o,kt}$, estimating equation 21 recovers:

$$\frac{\psi_{ikt} \psi_{i'kt'} \psi_{ik't'} \psi_{i'k't}}{\psi_{ikt'} \psi_{i'kt} \psi_{ik't} \psi_{i'k't'}}.$$

Denmark’s main island Zealand, was sailing through the Great Belt, between Zealand and Funen. Degn, 2017 makes a number of arguments against this being an issue for these data. The Danish king introduced a prohibition on sailing through the Belt, directed specifically against the Prussian towns, extending this prohibition later on to all foreign nations. This was precisely because the Great Belt potentially allowed merchants to circumvent the Sound Toll. Only expert skippers possessed the specific knowledge required to pass through the Great Belt, a route usually taken in case of storms. Degn, 2017 concludes that skippers only reluctantly avoided the Sound. Finally, toll records from the Great Belt reveal that the number of passages was small. There is also no reason to believe these were high trade value passages: according to Gøbel, 2010, the toll revenue from the Great Belt and the Little Belt combined equalled but a few percent of the Sound Toll revenues. Alternatively, trade over land faced a high number of tolls, too, in that many more borders would have to be crossed when compared to passing through the Sound, compounding already high transport costs over land.

My source for identifying cities in the trade data are two files provided by the Soundtoll team, which convert 90,737 original city names to 3,085 unified city names. My sample are coastal cities (less than 25 kilometres distance to the coast, or those along a major river up to 100 kilometres from the coast) between Ireland to the West, Saint Petersburg to the East, the Arctic Circle to the North, and Bayonne to the South. 1,425 cities are in this sample, though only 676 are actively trading during the period I study.

B.1 Classification of Areas

The trade regressions in Section 3 include area-time fixed effects. These areas are not just one level of European NUTS regions as their sizes vary greatly and as Russia and Norway are not EU members. NUTS 1 is too aggregate for Northern Europe, with Finland and Denmark as each only one region. NUTS 2 is sufficiently granular in Northern Europe, but too granular in Western Europe, prompting a sixth of areas to only contain one port city. At NUTS 3, a third of all areas contain only one port.

Therefore, I construct similarly-sized areas: I use NUTS 2 (or a close equivalent) throughout, other than for the United Kingdom, France, Germany, and Spain, where I use NUTS 1. When aggregating areas below, this is always to ensure that no area has only one port city. Table 11 provides an overview of the number of units and their NUTS correspondance by country. Figure 18 shows that the export expansion result goes through also when including NUTS 1-time and NUTS 2-time fixed effects instead of the area-time fixed effects proposed here.

The United Kingdom is disaggregated into Scotland and Wales, and England is disaggregated into her nine regions, where I pool the East Midlands with the East of England, London with the South East, and the Isle of Man with the North West, and add the Channel Islands as an area. Ireland’s four provinces are aggregated to three together with Northern Ireland, with Connacht pooled with Ulster. The Netherlands are made up of their 5 provinces by the North Sea. Belgium’s areas are her three provinces by the North Sea, where Antwerpen is grouped with East Flanders. Denmark has 5 regions and the Faroe Islands are pooled with the main country. For Finland, I pick the provinces of Finland as of 1997, pooling Oulu and Lapland as Northern Finland and Åland with Western Finland. Only Southern and Western Finland show up in the data, however. France’s areas are the five regions on the Atlantic Coast and Channel. For Germany, I pick the current 16 states, of which 5 show up in the data. The three Baltic states are

Table 11: Areas and NUTS correspondence

Country	Unit type	# (in data)	NUTS level
Belgium	Provinces	2	NUTS 2
Denmark	Regions	5	NUTS 2
England	Regions (pooled)	7	NUTS 1
Estonia	Country	1	NUTS 2
Finland	Provinces (1997)	2	~NUTS 2
France	Regions	5	NUTS 1
Germany	States	5	NUTS 1
Ireland	Provinces (incl. NI)	3	~NUTS2
Latvia	Country	1	NUTS 2
Lithuania	Country	1	NUTS 2
Netherlands	Province	5	NUTS 2
Norway	Traditional regions	5	~NUTS 2
Poland	Voivodeships	3	NUTS 2
Russia	Oblasts	2	~NUTS 2
Scotland	Country	1	NUTS 1
Spain	Pooled	1	NUTS 1
Sweden	National areas (pooled)	6	NUTS 2
Wales	Country	1	NUTS 1

separate areas. Norway is disaggregated into five traditional regions that show up in the data. For Russia, I choose the two Baltic oblasts, pooling Arkhangelsk with Leningradskaya Oblast. Eight national areas are chosen in Sweden, corresponding to the NUTS 2 areas, though only 6 show up in the data. The few ports in Northern Spain, specifically in Asturias and Galicia, are pooled as one area. Finally, the areas in modern-day Poland are the three voivodeships on the Baltic Sea.

B.2 Additional Plague Results

Figure 8 shows digitised army marching routes. As armies spread the plague across Europe, regions' proximity to these routes is used in Table 14 to predict mortality rates when this information is not known. Table 12 shows information on plague outbreaks in my sample and provides an overview of timing and mortality rates. For plagued cities whose mortality rates were not observed, I form predicted mortality rates from regressing geographical covariates, the timing of the plague, and the proximity of army marching routes on observed mortality rates. These results are shown in Table 14 and used to form predicted mortality rates. Army marching are digitised based on maps by Spruner and Menke, 1880 and Barraclough, 1997. Table 14 regresses geographical controls, the timing of the plague, and proximity to army marching routes on observed mortality rates for the subset of plagued cities. For half of these cities, I form predicted mortality rates based on the results presented in this table, as no mortality estimates are available.

Table 12: List of plagued cities and mortality estimates

City	Modern Country	Plague	Time	Population	Mortality	Source
København/Copenhagen	Denmark	1	1711	60000	.283	Kroll and Grabinsky, 2007, Kroll, 2006
Helsingør	Denmark	1	1710 - 1711	4000	.408	Frandsen, 2009
Flensburg	Germany	1	not specified			Ulbricht, 2004
Frederiksort	Germany	1	1712			Ulbricht, 2004
Kiel	Germany	1	1712			Frandsen, 2009
Rendsburg	Germany	1	1712			Ulbricht, 2004
Altona	Germany	1	1712			Frandsen, 2009, Winkle, 1983
Glückstadt	Germany	1	1712			Frandsen, 2009
Pinneberg	Germany	1	1712			Winkle, 1983
Itzehoe	Germany	1	1712			Frandsen, 2009
Schleswig	Germany	1	1712			Ulbricht, 2004
Helsinki	Finland	1	1710		.66	Engström, 1994
Rauma	Finland	1	1710 - 1711			Vourinen, 2007
Turku	Finland	1	1710 - 1711	6000	.33	Vourinen, 2007
Raseborg	Finland	1	1710			Kroll and Grabinsky, 2007, Kroll, 2006
Pori	Finland	1	1710 - 1711			Vourinen, 2007
Oulu	Finland	1	1710 - 1711			Vourinen, 2007
Borgå	Finland	1	1710			Vourinen, 2007
Pietarsaari	Finland	1	1710 - 1711			Vourinen, 2007
Kokkola	Finland	1	1710 - 1711			Vourinen, 2007
Uusikaupunki	Finland	1	1710 - 1711			Vourinen, 2007
Enköping	Sweden	1	1710			Frandsen, 2009
Stockholm	Sweden	1	1710 - 1711	53750	.378	Kroll and Grabinsky, 2007, Kroll, 2006
Visby	Sweden	1	1710 - 1711	2375	.233	Kroll and Grabinsky, 2007, Kroll, 2006
Karlskrona	Sweden	1	1710 - 1712			Persson, 2011
Karlshamn	Sweden	1	1710 - 1712			Persson, 2011
Jönköping	Sweden	1	1710 - 1711	2500	.374	Kroll and Grabinsky, 2007, Kroll, 2006
Ystad	Sweden	1	1712	1950	.385	Kroll and Grabinsky, 2007, Kroll, 2006
Blekinge	Sweden	1	1710			Persson, 2011
Malmö	Sweden	1	1712	5000	.35	Kroll and Grabinsky, 2007, Kroll, 2006
Linköping	Sweden	1	1710 - 1711	1500	.295	Kroll and Grabinsky, 2007, Kroll, 2006
Domsten	Sweden	1	1711			Kroll and Grabinsky, 2007, Kroll, 2006
Narva	Estonia	1	1710 - 1711	3000		Kroll and Grabinsky, 2007, Kroll, 2006
Tallinn/Reval	Estonia	1	1710	9900	.661	Kroll and Grabinsky, 2007, Kroll, 2006
Kuressaare	Estonia	1	1710			Frandsen, 2009
Saaremaa	Estonia	1	1710			Frandsen, 2009
Riga	Latvia	1	1710 - 1711	10477	.651	Kroll and Grabinsky, 2007, Kroll, 2006
Pärnu	Estonia	1	1710	2350	.529	Kroll and Grabinsky, 2007, Kroll, 2006
Kaliningrad/Königsberg	Russia	1	1709 - 1710	36250	.239	Kroll and Grabinsky, 2007, Kroll, 2006
Klaipeda/Memel	Lithuania	1	1709 - 1710			Kroll and Grabinsky, 2007, Kroll, 2006
Sovjetsk/Tilsit	Russia	1	1709 - 1710			Kroll and Grabinsky, 2007, Kroll, 2006
Gdansk/Danzig	Poland	1	1709	50000	.533	Kroll and Grabinsky, 2007, Kroll, 2006
Elblag/Elbing	Poland	1	1709 - 1710		.3	Frandsen, 2009
Kamien Pomorski/Cammin	Poland	1	not specified			Wieden, 1999
Stargard	Poland	1	1710 - 1711	7000	.041	Kroll and Grabinsky, 2007, Kroll, 2006
Szczecin/Stettin	Poland	1	1709 - 1711	11250	.171	Kroll and Grabinsky, 2007, Kroll, 2006
Wolin	Poland	1	1710 - 1711			Wieden, 1999
Anklam	Germany	1	not specified			Wieden, 1999
Greifswald	Germany	1	1711			Wieden, 1999
Wolgast	Germany	1	1710 - 1711		.4	Wieden, 1999
Stralsund	Germany	1	1710 - 1711	7250	.314	Kroll and Grabinsky, 2007, Kroll, 2006
Goleniow/Gollnow	Poland	1	1709			Schöning, 1837
Hamburg	Germany	1	1712 - 1714	70000	.130	Kroll and Grabinsky, 2007, Kroll, 2006
Stade	Germany	1	1712			Frandsen, 2009, Winkle, 1983
Bremen	Germany	1	1712 - 1713	28000	.007	Frandsen, 2009

Notes: Sources are indicated in the last column. Population and mortality estimates, in the case of several available sources, denote the mean estimate. The modern names and countries of cities are used here.

Table 13: List of besieged cities

City	Modern Country	Siege
København	Denmark	1
Tönning	Germany	1
Hamburg	Germany	1
Halden	Norway	1
Gävle	Sweden	1
Härnösand	Sweden	1
Hudiksvall	Sweden	1
Piteå	Sweden	1
Söderhamn	Sweden	1
Umeå	Sweden	1
Södertälje	Sweden	1
Norrköping	Sweden	1
Nyköping	Sweden	1
Trosa	Sweden	1
Sundsvall	Sweden	1
Vyborg	Russia	1
Narva	Estonia	1
Riga	Latvia	1
Szczecin	Poland	1
Wolgast	Germany	1
Stralsund	Germany	1
Wismar	Germany	1

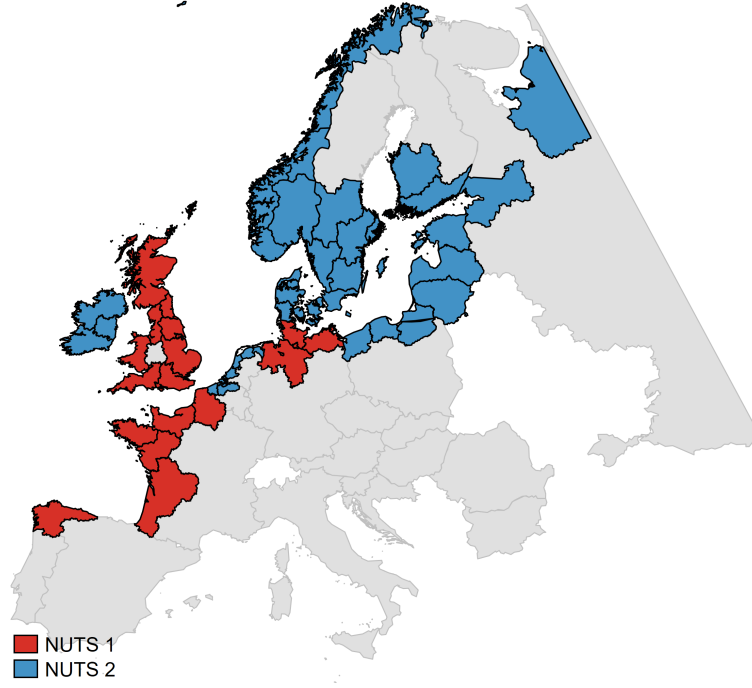
Notes: Six cities are both besieged and plagued. The six cities are Copenhagen (København), Narva, Riga, Stettin (Szczecin), Stralsund, and Wolgast. For Copenhagen and the first siege of Riga, there was no association with the plague, as these sieges occurred at the very beginning of the war in 1700 before the plague spread.

Table 14: Determinants of urban mortality

	Mortality rate
	(1)
Year of plague outbreak	-0.000 (0.001)
Latitude	0.010 (0.022)
Longitude	0.017** (0.007)
Distance to closest army	-0.062** (0.023)
East of the Sound	0.160** (0.055)
Observations	19

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the median urban mortality estimate. The independent variables are the year of the plague outbreak, latitude, longitude, distance to the closest army route between 1706 and 1714, and a dummy for the city being to the East of the Sound. The rationale for including army marching routes is that armies spread the plague.

Figure 7: Areas



Notes: This map shows the 56 areas into which regions are aggregated. For the UK, Germany, France, and Spain, I choose NUTS 1 regions, and for the rest of the sample areas that are equivalent to NUTS 2 regions. This avoids areas with an individual port city and generates areas of comparable sizes.

B.3 Demography

Table 15 shows that plagued cities were not larger or smaller than other cities, taking into account time trends and locational fundamentals by including time (or country x time) and city fixed effects. The source for these city-level populations is Buringh, 2021. As the closest data point after the plague is in 1750, I conclude that cities' populations had recovered to the trend after four decades. For regional rather than urban populations, I draw on Swedish parish-level population data (Andersson Palm, 2025). In Table 16, I find that by 1750, the populations of plagued Swedish parishes had recovered.

Figure 8: Army marching routes, 1706-1714



Notes: Map of army marching routes between 1706 and 1714, when the plague was recorded to have spread across Eastern and later Northern and Central Europe. No distinction is made between armies, as both allied and Russian troops are known to have spread the plague. I base these routes on maps by Spruner and Menke, 1880 and Barracough, 1997 which I digitised.

Table 15: Impact of plague on city population

	Population		Log Population	
	(1)	(2)	(3)	(4)
Plague x Year=800	0.0749 (11.86)	-0.0242 (10.77)	0.224 (0.811)	0.154 (0.584)
Plague x Year=900	0.0715 (11.86)	-0.0998 (10.77)	0.141 (0.810)	0.0493 (0.625)
Plague x Year=1000	0.251 (11.86)	-0.0178 (10.76)	0.0817 (0.737)	0.0286 (0.531)
Plague x Year=1100	0.161 (11.84)	-0.232 (10.74)	0.112 (0.710)	0.0373 (0.479)
Plague x Year=1200	0.424 (11.82)	-0.163 (10.72)	0.130 (0.697)	0.0898 (0.469)
Plague x Year=1300	1.339 (11.78)	0.352 (10.68)	0.366 (0.691)	0.282 (0.460)
Plague x Year=1400	1.772 (11.74)	1.043 (10.64)	0.460 (0.690)	0.468 (0.458)
Plague x Year=1500	3.816 (11.35)	2.709 (10.34)	0.597 (0.688)	0.592 (0.455)
Plague x Year=1550	4.758 (11.24)	3.489 (10.23)	0.622 (0.686)	0.637 (0.454)
Plague x Year=1600	6.464 (11.05)	4.798 (10.02)	0.684 (0.686)	0.704 (0.455)
Plague x Year=1650	7.789 (10.81)	5.332 (9.784)	0.783 (0.685)	0.747* (0.453)
Plague x Year=1700	9.645 (10.52)	6.381 (9.499)	0.800 (0.687)	0.737 (0.455)
Plague x Year=1750	11.20 (10.29)	7.018 (9.271)	0.840 (0.686)	0.668 (0.454)
Plague x Year=1800	14.78 (9.583)	8.766 (8.657)	0.954 (0.687)	0.697 (0.454)
Plague x Year=1850	21.37** (8.999)	10.64 (8.147)	0.844 (0.687)	0.692 (0.454)
Plague x Year=1900	63.51*** (19.85)	44.93** (18.64)	0.767 (0.691)	0.948** (0.460)
Plague x Year=1950	107.4** (42.88)	87.99** (38.62)	0.548 (0.695)	0.907* (0.465)
Plague x Year=2000	162.9*** (59.85)	153.2*** (54.14)	0.438 (0.694)	0.999** (0.467)
Constant	19.54*** (0.940)	19.69*** (0.947)	1.640*** (0.0266)	1.637*** (0.0181)
<i>Fixed Effects:</i>				
– City	✓		✓	
– Country x Year	✓	✓	✓	✓
– Year		✓		✓
Observations	22,470	22,489	17,027	17,059

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is (log) city population. The independent variable is a plague dummy interacted with year dummies. The area is restricted to the Baltic Sea and North Sea areas, specifically to cities between the tip of Norway, the Northern end of the Alps, the Irish West Coast and Arkhangelsk.

Table 16: Parish-level population recovery after the plague

	Population	Log Population
	(1)	(2)
Plague & Year=1750	25.51 (28.84)	0.0337*** (0.0125)
Plague & Year= 1810	-28.01 (59.52)	-0.0324** (0.0151)
<i>Fixed Effects:</i>		
– Parish	✓	✓
– Year	✓	✓
Observations	7,032	7,032

Notes: Standard errors clustered at the parish level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are parish-level (logarithmic) populations in 1690, 1750, and 1810. The independent variable is a post-plague dummy.

Demographic Response

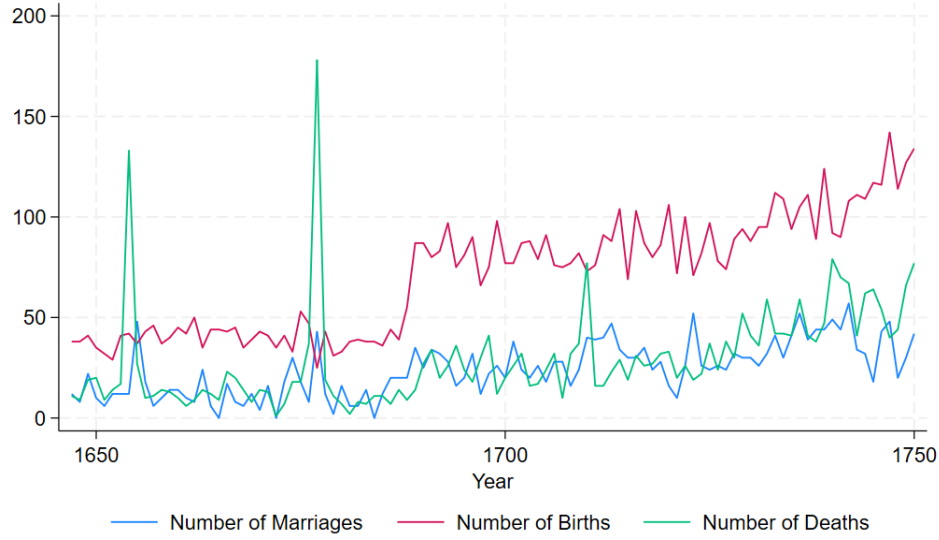
What was the demographic response to this plague outbreak? Theoretically, three main channels could contribute to population recovery. First, an increase in fertility. Guinnane, 2011 argues for a positive elasticity of fertility with respect to income. However, there is reason to doubt the strength and direction of this relationship in the particular area studied here. There is evidence for delayed marriage (De Moor and Zanden, 2009) after the Black Death for the North Sea region, which may suggest falling fertility. While there is no direct evidence on Baltic cities, this area falls under the European Marriage Pattern (Hajnal, 1965).

Population growth may still have accelerated due to lowered mortality, the second channel. Waldinger, 2022 shows that higher agricultural productivity was associated with lower mortality. Migration is the third channel through which urban populations may have recovered. Waldinger, 2022 also shows for England that warming increased agricultural productivity and drew in more migrants. While there is limited evidence for such migration across regions, I argue that rural-to-urban migration within regions may be sufficient to produce urban population recovery.

The Scanian Economic Demographic Database provides a uniquely granular source to answer this question (Bengtsson et al., 2014). Figure 9 shows annual births, deaths, and marriages for all of Scania. The 1710 peak in deaths is the plague outbreak studied in this paper. It was followed by a slight increase in marriages and births. This suggests a positive fertility response.

I also present results on the fertility channel at the parish level in Table 17. I find no significant fertility response at the parish level. As about a third of inhabitants of these parishes vanished, this implies a positive fertility response. However, this result is based on 59 parishes in one particular region of Sweden, limiting the external validity of this finding.

Figure 9: Number of births, deaths, and marriages in Scania, 1647-1750



Notes: The 1650s peak in deaths marks the Second Northern War and the 1670s peak the Scanian War.

Table 17: Impact of plague on number of births in 59 Scania parishes

	Births		Log Births	
	(1)	(2)	(3)	(4)
Post Plague	0.174 (0.725)	-0.376 (0.735)	-0.065 (0.068)	-0.104 (0.083)
<i>Fixed Effects:</i>				
– Origin	✓	✓	✓	✓
– Year	✓	✓	✓	✓
Years	All	1647-1750	All	1647-1750
Observations	1,015	591	973	578

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is a the (log) number of births per year and parish. The independent variable is a plague dummy, equal to one after the parish suffered a plague outbreak.

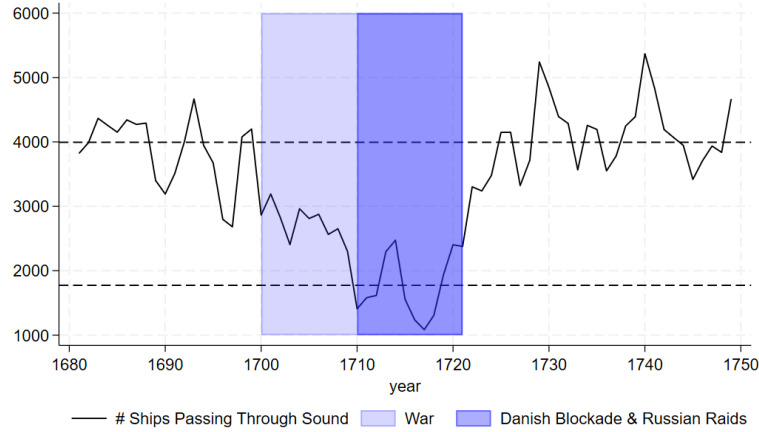
Insights from Other Plague Outbreaks

The Great Northern War plague outbreak's mortality rate of 36% is similar to the 40% mortality rate of the Black Death (Jedwab, Johnson, and Koyama, 2022). A key difference is that the Great Northern War plague outbreak was geographically very concentrated. This is a feature of later plague outbreaks also in Southern Europe (Alfani, 2023). Several reasons may have contributed to this, most importantly environmental and epidemiological changes affecting both immunity and transmissibility (Alfani and Murphy, 2017). It is therefore not atypical for such a late plague outbreak to be geographically concentrated.

The economic consequences of plague outbreaks depend crucially on how mortality interacts with sex, age, and socioeconomic status. Alfani and Murphy, 2017 summarises the existing literature on other plague outbreaks. There is no evidence that mortality differed by sex and ambiguous evidence on age. For the original Black Death and some later plague outbreaks, there is clear evidence that elites and lower-class citizens alike suffered high mortality.⁵⁹ Raster, 2023 analyses plague mortality for the Great Northern War plague outbreak specifically. While geographically constrained to Estonia, he finds no significant differences in mortality rates. Frandsen, 2009 shows qualitative evidence confirming this finding. Thus, the compositional effects of the plague outbreak should have been very limited.

⁵⁹Pullan, 1992 describes that 17% of the members of the Great Council were killed in the 1630 Venice plague and that 40% of the members of the Great and Low Councils died in the 1656-57 Genova plague outbreak.

Figure 10: Number of ships passing the Sound per year



Notes: This Figure displays the number of ships passing the Sound each year. In light blue, the war years of 1700 to 1721 are shaded; in dark blue, the years of 1710 to 1721 are shaded, during which the Danish navy blocked ships between Sweden and its Northern German possessions and the Russian army raided Baltic coastal areas and frequently intercepted ships. During this second half of the conflict, the number of ships passing the Sound collapsed to about a third, reducing observed trade for both plagued and un plagued ports.

C Trade Appendix

This Appendix holds all additional and robustness results for trade. The structure of this Appendix follows the structure of the trade results in the main paper. Figure 10 shows how blockades and naval raids affected the number of ships passing through the Sound. Table 18 lists blocked and raided Baltic ports, which is used in Table 19 to show that proximity to the naval base from which the Danish blockade was enforced predicts significantly lower exports, beyond blockades and raids.

Table 18: Naval raids and blockades affecting Baltic ports, 1709-1721

City	Event	Year(s)	Source
Christiansø	Danish base for blockade	1709–1720	Andersen, 2004
Karlskrona	Allied blockade	1716–1720	Aldridge, 2009
Gothenburg	Danish blockade	1717–1719	Andersen, 2004
Marstrand	Danish blockade	1719	Andersen, 2004
Strömstad	Danish blockade	1719	Andersen, 2004
Öregrund	Russian coastal raid	1719	Ericson Wolke, 2012
Gotland	Russian coastal raid	1717	Ericson Wolke, 2012
Norrtälje	Russian coastal raid	1719	Ericson Wolke, 2012
Södertälje	Russian coastal raid	1719	Ericson Wolke, 2012
Stockholm archipelago	Russian coastal raid	1719	Ericson Wolke, 2012
Trosa	Russian coastal raid	1719	Ericson Wolke, 2012
Nyköping	Russian coastal raid	1719	Ericson Wolke, 2012
Norrköping	Russian coastal raid	1719	Ericson Wolke, 2012
Gävle	Russian coastal raid	1719	Ericson Wolke, 2012
Umeå	Russian coastal raid	1720	Ericson Wolke, 2012
Söderhamn	Russian coastal raid	1721	Ericson Wolke, 2012
Hudiksvall	Russian coastal raid	1721	Ericson Wolke, 2012
Sundsvall	Russian coastal raid	1721	Ericson Wolke, 2012
Härnösand	Russian coastal raid	1721	Ericson Wolke, 2012
Piteå	Russian coastal raid	1721	Ericson Wolke, 2012

Notes: Based on the specific sources in column 4 and on Christiansø Administration, 2020 and Frost, 2000.

Table 19: Proximity to Christiansø is associated with significantly lower exports

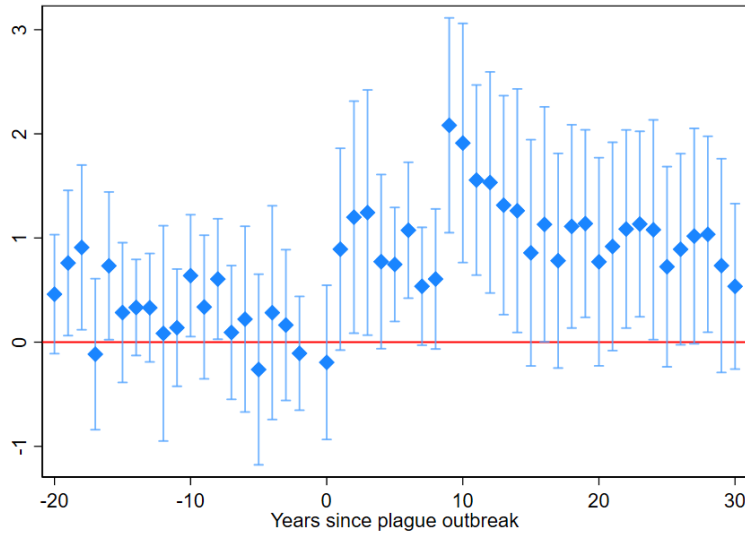
	Log Proximity to Christiansø			
	(1)	(2)	(3)	(4)
Logarithmic Proximity to Christiansø	-0.425** (0.166)	-0.581*** (0.210)	-0.455*** (0.172)	-0.604*** (0.217)
Logarithmic Proximity x Plague		0.022 (0.014)		0.021 (0.013)
Dummy for Danish Blockade			-0.160 (0.137)	-0.112 (0.130)
Dummy for Russian Raid			-0.443 (0.316)	-0.429 (0.319)
<i>Fixed Effects:</i>				
– Origin	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓
Observations	13,416	13,416	13,416	13,416

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the number of ships leaving an origin port. The independent variables are logarithmic proximity to the naval base from which the Danish blockade of Swedish ports was operated, defined as the product of a war period indicator (equal to one for 1709–1721) and the negative log of the great-circle distance in kilometres from each city to Christiansø, so that it takes larger values for cities that are geographically closer to the Danish blockade base during the war years and is zero in peacetime. Further, the interaction of this proximity measure with a plague dummy, a blockade dummy for additionally directly blocked ports, and a dummy for ports raided by Russian troops. All three are based on sources listed in Table 18.

C.1 Fact #1: Capital-intensive exports increase more than labour-intensive exports.

Figure 11 shows that reassigning four debatable goods (wine, tow, ash, and planks) as labour-intensive produces virtually identical results. Table 20 shows PPML and OLS results for Fact #1 in a difference-in-difference-in-difference set up. Table 21 repeats the specification while including army proximities and sieges. Figure 12 shows findings by factor intensity using the imputation estimator in Borusyak, Jaravel, and Spiess, 2021. Table 22 corroborates the increase in capital-intensive exports over labour-intensive exports by sector. Figure 13 shows the result by goods. Figure 14 shows that there is no shift into land-intensive exports after the plague. Finally, Figure 15 shows that the shift into capital-intensive exports lasts for almost 90 years.

Figure 11: Alternative assignment as capital-intensive: Fact #1



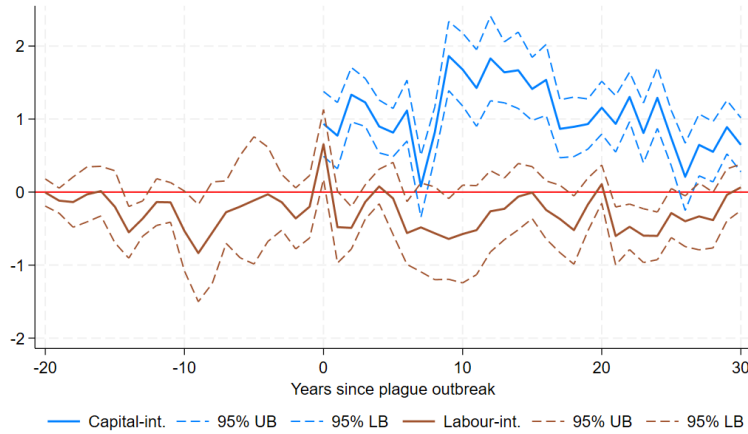
Notes: Among the top 30 goods (see Table 6), I have reassigned four debatable goods (wine and tow from CA to LA, planks and ash from CM to LM) and find almost identical results.

Table 20: Plague increases capital-intensive exports more than labour-intensive exports

	PPML		OLS	
	(1)	(2)	(3)	(4)
Post Plague	-1.219** (0.484)	-1.390*** (0.436)	-0.570*** (0.205)	-0.578*** (0.186)
Post Plague x Capital-Int.	0.904* (0.527)	1.385*** (0.229)	0.553* (0.289)	0.929*** (0.164)
<i>Fixed Effects:</i>				
– Origin x Sector	✓	✓	✓	✓
– Sector x Year	✓		✓	
– Year		✓		✓
Estimator	PPML	PPML	OLS	OLS
Observations	68,989	71,980	12,886	12,901

Notes: Difference-in-differences set up dropping the war years of 1700 to 1721. Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is annual exports by sector in levels (columns 1-2) and logs (columns 3-4). The independent variable is a post plague dummy, equal to one after the origin suffered a plague outbreak, and a dummy interacting the post plague dummy with a dummy for capital-intensive exports. Columns 1-2 show unweighted results, whereas in columns 3-4 the weight are exports in levels. Annual growing season temperatures are an additional control in all specifications.

Figure 12: Fact #1 using imputation estimator in Borusyak, Jaravel, and Spiess, 2021



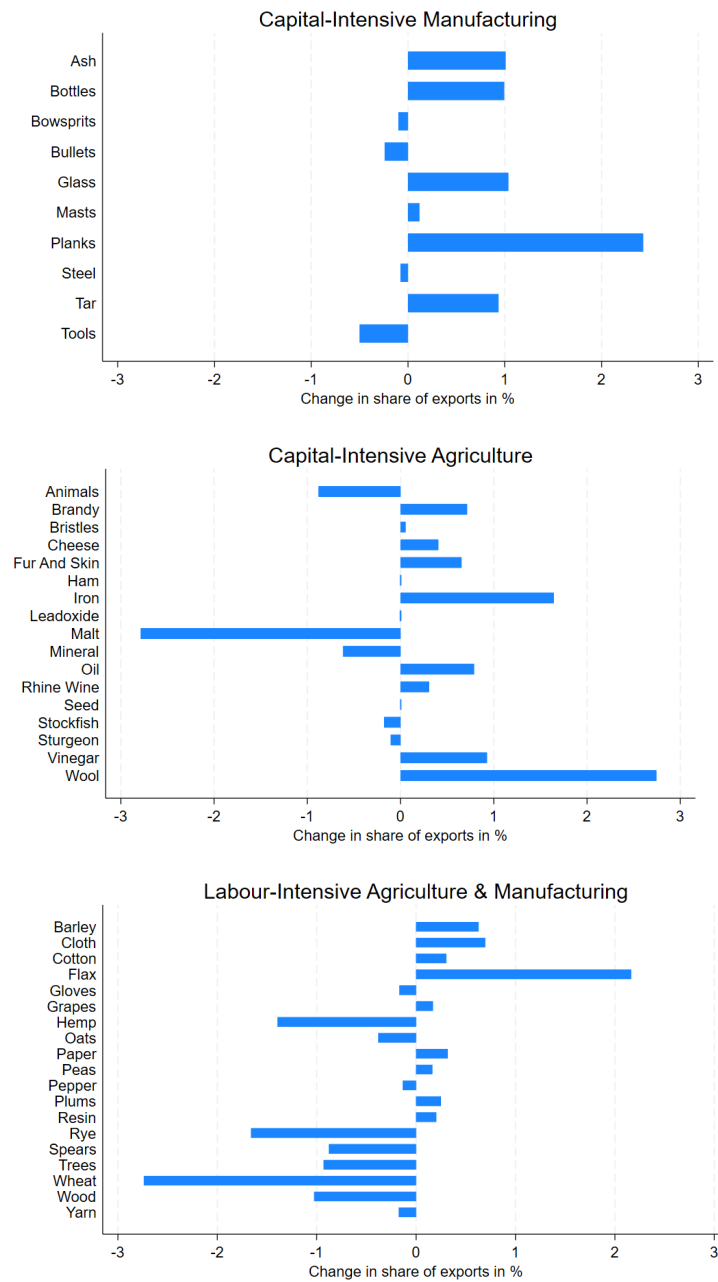
Notes: I estimate equation 1 by factor intensity using the imputation estimator in Borusyak, Jaravel, and Spiess, 2021. The package does not permit the estimation of pretrend heterogeneity, so these estimates are pooled across both factor intensities.

Table 21: Shift into capital-intensive exports by army proximity and siege

	PPML			OLS			PPML			OLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Army Distance x Post War	-0.060 (0.071)	-0.061 (0.057)	0.015 (0.031)	0.006 (0.025)								
Army Distance x Post War x Capital-Int.	0.061 (0.043)	0.047** (0.023)	0.008 (0.028)	-0.002 (0.016)								
Siege x Post War					0.715*** (0.098)	0.750*** (0.083)	0.579*** (0.090)	0.607*** (0.092)				
Siege x Post War x Capital-Int.					0.335 (0.273)	0.309 (0.315)	0.007 (0.229)	0.007 (0.239)				
Post Plague	-1.136** (0.510)	-1.256*** (0.419)	-0.424** (0.177)	-0.545*** (0.119)	-1.156*** (0.385)	-1.321*** (0.324)	-0.615*** (0.131)	-0.709*** (0.103)				
Post Plague x Capital-Int.	1.132** (0.530)	1.265*** (0.270)	0.514* (0.308)	0.842*** (0.193)	1.109** (0.563)	1.345*** (0.340)	0.573* (0.329)	0.923*** (0.216)				
<i>Fixed Effects:</i>												
- Origin x Sector	✓	✓	✓	✓	✓	✓	✓	✓				
- Sector x Year	✓		✓		✓		✓					
- Year		✓		✓		✓						
Estimator	PPML	PPML	OLS	OLS	PPML	PPML	OLS	OLS				
Observations	100,530	103,750	17,353	17,368	102,636	105,908	17,545	17,560				

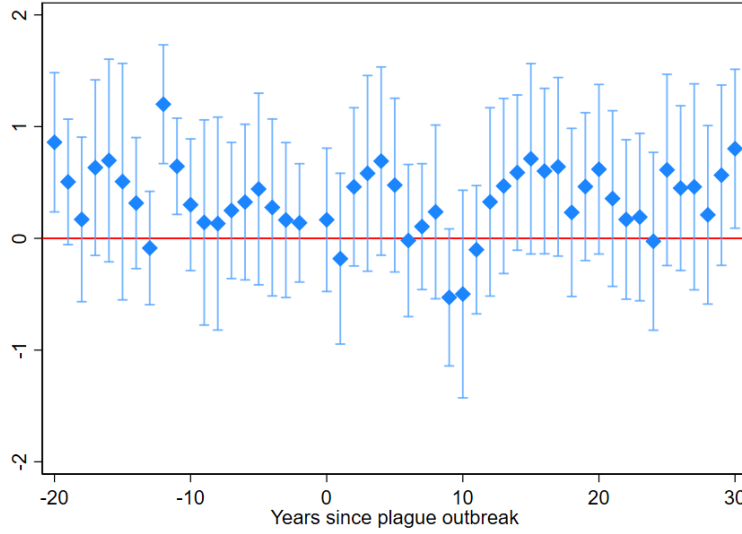
Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is annual exports by sector in levels (PPML) or logs (OLS). The independent variable is a post plague dummy, equal to one after the origin suffered a plague outbreak, and a dummy interacting the post plague dummy with a dummy for capital-intensive exports. Additionally, in columns 1-4 I include the smallest distance any army had to a region during the Great Northern War, interacted with a post-war dummy (equal to 1 after 1721), and this interactions interaction with a capital-intensity dummy. The PPML columns show unweighted results, whereas in the OLS columns the weight are exports in levels. Annual growing season temperatures are an additional control in all specifications. Table 13 shows besieged cities and Figure 8 shows army routes.

Figure 13: Fact #1: Shift into capital-intensive exports by good



Notes: Results from regressing a plague dummy on the share a good constitutes among origin-time exports, including origin and time fixed effects. Shown is a selection of point estimates that are significant at the 10% level.

Figure 14: No shift into land-intensive exports after the plague



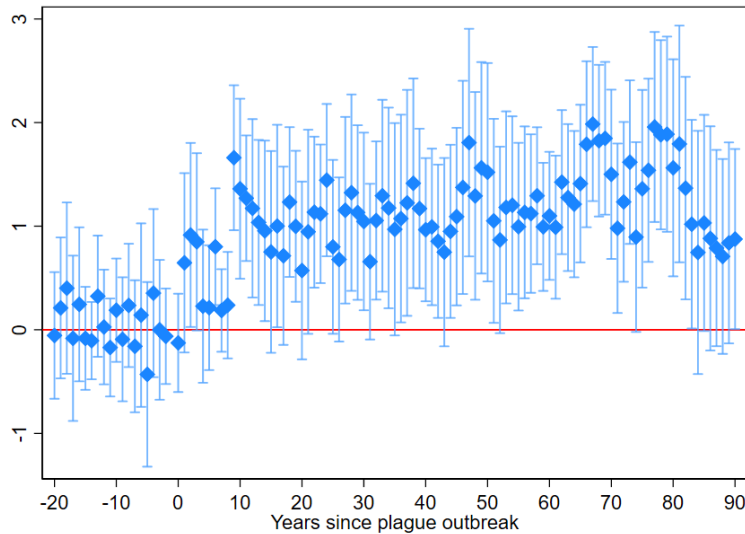
Notes: Distinguishing land-intensive (arable crops, livestock and animal products, plantation goods, raw forestry) from labour-intensive goods (crafted manufactures, iron and metalwork, textiles and clothing).

Table 22: Plague increases capital-intensive exports more than labour-intensive exports in both sectors

	Manufacturing		Agriculture	
	(1)	(2)	(3)	(4)
Post Plague	-1.491*** (0.498)	-0.547*** (0.121)	-1.136*** (0.355)	-0.566*** (0.183)
Post Plague x Capital-Int.	1.267*** (0.235)	0.787*** (0.142)	1.507** (0.647)	0.740* (0.379)
<i>Fixed Effects:</i>				
– Origin x Sector	✓	✓	✓	✓
– Year	✓	✓	✓	✓
Estimator	PPML	OLS	PPML	OLS
Observations	31,023	5,043	53,369	9,578

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is annual exports by sector in levels (columns 2 and 4) and logs (columns 1 and 3). The independent variable is a post plague dummy, equal to one after the origin suffered a plague outbreak, and a dummy interacting the post plague dummy with a dummy for capital-intensive exports. Columns 1 and 3 show unweighted results, whereas in columns 2 and 4 the weight are exports in levels. Annual growing season temperatures are an additional control in all specifications.

Figure 15: Capital-intensive exports expand relative to labour-intensive exports also in the long run

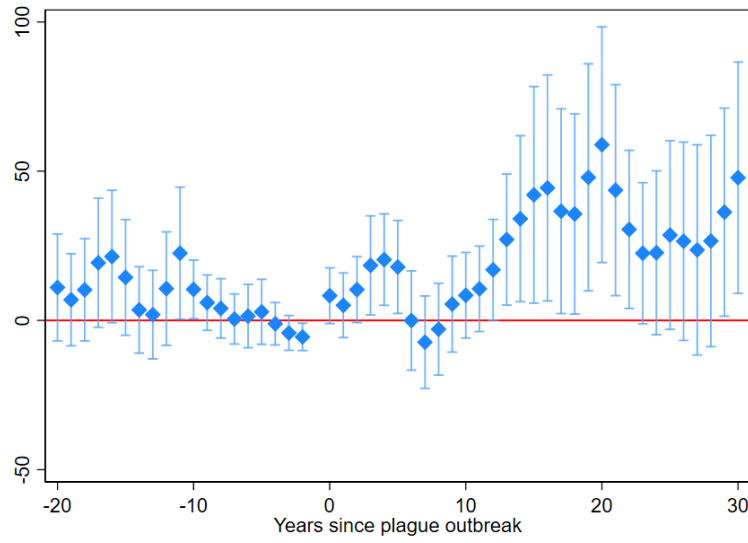


Notes: Estimation of equation 1 in the long run. Standard errors clustered at the origin level.

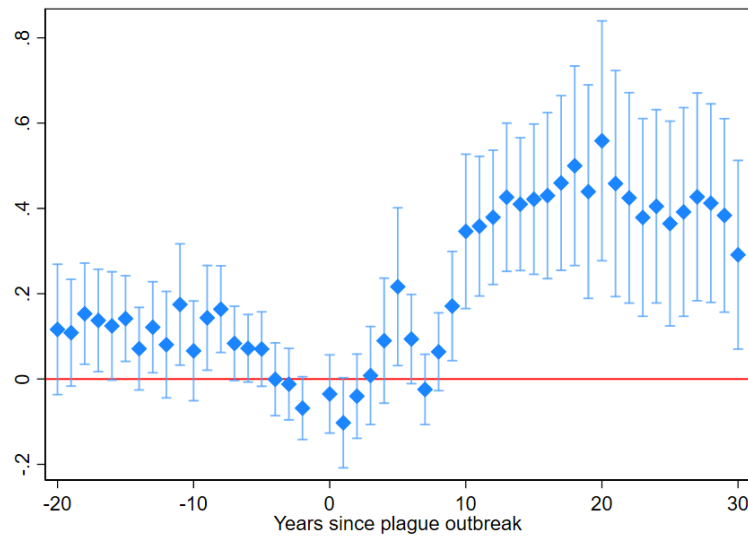
C.2 Fact #2: Plagued regions capture larger shares of destination markets.

I now present robustness results for the post-plague export expansion. Figure 16a shows that export volumes expand and Figure 16b shows the extensive margin expansion. Table 23 shows results from a difference-in-differences set up excluding the war years of 1700 to 1721. Figure 17a extends the pre-period to 30 years and again finds no pre-trends. Figure 20a adds lagged temperatures. I also show results from cleaned values following the decomposition of duty records into underlying value as outlined in Section A.4 in Figure 17b. Figure 18 finds similar results with NUTS 1-time and NUTS 2-time fixed effects instead of area-time fixed effects. Figure 20b repeats the main analysis on trade shares using the imputation estimator by Borusyak, Jaravel, and Spiess, 2021. Figure 19 decomposes the intensive margin expansion by sector and Figure 21 by factor intensity. Figure 22 shows the export volume expansion by good. Finally, Figures 23a and 23b show the export expansion in the long run.

Figure 16: Robustness results for Fact #2: Export volumes and the extensive margin expand



(a) Export volumes



(b) Number of exported goods

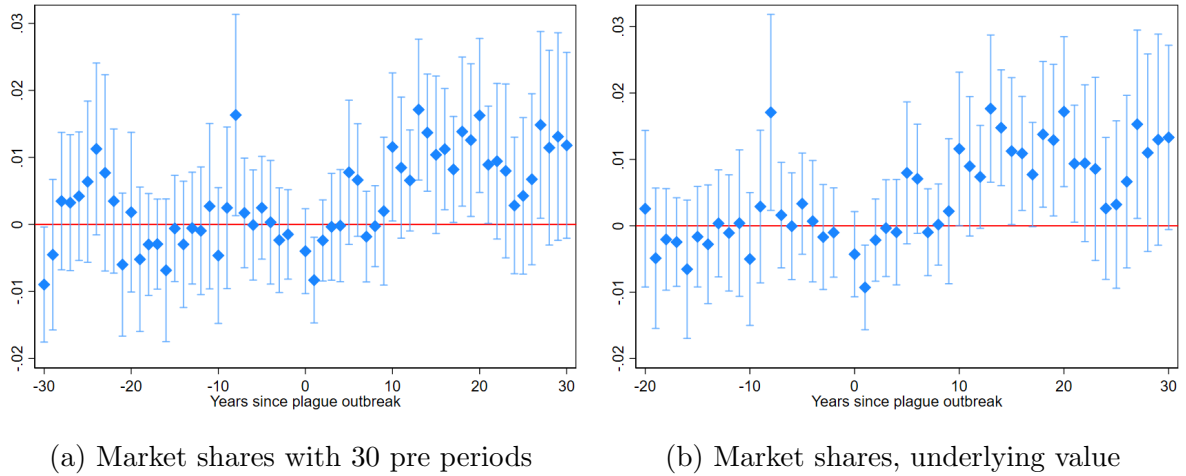
Notes: In Figure 16a, estimation of equation 2 on $T_{ijt} = x_{ijt}$. Standard errors clustered at the origin level. In Figure 16b, estimation of equation 2, where T_{ijt} is the number of exported goods.

Table 23: Plague increases exports at the intensive and extensive margins

	Volume		# Goods		# Novel Goods	
	(1)	(2)	(3)	(4)	(5)	(6)
Post Plague	0.503*** (0.086)	38.735*** (14.154)	0.206 (0.146)	0.267*** (0.072)	1.252*** (0.291)	0.090** (0.044)
<i>Fixed Effects:</i>						
– Area x Year	✓	✓	✓	✓	✓	✓
– Origin x Destination	✓	✓	✓	✓	✓	✓
– Destination x Year	✓	✓	✓	✓	✓	✓
Estimator	PPML	OLS	PPML	OLS	PPML	OLS
Observations	103,532	361,059	246,958	361,059	46,727	361,059

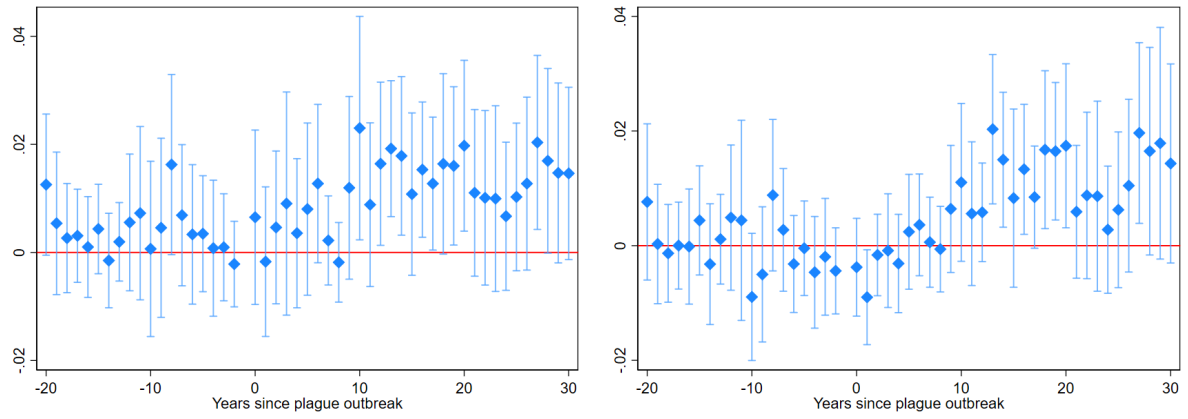
Notes: Difference-in-differences set up dropping the war years of 1700 to 1721. Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are annual exports, the number of exported goods, and the number of novel exported goods. The independent variable is a post plague dummy, equal to one after the origin suffered a plague outbreak. Annual growing season temperatures are an additional control in all specifications.

Figure 17: Robustness results for Fact #2: Thirty pre periods and underlying value



Notes: Figure 17a estimates equation 2 on shares, extending the pre-period to 30 years. Figure 17b shows regression results based on underlying value, as recovered in Appendix A.4. Standard errors clustered at the origin level.

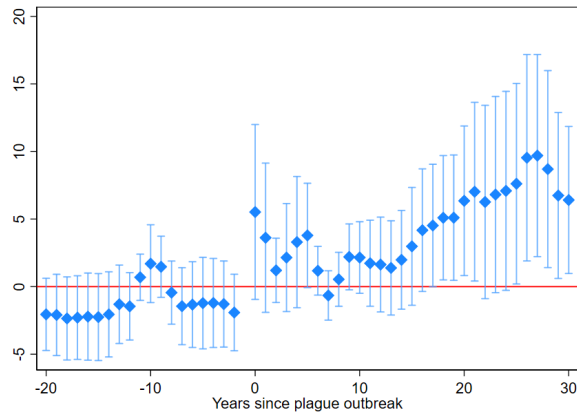
Figure 18: Robustness results for Fact #2: NUTS 1-time and NUTS 2-time fixed effects



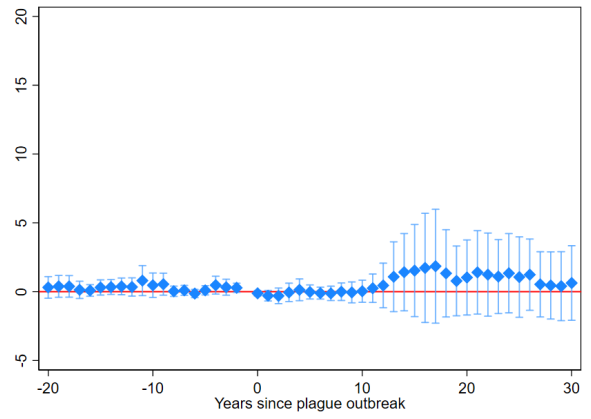
(a) Market shares with NUTS 1-time fixed effects (b) Market shares with NUTS 2-time fixed effects

Notes: These Figures estimate equation 2 on shares. Instead of area-time fixed effects, I use NUTS 1-time and NUTS 2-time fixed effects consistently. Standard errors clustered at the origin level.

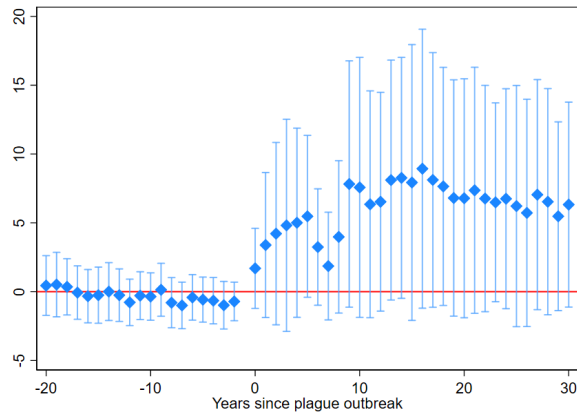
Figure 19: Intensive margin expansion, volumes by sector



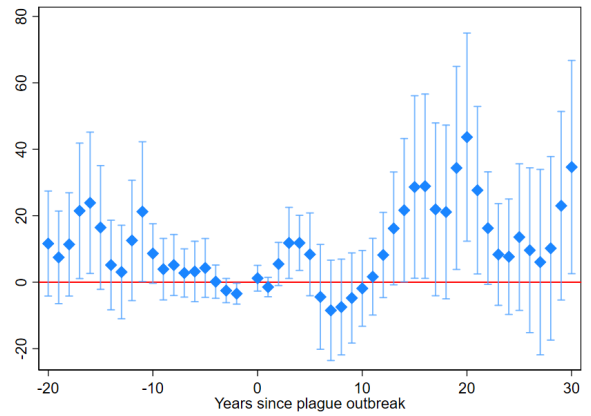
(a) Capital-int. manufacturing



(b) Labour-int. manufacturing



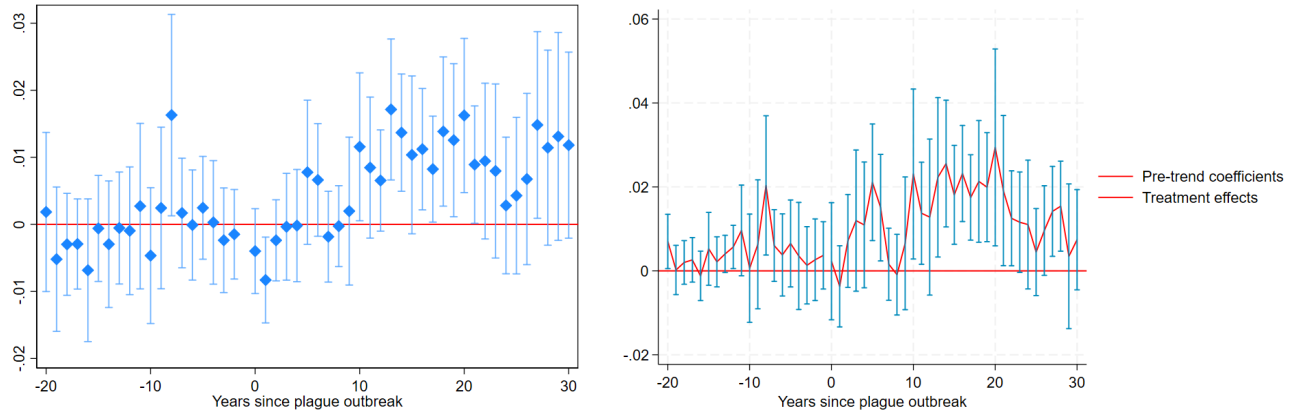
(c) Capital-int. agriculture



(d) Labour-int. agriculture

Notes: Estimation of equation 2 on volumes. The y-axis differs in Figure 19d to preserve the legibility of the other Figures. Between 1668-1750, 55% of traded value was in labour-intensive agriculture, 27% in capital-intensive agriculture, 3% in labour-intensive manufacturing and 13 % in capital-intensive manufacturing. Standard errors clustered at the origin level.

Figure 20: Lagged temperature and imputation estimator in Borusyak, Jaravel, and Spiess, 2021

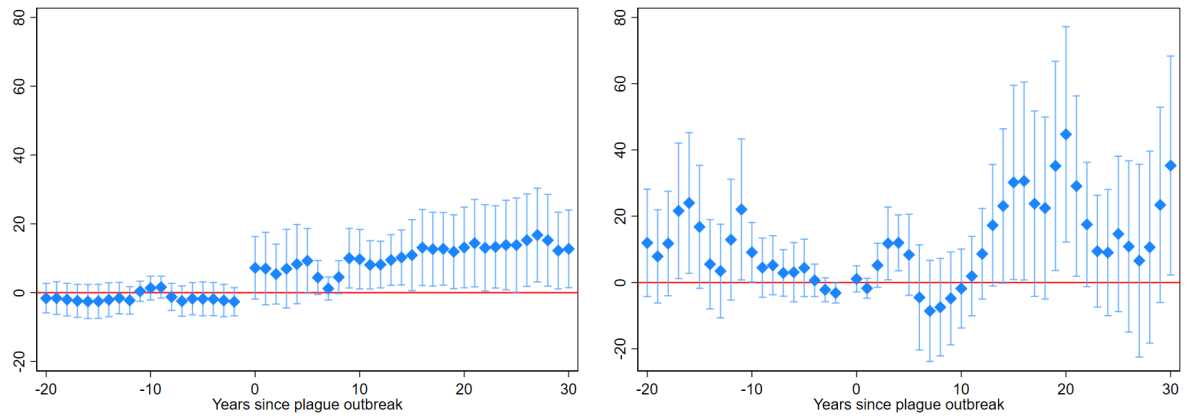


(a) Market shares, adding temperature lags of 10 and 20 years

(b) Market shares, imputation estimator

Notes: Figure 20b shows results from estimating equation 2 with 20 pre-trends and 30 post periods using the imputation estimator by Borusyak, Jaravel, and Spiess, 2021. Standard errors clustered at the origin level.

Figure 21: Intensive margin expansion, volumes by factor intensity

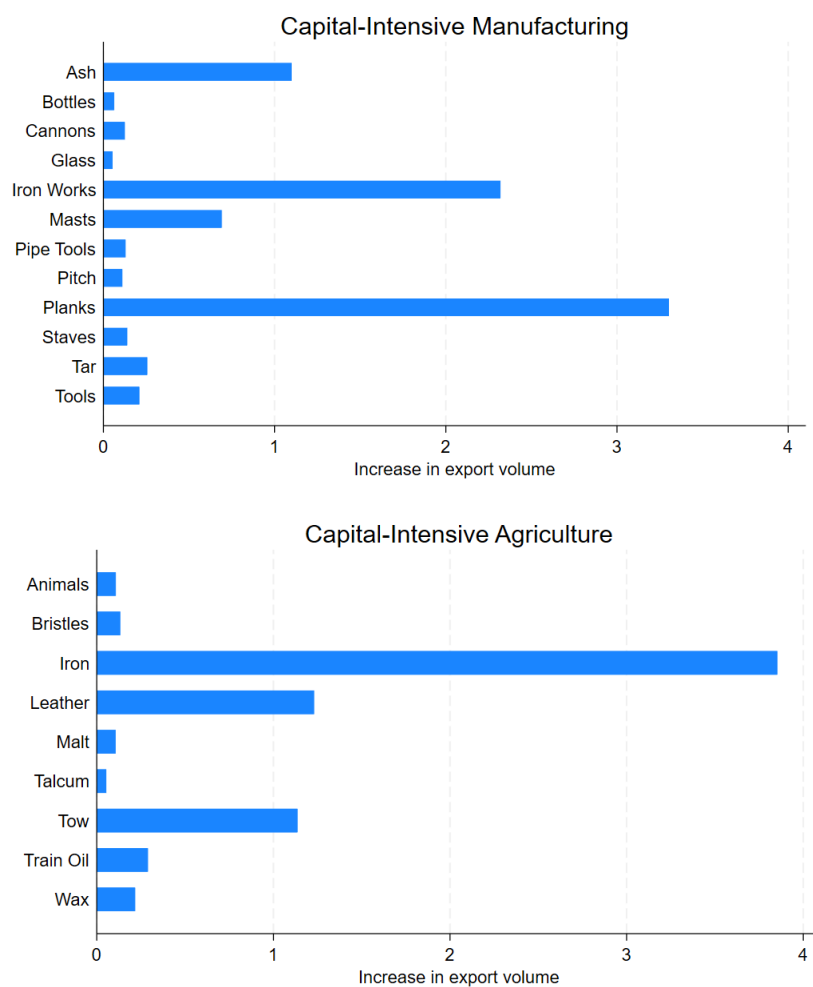


(a) Capital-intensive sectors

(b) Labour-intensive sectors

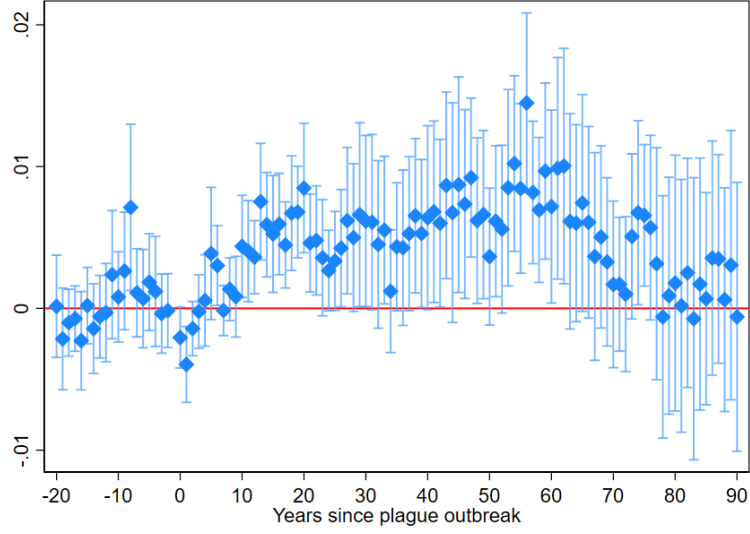
Notes: Estimation of equation 2 on volumes. Between 1668-1750, 58% of traded value was in labour-intensive sectors and 40% in capital-intensive sectors. Standard errors clustered at the origin level.

Figure 22: Fact #2: Volume expansion by good, capital-intensive sectors

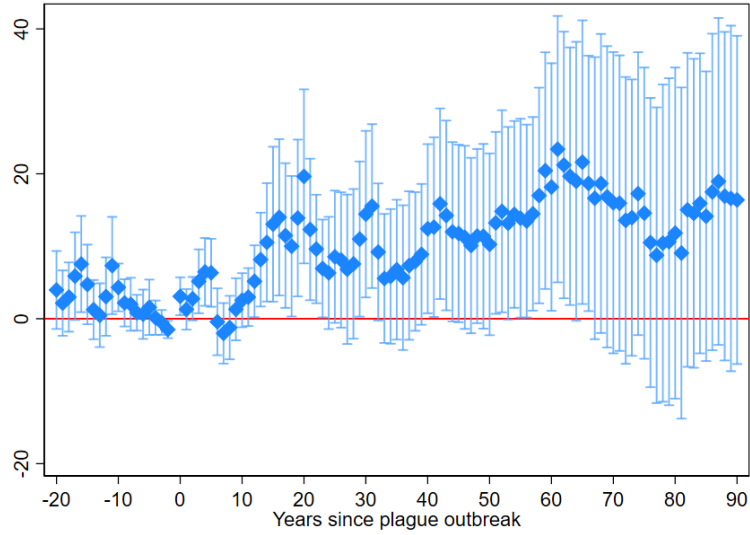


Notes: Results from regressing a plague dummy on i 's exports to j in good g in annual data, controlling for area $\times$ year, origin $\times$ destination, and destination $\times$ year fixed effects. Shown are only point estimates that are significant at the 5% level.

Figure 23: Market shares and export volumes expand in the long run



(a) Market shares



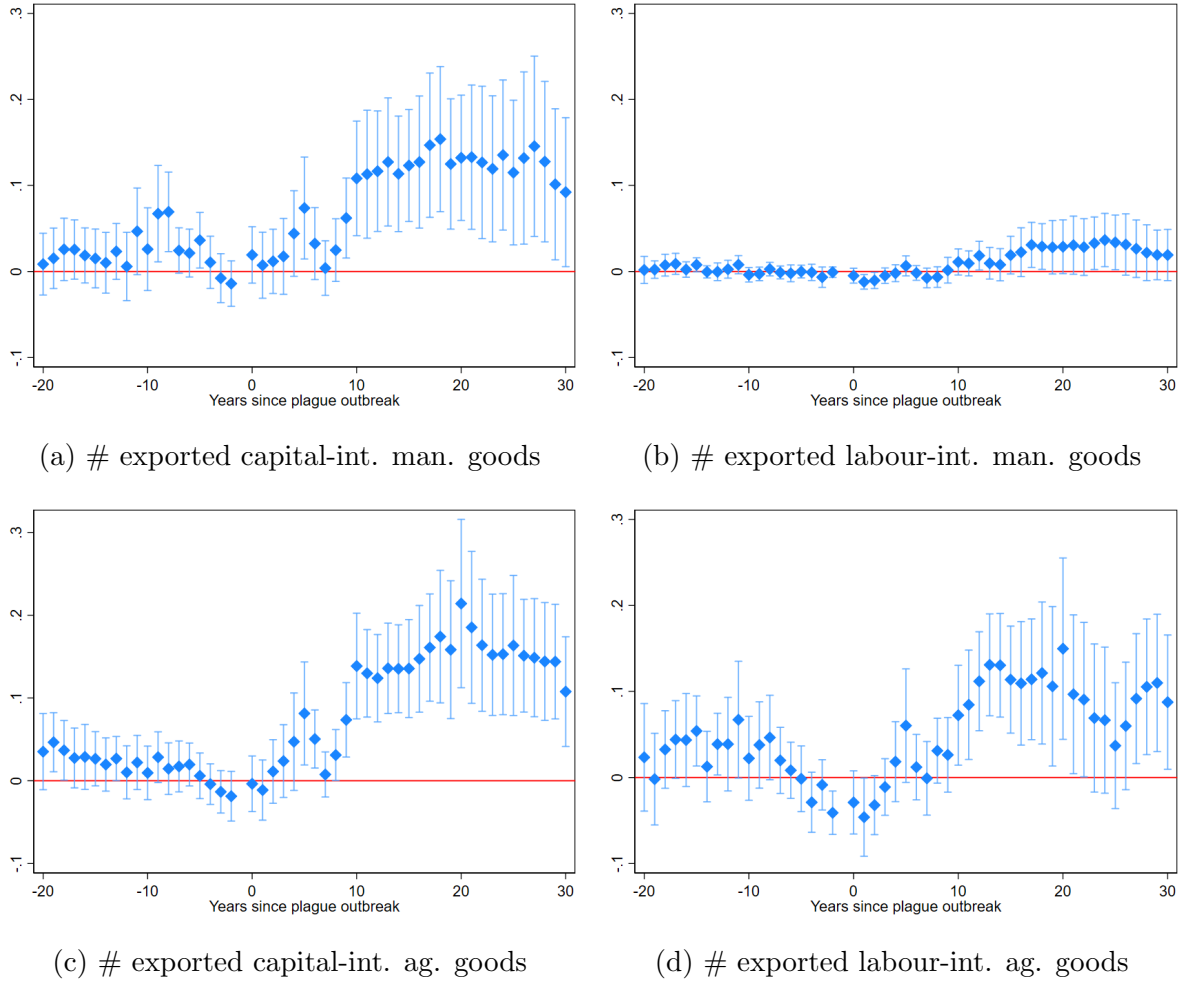
(b) Export volumes

Notes: In Figure 23a, estimation of equation 2 on i 's share of j 's imports, $T_{ijt} = \frac{x_{ijt}}{\sum_{i \in I} x_{ijt}}$. In Figure 23b, estimation of equation 2 on $T_{ijt} = x_{ijt}$. Standard errors clustered at the origin level.

C.3 Extensive Margin: Plagued regions export a larger variety of goods.

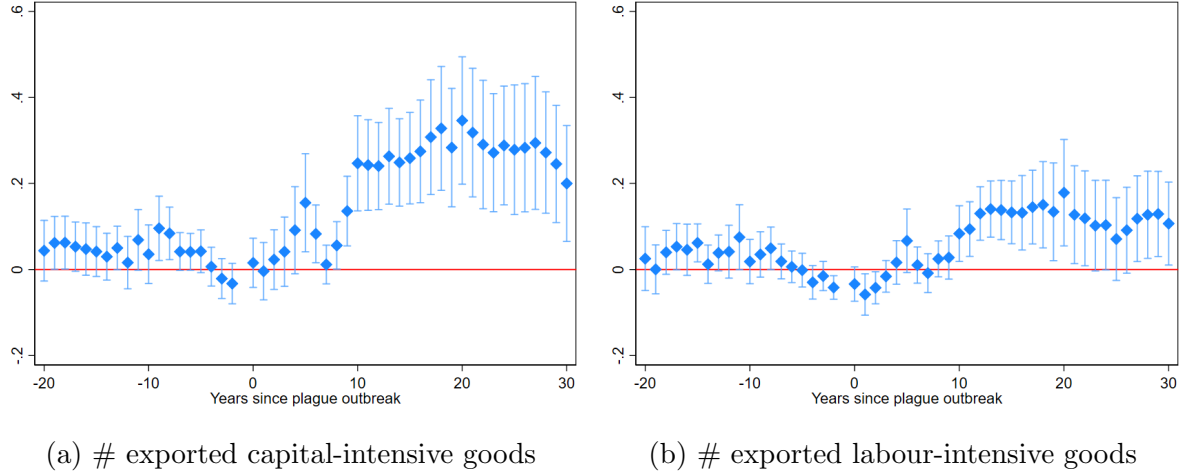
Robustness results for the extensive margin expansion follow. Figure 24 shows that all four sectors participated in the expansion, with labour-intensive manufacturing contributing the least. Figure 25 separates the extensive margin expansion by factor intensity. Figure 26 shows results at the goods level.

Figure 24: Extensive margin expansion, by sector



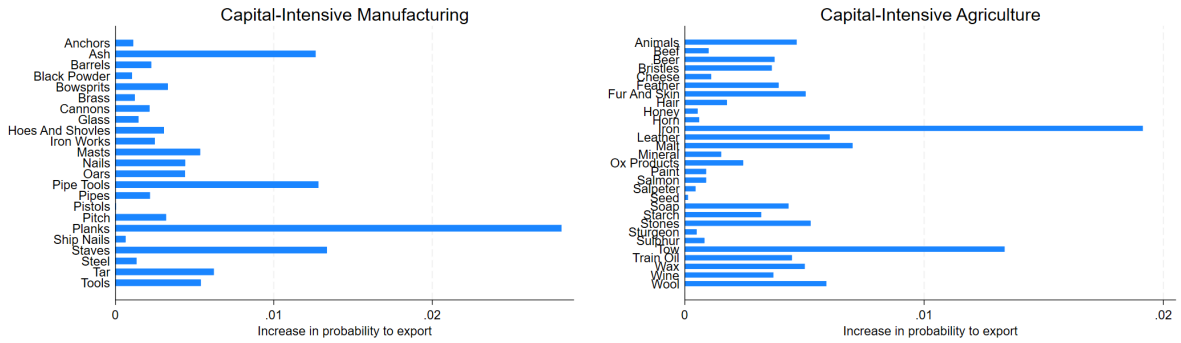
Notes: Estimation of equation 2 on the number of exported goods by sector and factor intensity. Between 1668-1750, 55% of traded value was in labour-intensive agriculture, 27% in capital-intensive agriculture, 3% in labour-intensive manufacturing and 13 % in capital-intensive manufacturing. Standard errors clustered at the origin level.

Figure 25: Extensive margin expansion, by factor intensity



Notes: Estimation of equation 2 on the number of exported goods by factor intensity. Between 1668-1750, 58% of traded value was in labour-intensive sectors and 40% in capital-intensive sectors. Standard errors clustered at the origin level.

Figure 26: Fact #3: Extensive margin expansion by good, capital-intensive sectors

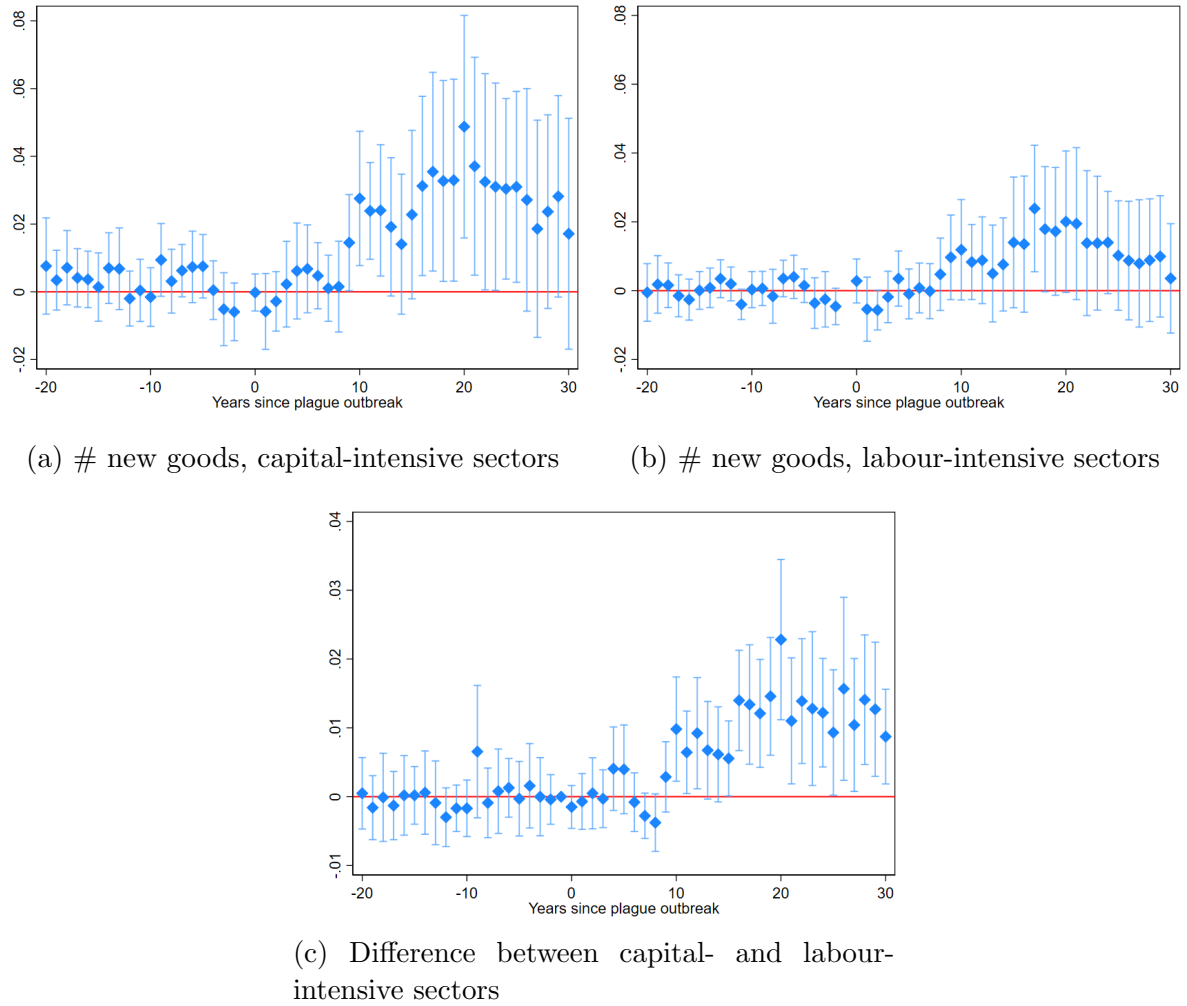


Notes: Results from regressing a plague dummy on a dummy for positive exports of good g from i to j in annual data, controlling for area $\times$ year, origin $\times$ destination, and destination $\times$ year fixed effects. Shown are only point estimates that are significant at the 5% level.

Innovation: Exports of new products increase after the plague.

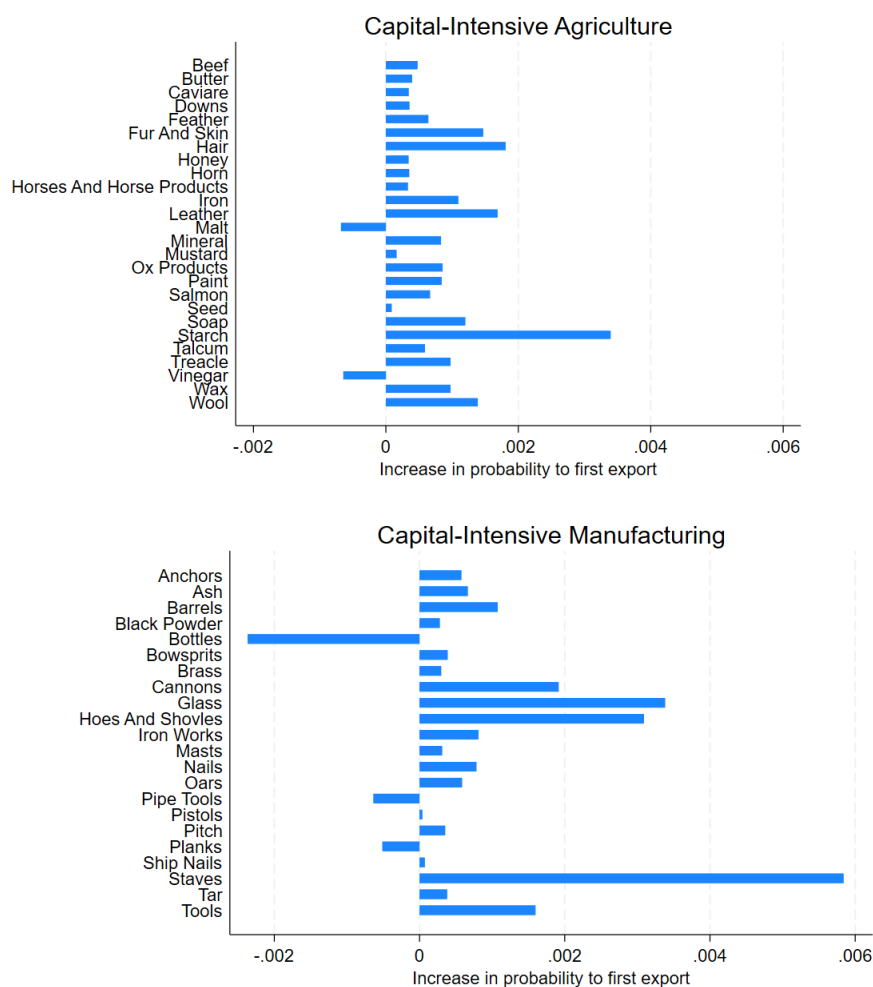
Figure 27 shows the extensive margin of new goods by factor intensity, while Figure 28 shows these results by good. Table 24 shows results from PPML panel regressions of a post plague dummy on the number of goods never exported before for each of the four sectors. Table 25 shows these results for the volume of goods never exported before, thus illustrating the contribution of new exports to the intensive margin results.

Figure 27: New exports at the extensive margin, by factor intensity



Notes: Estimation of equation 2 on the number of exported goods, by factor intensity. A new export is defined as a good first exported by a city between 1709, the onset of the plague in my study area, and 1732, 20 years after the last plague year in this region. Standard errors clustered at the origin level.

Figure 28: Fact #4: Increased probability of exporting a good for the first time, capital-intensive sectors



Notes: Results from regressing a plague dummy on a dummy for a new export of good g from i to j in annual data, controlling for area $\times$ year, origin $\times$ destination, and destination $\times$ year fixed effects. A new export is a good not exported before the plague. Shown are only point estimates that are significant at the 5% level.

Table 24: Impact of plague on number of goods never exported before, by factor intensity

	Overall	Labour-intensive ag.	Capital-intensive ag.	Labour-intensive man.	Capital-intensive man.
	(1)	(2)	(3)	(4)	(5)
Post Plague	0.962*** (0.242)	1.011*** (0.325)	0.413 (0.280)	1.349*** (0.266)	0.919*** (0.321)
<i>Fixed Effects:</i>					
– Area x Year	✓	✓	✓	✓	✓
– Origin x Destination	✓	✓	✓	✓	✓
– Destination x Year	✓	✓	✓	✓	✓
– Controls	✓	✓	✓	✓	✓
Estimator	PPML	PPML	PPML	PPML	PPML
Observations	74,452	17,270	40,554	4,140	14,185

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is the number of goods that a region has never exported before. The independent variable is a plague dummy, equal to one after the origin suffered a plague outbreak. The first 20 years of the data are dropped, as mechanically initially every good is new. Annual growing season temperatures are an additional control.

Table 25: Impact of plague on volume of goods never exported before, by factor intensity

	Overall	Labour-intensive ag.	Capital-intensive ag.	Labour-intensive man.	Capital-intensive man.
	(1)	(2)	(3)	(4)	(5)
Post Plague	1.583*** (0.467)	2.106*** (0.624)	1.260* (0.749)	1.038** (0.522)	1.217 (0.853)
<i>Fixed Effects:</i>					
– Area x Year	✓	✓	✓	✓	✓
– Origin x Destination	✓	✓	✓	✓	✓
– Destination x Year	✓	✓	✓	✓	✓
– Controls	✓	✓	✓	✓	✓
Estimator	PPML	PPML	PPML	PPML	PPML
Observations	50,472	13,365	30,255	3,284	12,149

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is the volume of goods that a region has never exported before. The independent variable is a plague dummy, equal to one after the origin suffered a plague outbreak. The first 20 years of the data are dropped, as mechanically initially every good is new. Annual growing season temperatures are an additional control.

C.4 Other Robustness Results

Results from gravity regressions with origin-time, destination-time, and origin-destination fixed effects are displayed in Table 26. Table 27 shows that the plague increased regions' probability to export. For all four facts, I present heterogeneity results in Table 28. In Table 29, I show that days at sea (a measure of travel time and handling times in harbours) are unrelated to both the plague and to capital intensity. Table 30 shows heterogeneity results by plagued regions' siege status. Figure 29 presents evidence that my finding is not driven by export hubs whose reduced local consumption pushed more goods into exports. I show that plagued regions expanded their imports, speaking against a local demand contraction. Finally, Table 31 shows in data by Baasch, 1929 that Hamburg saw an export expansion and a shift into a higher capital intensity of exports after the plague. This speaks against both mechanical explanations and selection issues.

Table 26: Gravity analysis of plague shock on bilateral trade

	Overall	Agriculture		Manufacturing	
		Labour-intensive	Capital-intensive	Labour-intensive	Capital-intensive
	(1)	(2)	(3)	(4)	(5)
Post Plague	0.674* (0.372)	0.718** (0.356)	0.470* (0.279)	1.232*** (0.454)	-0.602 (0.682)
<i>Fixed Effects:</i>					
– Origin x Year	✓	✓	✓	✓	✓
– Destination x Year	✓	✓	✓	✓	✓
– Origin x Destination	✓	✓	✓	✓	✓
Estimator	PPML	PPML	PPML	PPML	PPML
Observations	505,415	49,887	69,106	26,141	40,514

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is annual bilateral trade. The independent variable is a bilateral plague dummy, equal to one after either the origin, the destination, or both cities suffered a plague outbreak.

Table 27: Impact of plague on probability to export

	Marginal Effect		Average Marginal Effect			
	(1)	(2)	(3)	(4)	(5)	(6)
Post Plague	0.035*** (0.010)	0.007 (0.008)	0.032*** (0.008)	0.154*** (0.025)	0.057*** (0.015)	0.003 (0.004)
<i>Fixed Effects:</i>						
– Origin		✓		✓		✓
– Year		✓		✓		✓
Estimator	OLS	OLS	Logit	Logit	Tobit	Tobit
Observations	502,980	502,980	502,980	502,980	502,980	502,980

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is a dummy for whether a region exports in a given year. The independent variable is a plague dummy, equal to one after the origin suffered a plague outbreak. For Tobit, I model the exporter dummy as left truncated at 0. For Logit and Tobit, average marginal effects are displayed. In the balanced trade panel, the share of active exporters in a given year averages 10.4%.

Table 28: Heterogeneity of trade findings by pre-plague export levels

	Export Volume		Export Share		# Exported Goods		# New Exported Goods	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post Plague	-0.98035050** (0.41571542)	-0.57260670 (0.44376456)	0.01024532*** (0.00274101)	0.00405861 (0.00311360)	0.19697625*** (0.05694356)	0.11466535** (0.05151306)	0.06830435** (0.03415583)	0.02939703 (0.02897934)
Post Plague x Cap-Int.	0.96971886* (0.52203372)	1.14320257*** (0.41610229)						
Post Plague x Pre-Plague Exports		-0.00000115*** (0.00000037)		0.00000006*** (0.00000001)		0.00004342* (0.00002343)		0.00002052** (0.00000889)
Post Plague x Cap-Int. x Pre-Plague Exports		-0.00000154*** (0.00000042)						
<i>Fixed Effects:</i>								
- Origin x Sector	✓	✓						
- Sector x Time	✓	✓						
- Area x Year			✓	✓	✓	✓	✓	✓
- Origin x Destination			✓	✓	✓	✓	✓	✓
- Destination x Year			✓	✓	✓	✓	✓	✓
Estimator	PPML	PPML	OLS	OLS	OLS	OLS	OLS	OLS
Observations	102,636	102,636	491,277	491,277	491,277	491,277	491,277	491,277

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable in columns 1-2 is the volume of exports by capital intensity. In columns 3-4, it is the market share origin i captures in j. In columns 5-6, it is the number of exported goods. In columns 7-8, it is the number of new exported goods. The independent variables for all four dependent variables are first a post plague dummy and then additionally its interaction with the level of exports before 1709 (number of exported goods) for columns 1-4 (5-8). In columns 1-2, the independent variables also include a post-plague x capint dummy and its interaction with pre-plague exports.

Table 29: Travel times are unaffected by the plague

	Days at Sea			
	(1)	(2)	(3)	(4)
Value of Cap.-Int. Goods	-0.022 (0.014)	-0.010 (0.016)		
Total Value	0.032** (0.015)	0.031 (0.025)		
Post Plague x Value of Cap.-Int. Goods		-0.040 (0.035)		
Post Plague x Total Value		0.002 (0.031)		
Weight of Cap.-Int. Goods			-0.000 (0.000)	0.000 (0.000)
Total Weight			0.000 (0.000)	-0.000 (0.000)
Post Plague x Weight of Cap.-Int. Goods				-0.000 (0.000)
Post Plague x Total Weight				0.000 (0.000)
<i>Fixed Effects:</i>				
– Captain	✓	✓	✓	✓
– Origin x Year	✓	✓	✓	✓
Measure	Value	Value	Weight	Weight
Observations	66,241	66,241	66,241	66,241

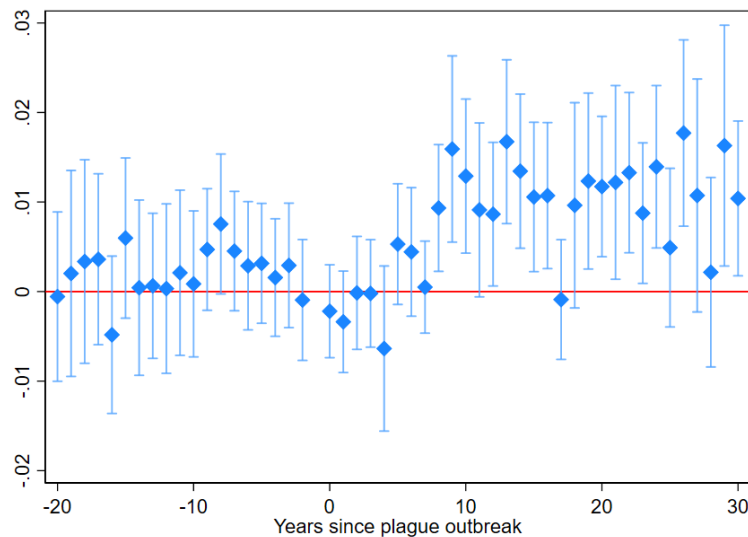
Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sample are trips by the same captain (identified by full name and home town) to and from the Sound, where I restrict to trips where the initial destination is the same as the current origin to avoid including trips with unobserved trips within the Baltic or within the North Sea. To the same end, I exclude observations where more than a year has passed since a captain had last been seen at the Sound. The final sample includes 101,615 passages. Days at sea is the difference in days between two observations at the Sound of the same captain. I control for captain-specific and route-time-specific determinants of travel time with fixed effects.

Table 30: Heterogeneity of trade findings by siege

	Export Volume	Export Share	# Exported Goods	# New Exported Goods
	(1)	(2)	(3)	(4)
Post Plague	-0.948** (0.442)	0.004 (0.003)	0.128** (0.054)	0.027 (0.026)
Post Plague x Pre-Plague Exports	-0.000 (0.000)	0.000*** (0.000)	0.000* (0.000)	0.000** (0.000)
Post Plague x Cap-Int.	1.436*** (0.434)			
Post Plague x Cap-Int. x Pre-Plague Exports	-0.000*** (0.000)			
Post Plague x Siege	0.471*** (0.147)	0.000 (0.003)	-0.061 (0.087)	0.010 (0.026)
Post Plague x Cap-Int. x Siege	-0.085 (0.228)			
<i>Fixed Effects:</i>				
– Origin x Sector	✓			
– Sector x Time	✓			
– Area x Year		✓	✓	✓
– Origin x Destination		✓	✓	✓
– Destination x Year		✓	✓	✓
Estimator	PPML	OLS	OLS	OLS
Observations	102,636	491,277	491,277	491,277

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column 1 is the volume of exports by capital intensity. In column 2 it is the market share origin i captures in j . In column 3 it is the number of exported goods. In column 4 it is the number of new exported goods. The independent variables for all four dependent variables are a post plague dummy, its interaction with the level of exports before 1709 (or number of exported goods), and its interaction with a siege dummy. In column 1, the independent variables also includes a post-plague x capint dummy, its interaction with pre-plague exports, and its interaction with a siege dummy. Table 13 shows besieged cities.

Figure 29: Plagued regions also expanded their imports



Notes: Results arguing against mechanical explanations. Standard errors clustered at the level of destinations (as these are import results).

Table 31: Export expansion and shift into capital-intensive goods in post-plague Hamburg

	Absolute		Relative	
	(1)	(2)	(3)	(4)
Post Plague	0.545*** (0.107)	0.439 (0.579)		
Post Plague x Capital-Int.			0.569** (0.230)	0.554 (0.857)
<i>Fixed Effects:</i>				
– Year			✓	✓
– Sector	✓	✓	✓	✓
Estimator	OLS	PPML	OLS	PPML
Observations	243	304	243	304

Notes: Difference-in-differences set up using Hamburg export data to all destinations between 1702 and 1713, with the plague in 1711, based on Baasch (1929). 38 goods have been classified into sectors precisely as in the rest of the paper. Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is annual exports by sector in levels (columns 2 and 4) and logs (columns 1 and 3). The independent variable is a post plague dummy, equal to one after 1711 when Hamburg suffered a plague outbreak, and a dummy interacting the post plague dummy with a dummy for capital-intensive exports. Columns 2 and 4 show unweighted results, whereas in columns 1 and 3 the weight are exports in levels.

C.5 Continuous Plague Treatment

As an alternative to a plague dummy, I construct a continuous plague treatment variable, incorporating plague outbreaks across all of Europe and drawing on additional sources for 106 recorded plague outbreaks (Table 32 and Figure 30). A NUTS 3 region counts as plagued if there is at least one plagued town or region in it. The continuous plague treatment is constructed as

$$plague_{it}^{cont} = plague_{it}^{own} + \sum_j \frac{Population_j}{Distance_{ij}} plague_{jt},$$

where i are ports, j are regions in Europe at the NUTS 3 level, $plague_{it}^{own}$ denotes a region's own plague status at time t, and $plague_{jt}$ is a NUTS 3 region's plague status at time t. In the absence of population data at this resolution, I assume a constant population density and proxy with the size of a NUTS 3 region. Distances are computed as straight lines.⁶⁰ As a robustness check, I also construct a treatment that weighs all regions the same by dropping the population (or area) variable:

$$plague_{it}^{cont} = plague_{it}^{own} + \sum_j \frac{1}{Distance_{ij}} plague_{jt}.$$

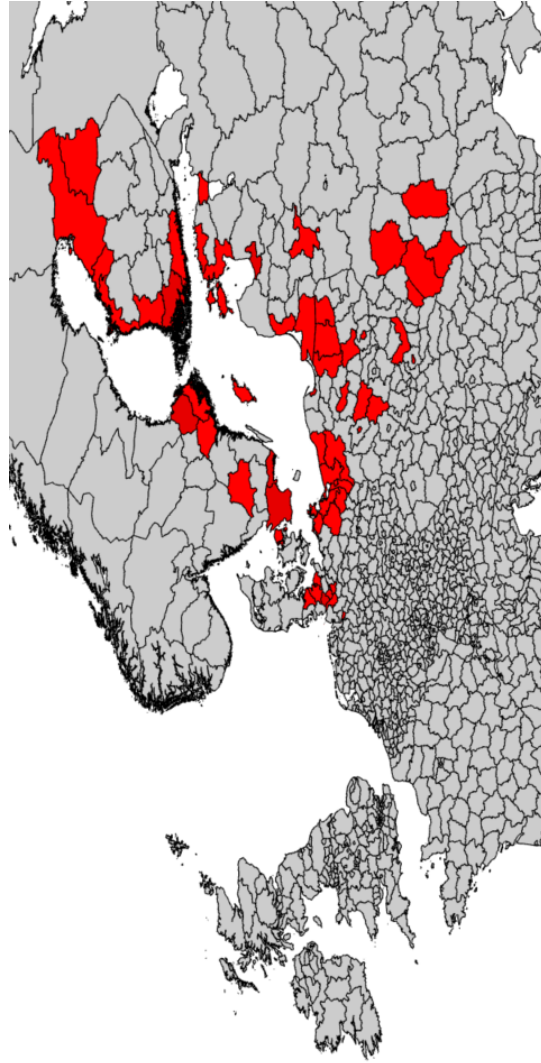
I show in Table 33 that all results hold when regressing on a continuous plague treatment variable. The positive coefficients for the shift into capital-intensive exports after an own plague outbreak is no longer significant, however.

⁶⁰In instances where a region, not a town, is mentioned in the source, I use the coordinates of the historical capital to compute distances.

Table 32: List of plagued cities and regions across all of Europe

Town or Region	Modern Country	Time	Historical Capital	Source
Płiećzów	Poland	1702		Frandsen, 2009
Ruthenia	Ukraine	1703–1706	Lwow	Frandsen, 2009
Podolia	Ukraine	1703–1706	Kamianets-Podilskyi	Frandsen, 2009
Volhynia	Ukraine	1703–1706	Lutsk	Frandsen, 2009
Lviv (Lemberg)	Ukraine	1704–1705		Frandsen, 2009
Kolomyja	Ukraine	1705–1706		Jaroslav Burchardt, Meissner, and Burchardt, 2009
Stanisławów	Ukraine	1705–1706		Jaroslav Burchardt, Meissner, and Burchardt, 2009
Stryj	Ukraine	1705–1706		Jaroslav Burchardt, Meissner, and Burchardt, 2009
Sambor	Ukraine	1705–1706		Jaroslav Burchardt, Meissner, and Burchardt, 2009
Przemysł	Poland	1705–1706		Jaroslav Burchardt, Meissner, and Burchardt, 2009
Jarosław	Poland	1705–1706		Jaroslav Burchardt, Meissner, and Burchardt, 2009
Kraków	Poland	1707		Frandsen, 2009
Lesser Poland	Poland	1707–1710	Krakow	Frandsen, 2009
Mazovia	Poland	1707–1710	Warsaw	Frandsen, 2009
Warsaw	Poland	1707–1710		Frandsen, 2009
Great Poland	Poland	1707–1709	Gniezno	Frandsen, 2009
Ostrów	Poland	1707–1709		Frandsen, 2009
Kalisz	Poland	1707–1709		Frandsen, 2009
Poznań	Poland	1707–1709		Frandsen, 2009
Toruń	Poland	1708		Frandsen, 2009
Piekielko	Poland	1708		Frandsen, 2009
Bialutten	Poland	1708		Frandsen, 2009
Hohenstein	Poland	1708		Salm, 1905
Masuria	Poland	1708–1710	Elk	Kossert, 2005
Königsberg	Russia	1709–1710		Frandsen, 2009
Danzig (Gdańsk)	Poland	1709		Kroll and Grabinsky, 2007
Elbing (Elbląg)	Poland	1709–1710		Frandsen, 2009
Insternburg	Prussia	1708–1711		Kossert, 2005
Menel	Prussia	1708–1711		Kossert, 2005
Ragnit	Prussia	1708–1711		Kossert, 2005
Tilsit	Prussia	1708–1711		Kossert, 2005
Pillupönen (Nevskoye)	Russia	1709		Frandsen, 2009
Stettin (Szczecin)	Poland	1709–1711		Thiede, 1849
Damm (Dabie)	Poland	1709		Wieden, 1999
Pasewalk	Germany	1709–1710		Wieden, 1999
Anklam	Germany	1709–1710		Wieden, 1999
Kammin (Kamień Pomorski)	Poland	1709–1710		Wieden, 1999
Belgard (Białogard)	Poland	1709–1710		Wieden, 1999
Stralsund	Germany	1710–1711		Wieden, 1999
Altentreptow	Germany	1710–1711		Wieden, 1999
Wolgast	Germany	1710–1711		Wieden, 1999
Wollin (Wolin)	Poland	1710–1711		Wieden, 1999
Stargard	Poland	1710–1711		Wieden, 1999
Bahn (Banie)	Poland	1710		Wieden, 1999
Neumark	Poland	1710	Soldin	Schwartz, 1901
Uckermark	Poland	1710	Prenzlau	Wieden, 1999
Prenzlau	Germany	1710		Wieden, 1999
Greifswald	Germany	1711		Wieden, 1999
Lithuania	Lithuania	1709–1713	Vilnius	Frandsen, 2009
Vilnius	Lithuania	1709–1713		Frandsen, 2009
Livonia	Latvia	1709–1711	Riga	Kroll, 2006
Rīga	Latvia	1710–1711		Kroll, 2006
Dünaburg (Daugavgrīva)	Latvia	1710		Frandsen, 2009
Estonia	Estonia	1709–1711	Reval	Frandsen, 2009
Reval (Tallinn)	Estonia	1710		Frandsen, 2009
Narva	Estonia	1710–1711		Kroll and Grabinsky, 2007
Pernau (Pärnu)	Estonia	1710		Kroll, 2006
Arensburg (Kuressaare)	Estonia	1710		Frandsen, 2009
Gotland	Sweden	1710–1712	Visby	Bohn, 1989
Visby	Sweden	1710–1712		Bohn, 1989
Stockholm	Sweden	1710–1711		Kroll, 2006
Uppland	Sweden	1710	Stockholm	Persson, 2011
Uppsala	Sweden	1710		Persson, 2011
Södermanland	Sweden	1710	Stockholm	Persson, 2011
Enköping	Sweden	1710		Persson, 2011
Jönköping	Sweden	1710–1711		Kroll and Grabinsky, 2007
Värmdö	Sweden	1711		Frandsen, 2009
Tillinge	Sweden	1711		Frandsen, 2009
Danmark parish	Sweden	1711		Frandsen, 2009
Helsingfors (Helsinki)	Finland	1710		Vourinen, 2007
Borgå (Porvoo)	Finland	1710		Vourinen, 2007
Ekenäs (Tammisaari)	Finland	1710		Vourinen, 2007
Åbo (Turku)	Finland	1710–1711		Vourinen, 2007
Nystad (Uusikaupunki)	Finland	1710–1711		Vourinen, 2007
Raumo (Rauma)	Finland	1710–1711		Vourinen, 2007
Björneborg (Pori)	Finland	1710–1711		Vourinen, 2007
Närendal (Naantali)	Finland	1710–1711		Vourinen, 2007
Jakobstad (Pietarsaari)	Finland	1710–1711		Vourinen, 2007
Gamlakarleby (Kokkola)	Finland	1710–1711		Vourinen, 2007
Uleåborg (Oulu)	Finland	1710–1711		Vourinen, 2007
Kajana (Kajaani)	Finland	1710–1711		Engström, 1994
Helsingør	Denmark	1710–1711		Frandsen, 2009
Copenhagen	Denmark	1711		Frandsen, 2009
Västana	Sweden	1711		Frandsen, 2009
Karlskrona	Sweden	1710		Persson, 2011
Karlshamn	Sweden	1710		Persson, 2011
Blekinge	Sweden	1710	Karlskrona	Persson, 2011
Domsten	Sweden	1711		Frandsen, 2009
Ystad	Sweden	1712		Frandsen, 2009
Malmö	Sweden	1712		Frandsen, 2009
Hamburg	Germany	1712–1714		Kroll, 2006
Altona	Germany	1713		Frandsen, 2009
Bremen	Germany	1712–1713		Frandsen, 2009
Stade	Germany	1712		Frandsen, 2009
Itzehoe	Germany	1712		Frandsen, 2009
Kropp	Germany	1712		Frandsen, 2009
Glickstadt	Germany	1712		Frandsen, 2009
Rendsburg	Germany	1712		Ulbricht, 2004
Laboe	Germany	1712		Ulbricht, 2004
Holstein	Germany	1712	Glickstadt	Frandsen, 2009
Schleswig	Germany	1712		Ulbricht, 2004
Flensburg	Germany	1712		Ulbricht, 2004
Friedrichsort	Germany	1712		Ulbricht, 2004
Pinneberg	Germany	1712		Winkle, 1983
Rellingen	Germany	1712		Winkle, 1983
Gröpeligen	Germany	1712		Frandsen, 2009

Figure 30: Map of Plagued NUTS 3 Regions



Notes: A map of plagued hinterlands based on Table 32. Red colour indicates 1 or more plagued locations within a NUTS 3 region.

Table 33: Impact of Plague on Trade, Accounting for Outbreaks in the Hinterland

	Plague (continuous treatment)				Plague (only accounting for distance)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Own Plague	70.19260 (48.95880)	0.01037*** (0.00285)	0.19819*** (0.05970)	0.07497** (0.03609)	68.78410 (53.02448)	0.01087*** (0.00295)	0.20180*** (0.06222)	0.07387** (0.03699)
Plague in Hinterland	-0.02843 (0.02638)	-0.00000 (0.00000)	-0.00002 (0.00010)	-0.00006 (0.00004)				
Own Plague x Capital-Int.	1.3e+02 (1.2e+02)				1.6e+02 (1.4e+02)			
Plague in Hinterland x Capital-Int.	0.01858* (0.01089)							
Plague in Hinterland, Only Distance								
Plague in Hinterland, Only Distance x Capital-Int.					-5.6e+01* (31.34625)	-0.00387** (0.00196)	-0.03760 (0.09991)	-0.03447 (0.02659)
<i>Fixed Effects:</i>								
– Origin x Sector	✓				✓			
– Year	✓				✓			
– Area x Year		✓	✓	✓		✓	✓	✓
– Origin x Destination		✓	✓	✓		✓	✓	✓
– Destination x Year		✓	✓	✓		✓	✓	✓
Estimator	246,095	489,617	489,617	489,617	246,095	489,617	489,617	489,617

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is sectoral exports in columns 1 and 5, trade between i and j in columns 2 and 6, a dummy for positive trade between i and j in columns 3 and 7, and the volume of new export between i and j in columns 4 and 8. The independent variables are an own plague dummy, a continuous plague variable accounting for the size and proximity of other Europe-wide plagued regions, a second such dummy accounting only for proximity of other plagued regions, and their interactions with a capital intensity dummy (only in columns 1 and 5). Annual growing season temperatures are an additional control in all specifications.

D Model Appendix

D.1 Recovering Sectoral Productivities

Costinot, Donaldson, and Komunjer, 2011 show that the following relationship between trade flows, sectoral productivities, and trade costs is true in their model:

$$\frac{X_{ijk}X_{i'jk'}}{X_{ijk'}X_{i'jk}} = \left(\frac{A_{ik}A_{i'k'}}{A_{ik'}A_{i'k}} \right)^\theta \left(\frac{d_{ijk}d_{i'jk'}}{d_{ijk'}d_{i'jk}} \right)^{-\theta}. \quad (22)$$

This equation takes ratios with a reference sector and region. Time variation will be introduced into gravity equation 10 as follows:

$$X_{ijkt} = \frac{\chi_k A_{ikt} (w_{ikt})^{-\gamma_k \theta} d_{ijkt}^{-\theta}}{CMA_{jkt}} \alpha_k Y_{jt}. \quad (23)$$

The following result can be established from equation 23:

$$\frac{X_{ijkt}X_{ijk't'}X_{i'jkt'}X_{i'jk't}}{X_{i'jkt}X_{i'jk't'}X_{ijk't}X_{ijk't}} = \frac{A_{ikt}A_{i'kt'}A_{ik't'}A_{i'k't}}{A_{ikt'}A_{i'kt}A_{ik't}A_{i'k't'}} \left(\frac{w_{ikt}w_{i'kt'}}{w_{i'kt}w_{ikt'}} \right)^{-\gamma_k \theta} \left(\frac{w_{ik't'}w_{i'k't}}{w_{i'k't'}w_{ik't}} \right)^{-\gamma_{k'} \theta}. \quad (24)$$

I assume that $d_{ijkt} = d_{ijk't'} \forall (ij, kt)$ or that $\frac{d_{ijkt}}{d_{i'jkt}} = \frac{d_{ijk't'}}{d_{i'jk't'}}$, such that trade costs cancel out. Therefore, an orthogonality assumption for trade costs is not required. As I am studying a relatively short period of time before the introduction of the steamship and it seems reasonable to further assume that trade costs were not changed by the plague, I consider this a reasonable assumption.

Three ratios are formed in equation 24. First, a ratio with respect to a reference sector k' to separate productivities from wages. Second, a ratio with respect to a reference origin i' , which leads to the demand side cancelling out. Third, a ratio with respect to a reference period t' , which means growth relative to a reference period is identified. This three way differencing is precisely what a three-way fixed model delivers (Olden and Møen, 2022). Accordingly, I regress trade flows as follows:

$$X_{ijkt} = \exp(\alpha + \delta_{ijt} + \delta_{jkt} + \delta_{ikt}) \times \epsilon_{ijkt}, \quad (25)$$

where δ_{ijt} is an importer-exporter-time fixed effect, δ_{ikt} an exporter-sector-time fixed effect and δ_{jkt} an importer-sector-time fixed effect. ϵ_{ijkt} is the error term. Note, first of all, that this specification nests all interaction terms of fixed effects and the model is therefore saturated (Angrist and Pischke, 2009). Second, note that the three-way fixed effect δ_{ikt} recovers differences-in-differences-in-differences (Olden and Møen, 2022). This is to say, δ_{ikt} recovers a time difference of what δ_{ik} recovered in Costinot, Donaldson, and Komunjer, 2011. Computing equation 24 confirms this finding, such that indeed δ_{ikt} captures:

$$\frac{\delta_{ikt}\delta_{i'kt'}\delta_{ik't'}\delta_{i'k't}}{\delta_{ikt'}\delta_{i'kt}\delta_{ik't}\delta_{i'k't'}} = \frac{A_{ikt}A_{i'kt'}A_{ik't'}A_{i'k't}}{A_{ikt'}A_{i'kt}A_{ik't}A_{i'k't'}} \left(\frac{w_{ikt}w_{i'kt'}}{w_{i'kt}w_{ikt'}} \right)^{-\gamma_k \theta} \left(\frac{w_{ik't'}w_{i'k't}}{w_{i'k't'}w_{ik't}} \right)^{-\gamma_{k'} \theta}. \quad (26)$$

To arrive at a productivity ratio, two assumptions are required:

1. All sectors have the same labour share, $\gamma_k = \gamma \forall k$.
2. The labour mobility friction does not vary within areas, such that $\phi_{ikt} = \phi_{jkt} \forall (k, t)$ and i, j within the same area.

This allows me to separate wages from productivities. Under serfdom, wages may vary between manufacturing and agriculture. I assume:

$$w_{iMt} = \begin{cases} w_{iAt}, & \text{without serfdom,} \\ (1 + \phi_{it})w_{iAt}, & \text{with serfdom.} \end{cases} \quad (27)$$

Unclassified goods will be the reference sector k' , which I assume to produce in the hinterland, paying wage w_{iAt} . Accordingly, for $k=LA, CA$, I can set $w_{ikt} = w_{ik't} \forall (i, t)$, such that the wage ratio drops out. Thus, δ_{ikt} identifies sectoral productivity growth for agricultural sectors. For manufacturing, $k=LM, CM$, I plug in wage equalisation assumption 27 to obtain:

$$\delta_{ikt} = \frac{A_{ikt}A_{i'kt'}A_{ik't'}A_{i'k't}}{A_{ik't'}A_{i'kt}A_{ik't}A_{i'k't'}} \left(\frac{(1 + \phi_{it})(1 + \phi_{i't'})}{(1 + \phi_{i't})(1 + \phi_{it'})} \right)^{-\gamma\theta}. \quad (28)$$

The ratio of ϕ parameters needs to be equal to 1 to separate productivities from labour mobility frictions. I assume that time variation in labour mobility and serfdom is constant within areas, such that area-sector-time fixed effects in equation 11 absorb the ratio of ϕ parameters though serfdom may vary over time.⁶¹ A special case is presented by Denmark (including large parts of modern-day Schleswig Holstein), which is the only country in this area that re-introduced serfdom in 1733 (Gary et al., 2022). However, as all areas in Denmark were affected by this change, this is inconsequential.

Section 2 outlined how the end of the Little Ice Age saw temperatures rise across Northern Europe. Regions will therefore have seen agricultural productivity growth (Waldinger, 2022). To account for exogenous temperature change, I model agricultural productivity growth as a function of temperature change. In plagued regions, I additionally allow for plague-induced productivity changes. I assume that for a non-plagued region i' :

$$\frac{A_{i',At}}{A_{i',At'}} = \left(\frac{temp_{i',t}}{temp_{i',t'}} \right)^{\epsilon_A}. \quad (29)$$

I assume a value of $\epsilon_A = 1$ and correct the recovered agricultural productivity ratios for both labour- and capital-intensive agriculture for this term. This value is reasonable given that most regions are in Northern Europe (see Liu, Mishra, and Ray, 2020) and represents an average over estimates for different crops.

A possible concern could be plague-induced reductions in exporter-specific trade costs that are picked up in the δ_{ikt} . However, in Table 29 I show that travel times (the sum

⁶¹There are two alternative conditions that do not require area-time fixed effects. First, if $\phi_{it} = \phi_{i't}$. At all points in time, the labour mobility friction would have to be the same across regions, which I refuse to assume. A second condition is: if $\phi_{it} = \phi_{i't'}$, the ϕ ratio also equals 1. This implies the absence of time variation in the labour mobility friction. Despite evidence for increased serfdom after the plague in Northern Estonia (Raster, 2023, the limits to labour mobility may have remained unchanged, and the fundamental hurdle of moving to a German-speaking city while not having been allowed to learn a trade remained in place. In my approach, the plague is allowed to have worsened serfdom; as long as these changes are symmetric within areas, the ϕ ratio will be absorbed by fixed effects.

of days at sea and handling times at harbours) are unrelated to either the plague or the capital intensity of exports, thus dispelling this concern.

Finally, instead of estimating with OLS as in Costinot, Donaldson, and Komunjer, 2011, I use PPML on a balanced panel, which is required for the equivalence between their equations (17) and (18). There are a number of reasons for preferring PPML. First of all, the model generates a gravity equation for trade, so all arguments for PPML in Santos-Silva and Tenreyro, 2006 apply. The issue of zeros is much more prevalent in my data than in the data by Costinot, Donaldson, and Komunjer, 2011.

D.2 Simulations

Figure 31 shows simulation results comparing simulated productivity growth between the hypothetical full sample and the observed sample. While in the observed sample only trade passing the Sound is registered, such that trade on only one side of the Sound is unobserved, in the theoretical, full sample I simulate trade volumes on all routes.

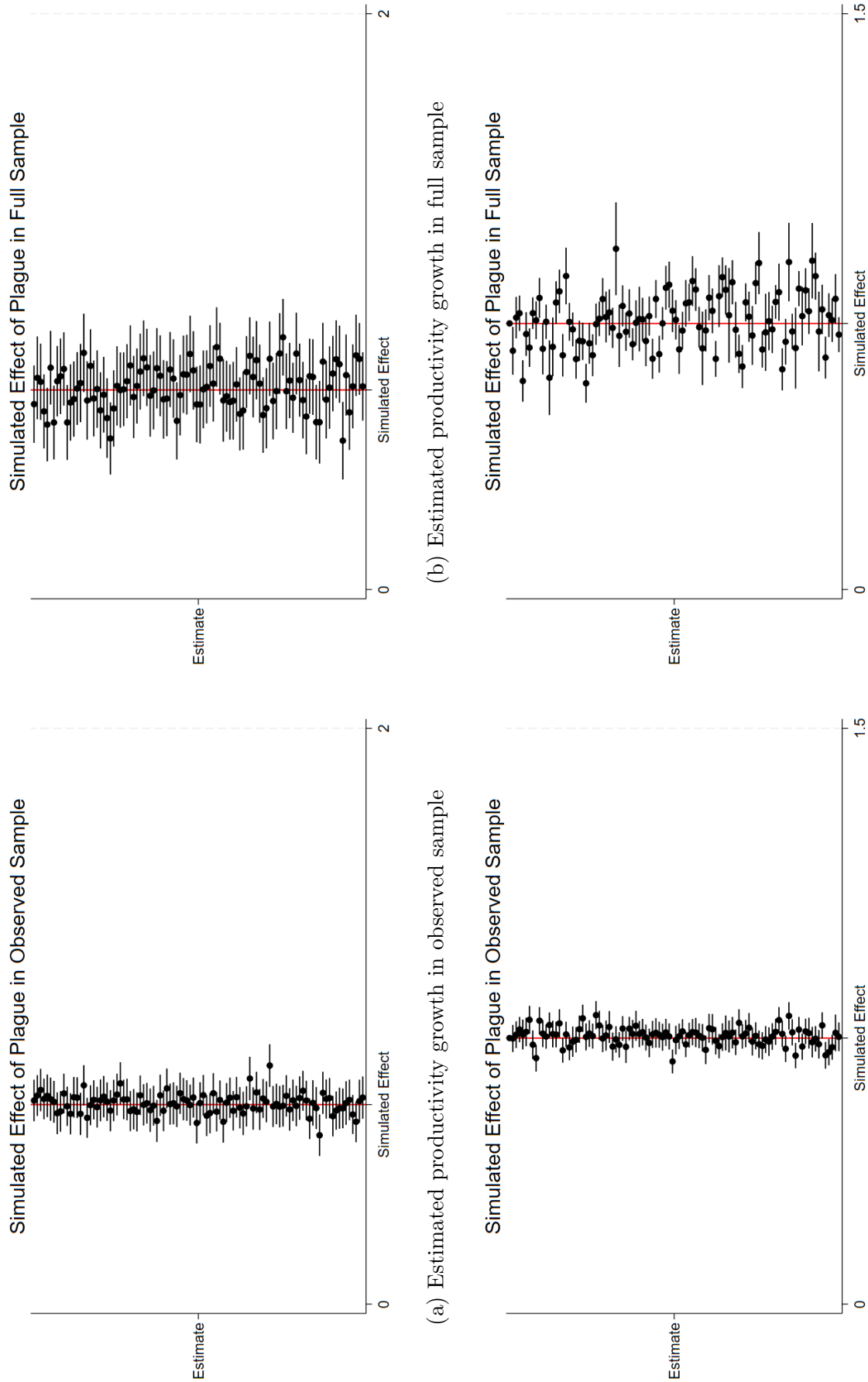
To simplify matters, I simulate two sectors: a labour-intensive and a capital-intensive one. I choose reference levels for wages, market access, χ_k , α_k , and Y_j and compute trade costs as $d_{ij}^{-\theta}$, where d_{ij} is distance over sea.⁶² I simulate two time periods, where in the first all regions have the same capital-intensive sector productivity levels. Plagued regions then experience simulated productivity growth in the capital-intensive sector and have higher levels in the second period, whereas non-plagued regions keep their capital-intensive productivity level. The simulated values imply that plagued regions' capital-intensive sector productivity growth is twice as high as that of non-plagued regions. Thus, the simulated effect is $\ln 2 = 0.693$. Below, I show results for a log-normal specification of noise, where I take the log of simulated trade values, add standard normally distributed noise, and then take the exponential of these noise-amended trade flows. Results with multiplicative and uniformly distributed noise display the same pattern. I then recover sectoral productivities as described in Section D.1. I regress a plague dummy on the recovered capital-intensive sector productivity growth and include region and time fixed effects.

The results are displayed in Figure 31. The simulated effect lies at 0.693, which is within 96 out of 100 confidence intervals for the regressions on the full sample. For the observed sample, 97 out of 100 confidence intervals contain the simulated effect. In all cases, point estimates are centred around the simulated effect.

I also simulate trade following equation 10 and find similar results when regressing simulated trade volumes in both samples on a plague dummy, including origin-destination and destination-time fixed effects. These results, presented in the bottom panels of Figure 31, show that a significant export increase is picked up in both samples. I run 100 simulations and show that the point estimates are very similar when comparing the full to the observed sample. In both samples, the estimates are centred around the simulated effect, suggesting no bias is introduced by restricting to the observed sample. I conclude that the sample restriction of only observing ships passing through the toll station is not introducing a bias in my productivity growth and trade estimates.

⁶²I compute cost distances for each city pair in the dataset using a raster approach and the CostDistance tool in ArcGIS similar to Bakker et al., 2021. My pixel resolution is 30 arc-seconds, corresponding to square cells of about 1 km side length (the longitudinal dimension is even less than that given the latitude of Northern Europe). I compute bilateral distances over sea between all cities. Including land transport is motivated by some large ports slightly inland, such as Thorn or Bordeaux.

Figure 31: Simulated productivity growth and trade increase in observed vs. full sample



E Mechanisms Appendix

Micro-Foundation of Equation 13

I propose a micro-foundation of equation 13 that rests on technology adoption choices being affected by changes in relative factor prices. I lean on Bustos, 2011, whose framework incorporates technology choice in a Melitz, 2003 model. Technology choice is between two technologies: under technology l, fixed costs are lower, but marginal costs are higher. Under technology h, firms need to incur higher fixed costs, yet benefit from lower marginal costs. My model differs from Bustos, 2011 in the following ways:

- First, Bustos, 2011 is a model of heterogeneous firms. I assume that firms produce varieties within sectors. The firm margin is not defined in my CRS model and thus I never observe firm outcomes. I will instead speak of producers.
- Second, Bustos, 2011 features only labour. I add capital as a factor of production. I make the choice between factor intensities discrete at the producer-level: producers choose between a labour- and a capital-intensive technology. This aggregates up to the sectoral capital-to-labour ratio, which remains continuous.
- Third, I turn the model into one of sectoral productivity variation. There is an exogenous component to producers' productivity, which is drawn from distribution $F_{ik}(z)$. Firms can choose to enhance their productivity by investing to operate under the capital-intensive technology, which increases their productivity multiplicatively by v_k . These productivity-enhancing effects are larger in capital-intensive sectors, $v_{k'} > v_k$ if k' is capital- and k labour-intensive.
- Fourth, I model fixed costs to be in units of capital and assume that the labour-intensive technology requires fixed cost if , whereas the capital-intensive technology requires fixed cost $i\rho f$, with $\rho > 1$. This fixed cost has to be paid every period.

I adjust the notation to my setting, where $B_{ik}(\omega)$ replaces ϕ as the exogenous productivity draw. B_{ik} will denote the scale parameter of the Fréchet distribution, which is called A_{ik} in the rest of the paper. In this microfoundation, A_{ik} will instead refer to the realised sectoral productivity incorporating both the exogenous and the endogenous components. This is to be consistent with A_{ik} denoting the measure of sectoral productivity that determines production. v_k replaces γ as the productivity advantage of the capital-intensive production method, and ρ replaces η as the fixed cost difference.

Firms choose between labour-intensive technology l and capital-intensive technology h. Thus, producers can increase their exogenous productivity draw through endogenous technology choice. Below, I drop the sector-origin subscript for legibility. The production functions are:

$$\begin{aligned} q_l(\omega) &= B(\omega) L_l^{\gamma_l} K_l^{1-\gamma_l} \\ q_h(\omega) &= v B(\omega) L_h^{\gamma_h} K_h^{1-\gamma_h}, \end{aligned}$$

where $\gamma_l > \gamma_h$ and $v > 1$. The marginal cost functions are:

$$\begin{aligned} MC_l &= \frac{w^{\gamma_l} r^{1-\gamma_l}}{B(\omega)} \\ MC_h &= \frac{w^{\gamma_h} r^{1-\gamma_h}}{v B(\omega)}. \end{aligned}$$

Producers adopt the capital-intensive technology if $(p - MC_h)q - r\rho f \geq (p - MC_l)q - rf$, where p is the price and r is the return rate of capital. This can be re-arranged to:

$$q \geq \frac{rf(\rho - 1)}{MC_l - MC_h}. \quad (30)$$

The marginal cost difference is given by:

$$MC_l - MC_h = \frac{vw^\gamma r^{1-\gamma} - w^{\gamma_h} r^{1-\gamma_h}}{vB(\omega)}. \quad (31)$$

Equation 31 is increasing in the factor price ratio $\frac{w}{r}$. The sign of the fraction equals the sign of the numerator, and as $\gamma_l > \gamma_h$ the numerator's first term grows faster in $\frac{w}{r}$ than the second term. Therefore, $\frac{\partial(MC_l - MC_h)}{\partial(\frac{w}{r})} > 0$. In sum, as wages rise relative to the price of capital, the relative marginal cost of the labour-intensive technology rises.

As a result, the cut-off productivity $B_{ik}^*(\omega)$ drops. To see this, note that quantity is increasing in productivity as per the production functions. Given equation 30, the cutoff quantity q^* drops as relative wages rise. As quantity increases with productivity, the productivity threshold $B_{ik}(\omega)^*$ also falls. More producers now adopt the capital-intensive technology. These are the most productive producers among those that have previously used labour-intensive technology. This aggregates up to sectoral productivity, now including both the exogenous and the endogenous productivity components, where I now re-introduce the origin-sector subscript:

$$A_{ik} = \int_{B_{ik}^{min}}^{B_{ik}^*(\omega)} B_{ik}(\omega) dF_{ik} + \int_{B_{ik}^*(\omega)}^{B_{ik}^{max}} v_k B_{ik}(\omega) dF_{ik}. \quad (32)$$

A_{ik} is increasing in $\frac{w}{r}$ as $B_{ik}^*(\omega)$ is decreasing in $\frac{w}{r}$: as relative wages rise, more producers adopt the capital-intensive technology, and are thus shifted into the second term of the sum, which increases the left-hand side. Similarly, the capital-to-labour ratio increases in $\frac{w}{i}$, which can be seen by aggregating producers' first order conditions:

$$\frac{K_{ik}}{L_{ik}} = \frac{\int_{\omega} K_{ik}(\omega)}{\int_{\omega} L_{ik}(\omega)}. \quad (33)$$

As both A_{ik} and $\frac{K_{ik}}{L_{ik}}$ are strictly increasing in $\frac{w}{r}$ and invertible, there exists a function:

$$A_{ik} = C \left(\frac{K_{ik}}{L_{ik}} \right)^{\beta_k}, \quad (34)$$

where constant C will be absorbed by fixed effects. Thus, equation 34 corresponds to equation 13. β_k captures several channels: the direct productivity-enhancing effect of technology adoption within producers, which varies by sector as v_k is larger in capital-intensive sectors; the fact that the most productive producers using the labour-intensive technology switch to the capital-intensive one; and the reallocation effect, as market shares are gained by more productive producers.

While I do not observe firm or producer outcomes, there is empirical evidence in support of this mechanism. Given fixed exporting costs in this framework, higher productivity from technology adoption leads more producers to profitably begin exporting. The extensive margin expansion documented in Figure 16b may reflect this producer

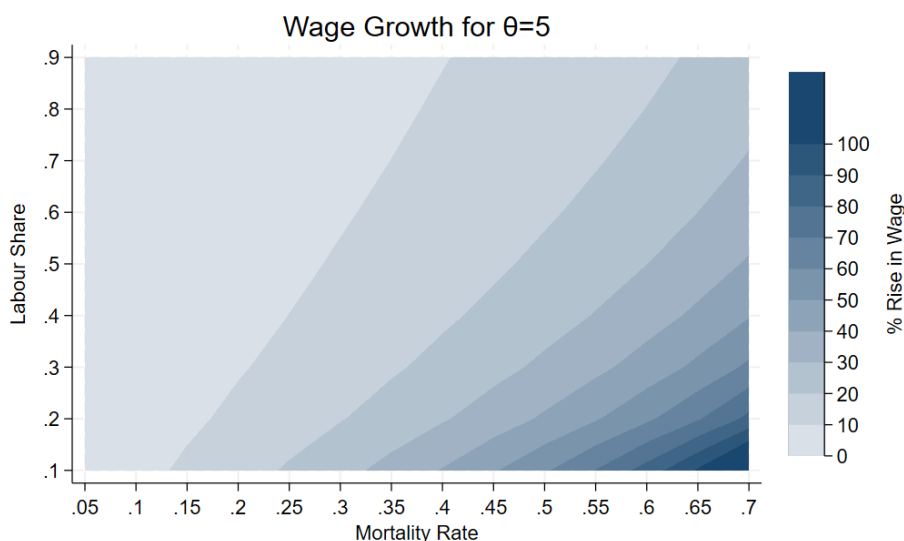
entry into exporting. This involves both new producers exporting goods previously exported by others and producers exporting novel goods. Figure 37 shows that exports of such novel export goods, never exported before by a region, expand after the plague.

The extensive margin expansion and addition of novel export goods are further stronger in capital-intensive sectors (Appendix Figures 25 and 27c). When interpreting this as more producers beginning to export, it implies that technology adoption leads in particular more producers in capital-intensive sectors to begin exporting. Given my assumptions on v_k being larger in those sectors, the productivity-enhancing effects of technology adoption and thus the shift into exporting are indeed larger in capital-intensive sectors. This is again consistent with the above mechanism.

Step #1: The Plague Induces Labour Scarcity

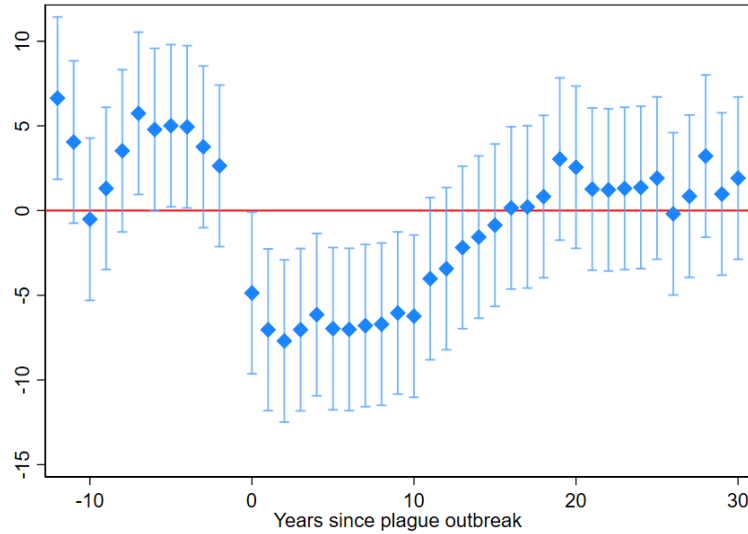
Figure 32 shows model-implied wage growth for the entire observed range of mortality rates and labour shares between 0.1 and 0.9. Figure 33 shows an event study for the number of captains living in a region before and after the plague. Table 34 shows that grain price-inferred interest rates in Danzig did not rise after the plague.

Figure 32: Model-implied wage growth by mortality rate and labour share



Notes: This figure shows model-implied wage growth for different mortality rates and labour shares.

Figure 33: Dynamics of population recovery after the plague: number of captains as proxy for population



Notes: Event study with area-year and region fixed effects on the number of unique captains' last names living in a region. Captains with the same first and last name from the same region in the same year are assumed to be duplicates and thus dropped.

Table 34: Grain price-inferred interest rates in Danzig do not increase

	Wheat		Rye		Barley	
	(1)	(2)	(3)	(4)	(5)	(6)
Post Plague	-0.042 (0.036)	0.006 (0.024)	-0.065** (0.030)	0.025 (0.025)	0.055 (0.053)	0.013 (0.018)
Sample	Full	r<0.1	Full	r<0.1	Full	r<0.1
Observations	197	160	259	199	172	116

Notes: Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variables are grain price-inferred monthly interest rates in Danzig between 1704 and 1793, using the methodology introduced by Keller, Shiue, and Wang, 2021. I assume that storage costs are constant. Grain price data by Allen and Unger, 2018. The independent variable is a post-plague dummy that takes the value 1 from 1709 onwards, when Danzig had been plagued. In some specifications, inferred interest rates above 10% have been dropped, as some underlying grain prices were distorted by outliers during war years.

E.1 Step #2: Production Becomes More Capital-Intensive

Table 35 lists examples of capital deepening by sector in this period and area. Table 36 shows that following a plague outbreak, ports significantly increased the number of ships they owned, which I interpret as a proxy for the capital stock. Figure 34 instead presents an event study on this proxy. Figure 35 shows that imports of capital goods increase

after the plague.⁶³ The only transiently increasing share of imports fits with the finding of stable relative imports in Section E.5 below. Figure 36 shows Scania's overall shift out of arable and into pastoral farming after the plague. Table 37 shows for individual farms that this shift was stronger the closer a farm was to plagued cities. Table 38 shows capital deepening in agriculture after the plague as more horses per capita are employed in plagued Swedish parishes. Table 39 shows that labour scarce farms with rich soils engage in stronger capital deepening. Figure 37 shows an event study for novel exports after the plague.

Table 36: Impact of plague on proxied capital stock

	# Ships							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Plague Dummy	166.218*** (45.161)	104.535*** (23.508)	164.057*** (45.329)	94.835*** (23.547)				
Mortality Rate					385.187*** (119.599)	248.952*** (64.118)	380.036*** (119.693)	226.573*** (63.366)
<i>Fixed Effects:</i>								
– Region		✓		✓		✓		✓
– Decade			✓	✓			✓	✓
Observations	41,400	41,400	41,400	41,400	41,253	41,253	41,253	41,253

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable is the number of ships registered in a city. It is assumed that ships do not sit idle and therefore that every owned ship appears once a decade. Further I assume that ships from the same city with the same captain in the same year are duplicates and drop these. The independent variable is first a plague dummy that equals 1 for plagued cities after the plague hit. The second independent variable is the mortality rate, which for half of regions is imputed as the predicted value from regression results presented in Appendix Table 14.

⁶³While this opens a door for embodied technology driving TFP growth through imports, the following findings of capital deepening and stronger TFP growth in sectors with more capital deepening support my proposed mechanism. Under technology spillovers, TFP should be expected to rise without requiring capital deepening.

Table 35: Examples of capital deepening by sector

Panel A: Labour-Intensive Agriculture

Example	Description
Iron ploughs	Replacing of wooden ploughs with iron-tipped or full-iron ploughs
Mechanical sowing	Replacing broadcast method, reducing seed waste
Horse-drawn harrows	Shift from manual to horse-drawn harrowing
Drainage infrastructure	Ditching, dyking, and polder construction
Grain storage	Fixed structures reducing post-harvest losses
Scythe	Replacing sickle, raising area harvested per worker
Wind-/watermills	Replacing hand-grinding

Panel B: Capital-Intensive Agriculture

Example	Description
Enclosure and fencing	Replacing herding labour with fixed capital
Livestock housing	Purpose-built winter housing extending productive season
Selective breeding	Improved cattle and sheep breeds for dairy/wool yield
Dairy processing	Butter churns and cheese presses raising output per worker
Hay/fodder storage	Hay barns and winter fodder infrastructure raising carrying capacity
Fish processing	Smokehouses, salt stores, drying racks for herring, cod, stockfish

Panel C: Labour-Intensive Manufacturing

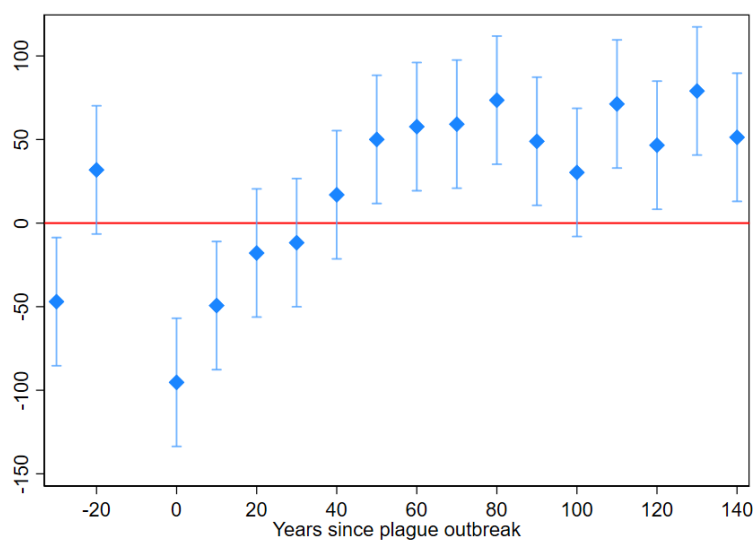
Example	Description
Dutch ribbon loom	Multi-ribbon loom weaving 12-24 ribbons simultaneously
Stocking frame	Replacing hand-knitting with frame
Fulling mills	Water-powered fulling replacing manual trampling of woollen cloth
Finishing equipment	Mangles, tentering frames, presses for cloth finishing
Bleachfields	Dedicated infrastructure for linen/cloth bleaching
Warping mills	Replacing labour with mechanical preparation of warp threads
Dyeing vats	Dyeing infrastructure with controlled heating

Panel D: Capital-Intensive Manufacturing

Example	Description
Coke-fired blast furnaces	Substitution of coke for charcoal in iron smelting (1709)
Larger blast furnaces	Scaling of charcoal furnaces with water-powered bellows
Slitting & rolling mills	Replacing hand-hammering with mechanical processing of bar iron
Water-powered sawmills	Replacing pit-sawing with mechanised timber processing
Dry docks	Fixed infrastructure for ship construction and repair
Shipyards cranes	Treadwheel and horse-powered cranes for heavy lifting
Ropewalks	Purpose-built long structures for rope production
Brick kilns & glassworks	Improved kiln and furnace design

Notes: These are illustrative examples of capital deepening by sector that are plausible for the Northern European and Baltic context around 1700.

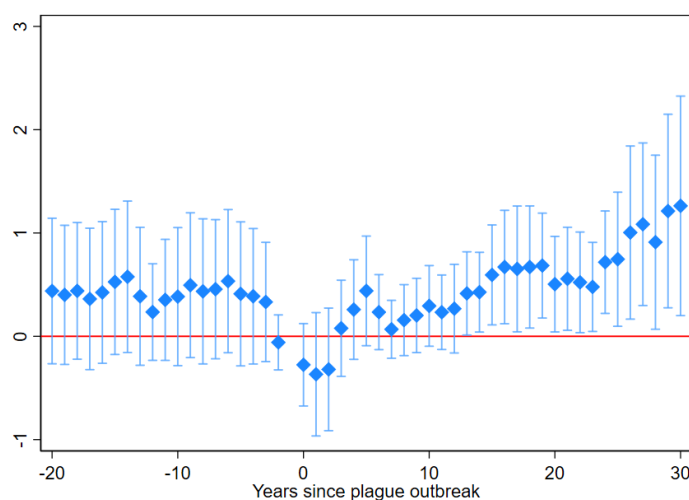
Figure 34: Dynamics of capital accumulation after the plague: number of ships as proxy for the capital stock



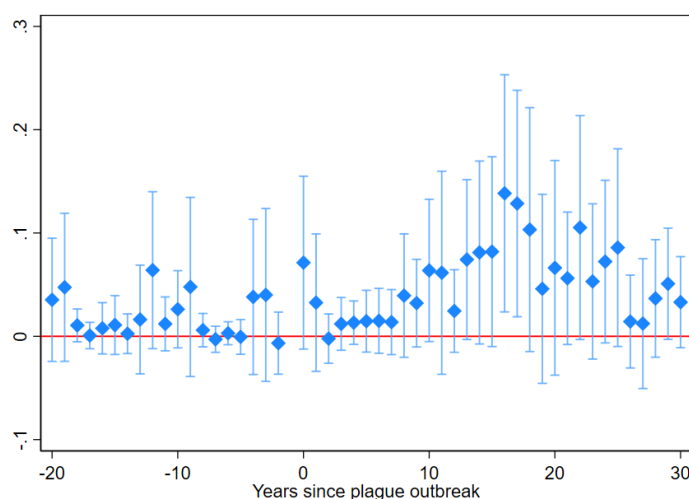
Notes: Event study with area-time, region, and decade fixed effects on the number of ships registered in a harbour in levels. Decade -1 is omitted as the reference decade.

Figure 35: Imports of capital goods increase after the plague, suggesting capital accumulation

(a) Imports of capital goods in levels

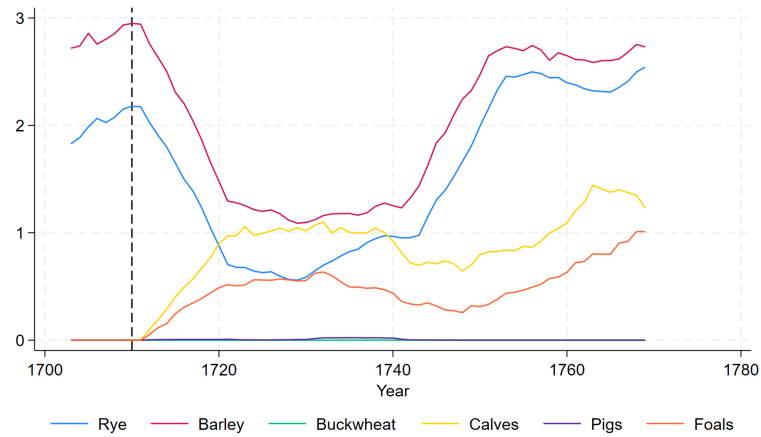


(b) Imports of capital goods as a share of imports



Notes: Capital goods are defined as metals (iron, iron works, steel, lead, copper, tin, sheet metal, brass, brass works), tools (tools, nails), and building materials (planks, tiles, stones, cement, chalk). In panel (a), I regress on imports in levels and include area-time, importer-exporter, and exporter-year fixed effects and control for growing season temperatures. In panel (b), I regress on ports' share of capital goods in imports, including importer and area-time fixed effects and controlling for growing season temperatures. In both panels, standard errors are clustered at the importer level.

Figure 36: Agricultural production in Scania becomes more capital-intensive



Notes: Figure 36 covers 119 Scanian farms and the composition of their production.

Table 37: Farm production and distance to plagued cities

	Calves	Pigs	Foals	Rye	Barley	Buckwheat
	(1)	(2)	(3)	(4)	(5)	(6)
Plague Distance x Post Plague	-0.00656** (0.00281)	-0.000395** (0.000156)	-0.00521*** (0.00171)	0.0182*** (0.00138)	0.0355*** (0.00310)	0.000228** (0.0000965)
<i>Fixed Effects:</i>						
– Year	✓	✓	✓	✓	✓	✓
– Farm	✓	✓	✓	✓	✓	✓
Observations	5,683	5,683	5,683	5,304	5,304	5,683

Notes: Standard errors clustered at the farm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are farm's production of calves, pigs, foals rye, barley, and buckwheat. The independent variable is the sum of distances multiplied by a plague dummy over the four closest cities in Scania province. Both Ystad and Malmö suffered a plague outbreak in 1712 with a mortality rate of 38% and 35%, respectively, whereas Landskrona and Helsingborg did not suffer plague outbreaks.

Table 38: Parish-level shift into higher capital intensity within agriculture

	Horses	Log Horses
	(1)	(2)
Plague & Year=1750	34.22** (13.71)	0.232*** (0.0527)
Plague & Year=1810	51.02*** (9.272)	0.334*** (0.0524)
Population	0.0858*** (0.0135)	
Log Population		0.157*** (0.0532)
<i>Fixed Effects:</i>		
– Parish	✓	✓
– Year	✓	✓
Observations	7,031	5,695

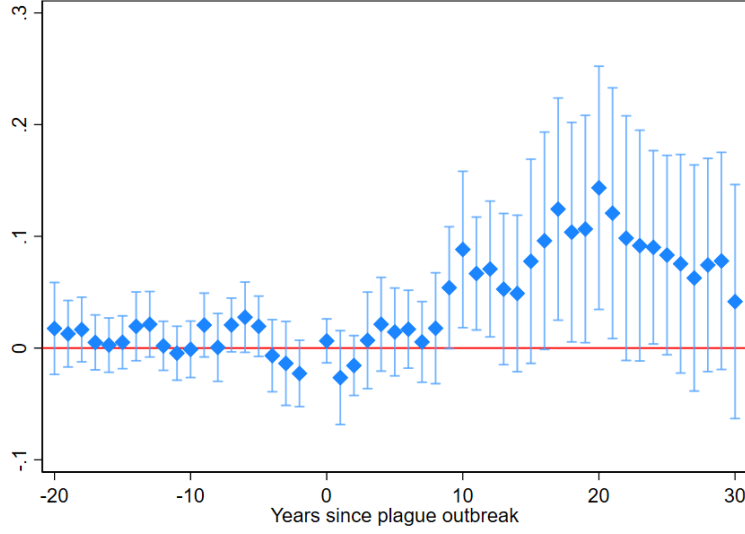
Notes: Standard errors clustered at the parish level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are parish-level logarithmic stocks of horses in 1690, 1750, and 1810. The independent variables are a post-plague dummy and parish-level population.

Table 39: Farm production, distance to plagued cities, and soil quality

	Calves	Pigs	Foals	Rye	Barley	Buckwheat
	(1)	(2)	(3)	(4)	(5)	(6)
Plague Distance x Post Plague	-0.00598 (0.00924)	0.0000460 (0.000181)	0.00425 (0.00575)	-0.0117 (0.00894)	0.0405*** (0.0105)	-0.000353 (0.000482)
Soil Quality x Post Plague	0.183 (0.260)	0.0113 (0.00798)	0.401** (0.157)	-1.156*** (0.258)	0.451 (0.270)	-0.00754 (0.0130)
Plague Distance x Soil Quality x Post Plague	-0.000356 (0.00262)	-0.000124 (0.0000845)	-0.00284* (0.00149)	0.00913*** (0.00269)	-0.00160 (0.00293)	0.000155 (0.000139)
<i>Fixed Effects:</i>						
– Year	✓	✓	✓	✓	✓	✓
– Farm	✓	✓	✓	✓	✓	✓
Observations	5,683	5,683	5,683	5,683	5,683	5,683

Notes: Standard errors clustered at the farm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are farm's production of calves, pigs, foals, rye, barley, and buckwheat. The specification interacts a post-plague indicator with (i) plague distance - the sum of distances to the four closest cities in Scania weighted by a plague dummy - (ii) average soil quality - a weighted index over ten soil-quality categories recorded in the land surveys, ranging from 1 (worst) to 10 (best) - and (iii) the product of plague distance and soil quality. Both Ystad and Malmö suffered a plague outbreak in 1712 with a mortality rate of 38% and 35%, respectively, whereas Landskrona and Helsingborg did not suffer plague outbreaks.

Figure 37: New exports at the extensive margin

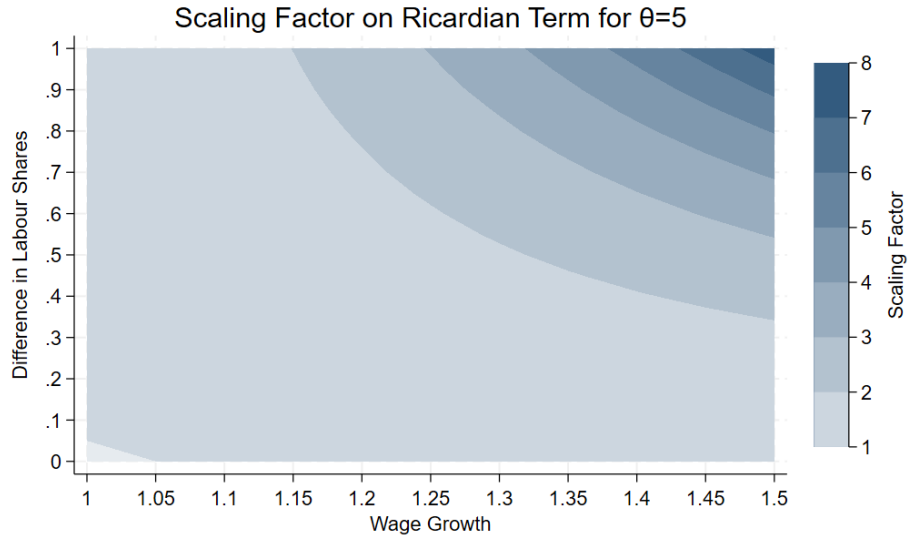


Notes: Estimation of equation 2 on the number of exported goods. A new export is defined as a good first exported by a city between 1689, 20 years before the onset of the plague in my study area, and 1732, 20 years after the last plague year in this region. Standard errors clustered at the origin level.

E.2 Step #3: Productivity Grows More in High β_k Sectors

Figure 38 shows the factor by which the Ricardian term is scaled up in case my assumption on equal labour shares is violated. If $\gamma_k \neq \gamma_{k'}$, δ_{ikt} recovers the product of relative productivity growth and relative wage growth. Specifically, relative productivity growth is multiplied by $\left(\frac{w_{ikt}w_{i'kt'}}{w_{i'kt}w_{ikt'}}\right)^{\theta(\gamma_{k'}-\gamma_k)}$. The second term drops out if $\gamma_k = \gamma_{k'}$; for $\gamma_k < \gamma_{k'}$, the wage ratio leads to an inflation in δ_{ikt} that is assigned to productivity. I put bounds on this potential bias. For model-implied wage growth of 6.6%, $\theta = 5$ and the highest possible $\gamma_k - \gamma_{k'}$ of 1, relative productivity growth is scaled up by 37.6%. Thus, when comparing a sector using only capital to a sector using only labour under the assumption of equal labour shares, the difference in point estimates by factor intensity is 0.57 in manufacturing (from 0.79 without the bias bounds correction) and 0.6 in agriculture (from 0.82). For a more reasonable difference in labour shares of 0.5, the scaling up amounts to 17.3%. Figure 38 shows the full range of this scaling factor: for most reasonable wage growth figures (note the model-implied one is 1.066), the scaling factor is below 2.

Figure 38: Scaling factor if labour shares assumption is violated

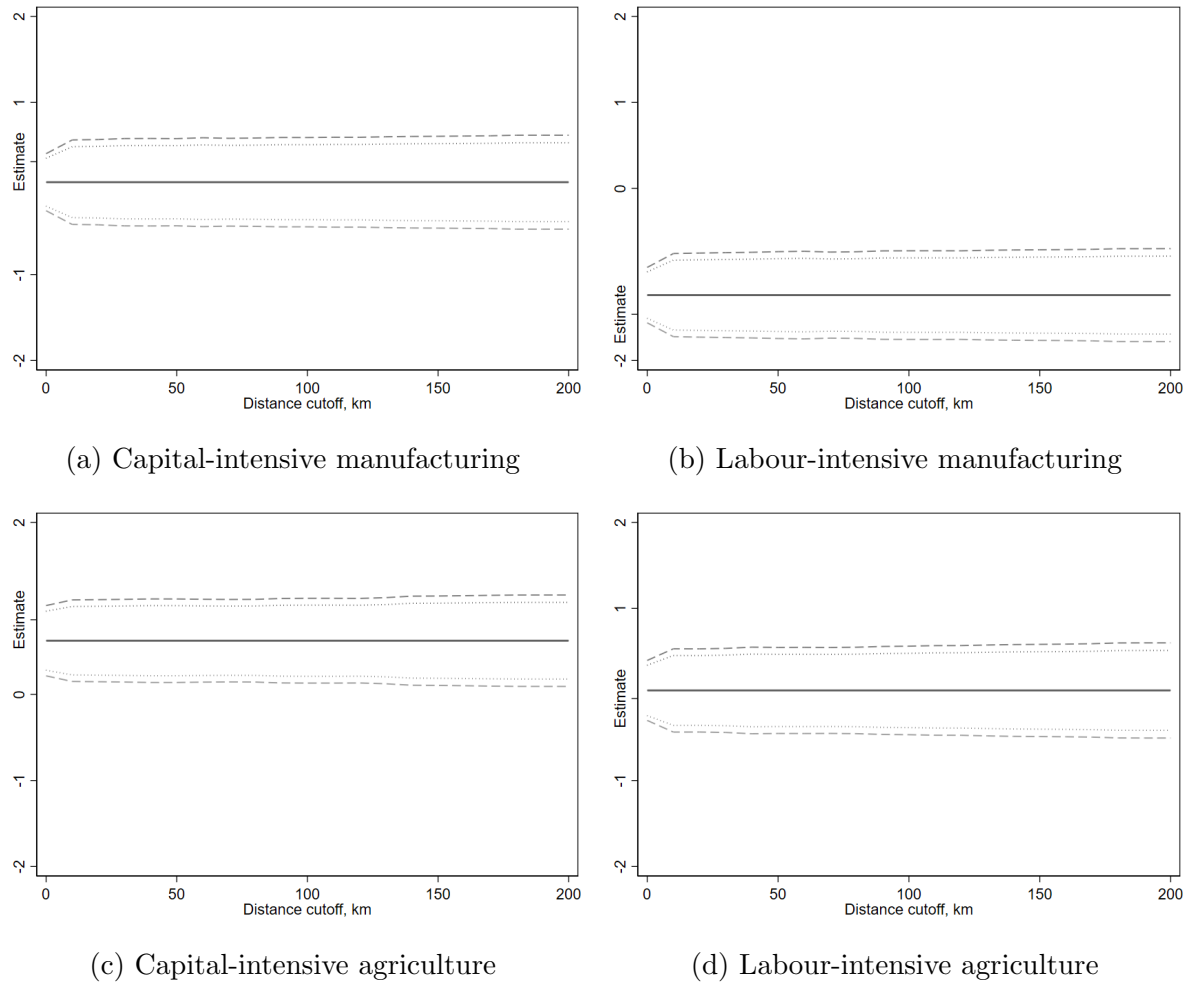


Notes: This figure shows the factor by which Ricardian differences in productivity growth are scaled up in case my assumption on equal labour shares is violated. Note that model-implied wage growth sits at 1.066.

Figure 39 repeats Table 3 using Conley standard errors (Conley, 1999) and a range of different distance cut-offs. Table 40 shows the long run response of sectoral productivities.

Figure 40 shows output per farm in Scania, adjusting for farm size, and documents only a small and brief dip in output after the plague. Figure 41 decomposes the wharves producing ships for the Swedish East India Company. Ships that were built in wharves founded after the plague in plagued regions account for the majority on all accounts.

Figure 39: The Plague's Productivity Effect by Sector, using Conley Standard Errors



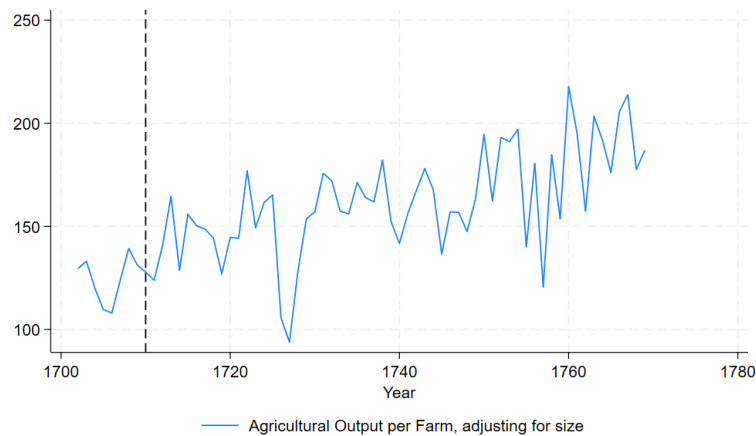
Notes: These Figures repeat columns 1, 3, 5, and 7 of Table 3 with Conley standard errors and distance cut-offs between 0 and 200 km in steps of 10 km. Distance cut-offs are denoted on the x axis and point estimates (solid line), 95% confidence intervals (dashed lines), and 90% confidence intervals (dotted lines) on the y axis. Implemented using Stata packages by Baum-Snow and Han, 2024, Thiemo Fetzter, and Hsiang, 2010.

Table 40: Impact of plague on sectoral productivity growth in the long run

	Labour-intensive ag.	Capital-intensive ag.	Labour-intensive man.	Capital-intensive man.
	(1)	(2)	(3)	(4)
0-9 Years Post Plague	-0.553** (0.246)	0.457 (0.292)	-0.732*** (0.229)	-0.226 (0.252)
10-19 Years Post Plague	0.165 (0.246)	0.360 (0.292)	-1.576*** (0.229)	0.178 (0.252)
20-29 Years Post Plague	-0.141 (0.246)	1.026*** (0.292)	-1.395*** (0.229)	0.298 (0.252)
30-39 Years Post Plague	-0.289 (0.246)	0.826*** (0.292)	-1.697*** (0.229)	1.131*** (0.252)
40-49 Years Post Plague	0.248 (0.246)	0.550* (0.292)	-1.328*** (0.229)	1.616*** (0.252)
50-59 Years Post Plague	0.647*** (0.246)	1.639*** (0.292)	-1.694*** (0.229)	1.773*** (0.252)
60-69 Years Post Plague	1.182*** (0.246)	1.558*** (0.292)	-1.125*** (0.229)	1.986*** (0.252)
70-79 Years Post Plague	1.287*** (0.246)	1.717*** (0.292)	-1.587*** (0.229)	0.629** (0.252)
80-89 Years Post Plague	1.706*** (0.246)	1.816*** (0.292)	-1.641*** (0.229)	0.308 (0.252)
90-99 Years Post Plague	1.074*** (0.246)	0.912*** (0.292)	0.110 (0.229)	1.295*** (0.252)
<i>Fixed Effects:</i>				
– Region	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓
Observations	170,403	170,430	170,430	170,430

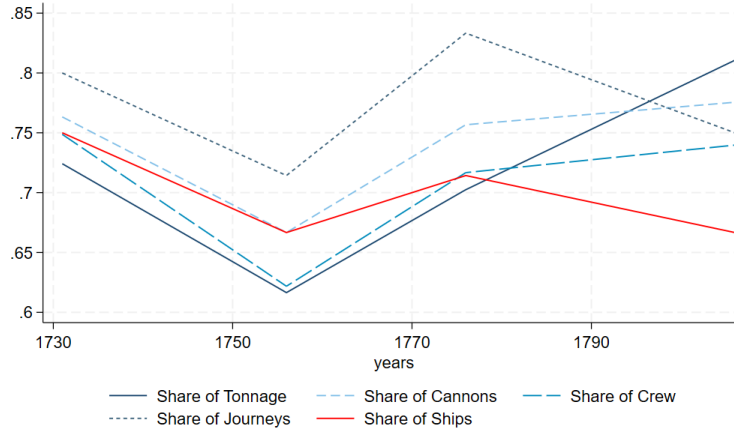
Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is log sectoral productivity growth. The independent variable is a plague dummy that equals 1 for plagued regions after the plague hit interacted with decade dummies after the plague. Denmark has been dropped from the sample as it reintroduced serfdom in 1733.

Figure 40: Total value of agricultural production per farm, controlling for farm size



Notes: Figure 40 includes all farms in Scania covered by the data and their total output.

Figure 41: Share of Swedish Wharf Output by Plague Status



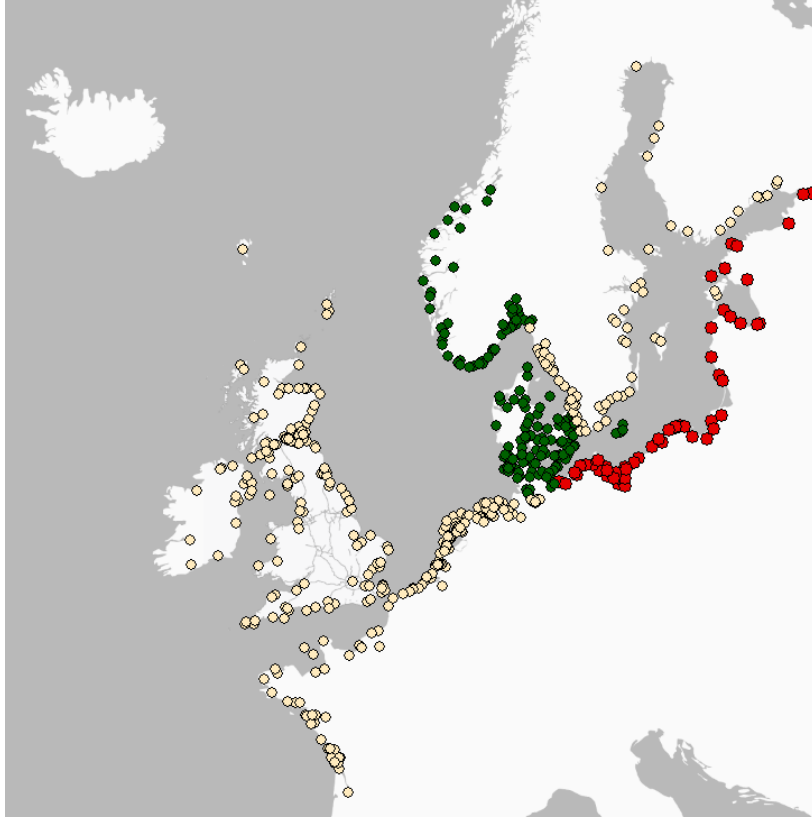
Notes: Based on data on the production location of all ships ($n=30$) operated by the Swedish East India Company. Wharves founded in plagued regions after 1714 are classified as located in plagued regions. The y axis denotes the share of tonnage, journeys, cannons, crew and ships for four points in time produced in wharves founded after the plague in plagued regions.

E.3 Evidence from Factor Frictions

E.3.1 Labour Frictions

There is disagreement between Raster, 2023 and Peters, 2022 about the presence of serfdom in Estonia, Livonia, and Pomerania. I follow Raster, 2023 in coding these areas as featuring serfdom for the following reasons. Raster, 2023 presents evidence for significant labour coercion in Estonia and Livonia. The German peasant-owning nobility successfully resisted Swedish attempts to abolish serfdom in 1681 (Tammisto, 2020, Seppel, 2019). Laws were passed in the early 1800s that replaced serfdom with villeinage, ascribing peasants to a parish rather than an individual landowner. This, however, still left peasants without the ability to move. By 1819, Peasant Laws had come into force, abolishing serfdom and permitting freedom of movement (Blūzma, 2019). There is evidence that even for several decades after this, mobility restrictions remained in place. (Merkel, 1800) Therefore, when judging serfdom from the perspective of mobility restrictions, Estonia and Livonia certainly were areas with serfdom. For Pomerania, Millward, 1982 notes that during the 17th century, all peasants were liable for unlimited service. In Swedish Pomerania, legislation permitted the sale of landless serfs as late as the 1780s, suggesting that serfdom was present in Pomerania, too. Figure 42 shows regions by their serfdom status.

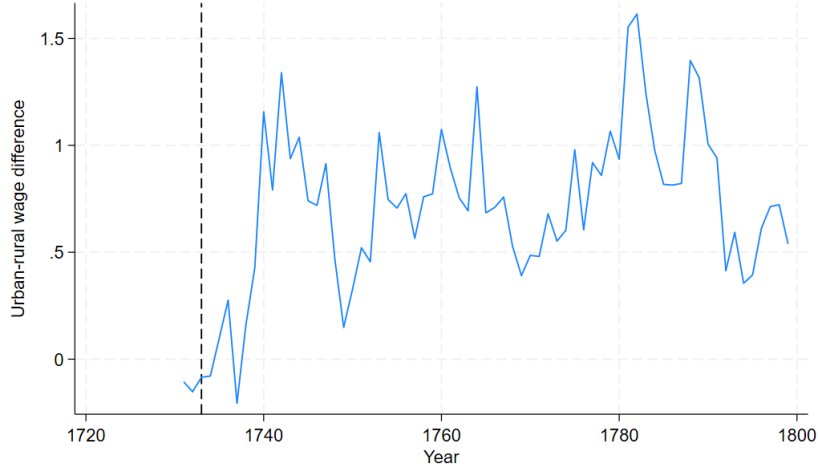
Figure 42: Regions by serfdom status



Notes: This map classifies regions by their serfdom status according to the above discussion. Yellow regions did not feature serfdom before and after the plague, whereas red regions did. Green areas re-introduced serfdom in 1733.

I use data by Gary et al., 2022 to show that there was no wage gap between unskilled urban and rural workers before the re-introduction of serfdom in Denmark in 1733, consistent with free movement between these areas. After serfdom was brought back, a wage gap appears, with scarcer urban unskilled labour compensated significantly higher. Figure 43 compares nominal day wages in Danish skilling for farmhands and unskilled workers in Copenhagen. Specifically, this figure displays the ratio $wage_{ratio} = \frac{wage_{urban} - wage_{farm}}{wage_{farm}}$. While few observations are available before 1733, the picture that emerges is one of a significant post-serfdom urban wage premium, averaging 0.7 farm day wages. With full labour mobility, peasants would have moved to Copenhagen to benefit from these higher wages. Therefore, restricted labour mobility translates into an urban-rural wage wedge. I conceptualise this by assuming a wedge of ϕ_i between urban and rural wages: $w_{iM} = (1 + \phi_i)w_{iA}$.

Figure 43: Urban-rural wage differential in Copenhagen



Notes: Difference between urban and farm day wages divided by farm day wages. Data by Gary et al., 2022. The dotted line signifies the 1733 re-introduction of serfdom.

Comparing two plagued regions, one with and one without serfdom, I expect the share of workers in agriculture to be higher in the region with serfdom and the share of workers in manufacturing to be higher in the region without serfdom. Plagued cities in regions with serfdom remain relatively scarcer in labour, whereas their rural counterparts become relatively abundant in labour. The strength of the shift into capital-intensive exports therefore depends on the presence of serfdom. I revisit the result from Step #3 above for both sectors:

$$\begin{aligned}\frac{X_{ijCM}}{X_{ijLM}} &= \xi w_{iM}^{-\theta(\gamma_{CM}-\gamma_{LM})} w_{iM}^{\beta_{CM}-\beta_{LM}} \frac{CMA_{jLM}}{CMA_{jCM}}, \\ \frac{X_{ijCA}}{X_{ijLA}} &= \xi w_{iA}^{-\theta(\gamma_{CA}-\gamma_{LA})} w_{iA}^{\beta_{CA}-\beta_{LA}} \frac{CMA_{jLA}}{CMA_{jCA}}.\end{aligned}$$

With urban wages increasing more with serfdom than without, the ratio $\frac{X_{ijCM}}{X_{ijLM}}$ should increase more in regions with serfdom. Agriculture becomes relatively abundant in labour, leading to lower agricultural wage increases in areas with serfdom than in those without. Accordingly, the ratio $\frac{X_{ijCA}}{X_{ijLA}}$ should increase less in regions with serfdom. I put this prediction to the test and estimate:

$$\begin{aligned}T_{ift} &= \delta_p plague_{it} + \delta_{ps} serfdom_i \times plague_{it} + \delta_{pf} factor_f \times plague_{it} \\ &\quad + \delta_{pfs} serfdom_i \times plague_{it} \times factor_f + \alpha_{if} + \alpha_{ft} + \gamma x_{it} + \epsilon_{ijkt},\end{aligned}$$

where T_{ift} are exports by factor intensity, α_{if} are origin-factor fixed effects and α_{ft} are factor-time fixed effects. ϵ_{ift} denotes the error term. δ are the coefficients of interest displayed in Table 41. δ_{pfs} is the most interesting coefficient as it measures how the effects of labour scarcity on capital-intensive exports vary by the presence of second serfdom.

The results in Table 41 confirm this expectation. The plague's effect on the capital intensity of exports is significantly higher in manufacturing under serfdom. This reflects *relative* labour scarcity.

Table 41: Impact of plague on shift into capital-intensive exports by sector & second serfdom

	Manufacturing		Agriculture	
	(1)	(2)	(3)	(4)
Plague	1.232* (0.707)	0.627 (0.398)	1.205* (0.681)	0.850 (0.529)
Plague x Serfdom	-2.498*** (0.419)	-0.930** (0.409)	-2.187*** (0.591)	-1.325** (0.526)
Plague x Capital-Int.	-1.855*** (0.546)	-0.809 (0.528)	0.653 (0.915)	0.868 (0.922)
Plague x Capital-Int. x Second Serfdom	2.544*** (0.539)	0.930* (0.525)	-0.478 (0.849)	-0.715 (0.911)
<i>Fixed Effects:</i>				
– Origin x Sector	✓	✓	✓	✓
– Sector x Year	✓	✓	✓	✓
Estimator	PPML	OLS	PPML	OLS
Observations	29,798	5,040	53,088	9,576

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable are sectoral exports in levels (columns 1 and 3) and logs (columns 2 and 4). In columns 2 and 4, values are weighted by exports in levels. The independent variables are a plague dummy, a second serfdom dummy, a capital-intensive sector dummy, and their interactions. I additionally control for growing season temperatures.

Both manufacturing sectors should see faster productivity growth under serfdom, as they are artificially labour-constrained and see larger increases in the capital-to-labour ratio. The opposite is true for agriculture, where the incentives for employing more capital are smaller under serfdom as labour is artificially abundant. Table 42 tests this prediction and finds no evidence supporting this conjecture. As the shift into higher capital intensity takes place, I speculate that the missing element are learning and scale effects, hampered by insufficient population mobility.

Table 42: Impact of plague on sectoral productivity growth, by second serfdom

	Labour-intensive ag.		Capital-intensive ag.	
	(1)	(2)	(3)	(4)
Plague	-0.428 (0.265)	0.948*** (0.311)	-1.169*** (0.244)	0.145 (0.252)
Plague & Serfdom	0.822** (0.340)	-0.560 (0.399)	-0.122 (0.314)	-0.125 (0.323)
<i>Fixed Effects:</i>				
– Region	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓
Observations	41,072	41,082	41,082	41,082

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is log sectoral productivity growth. The independent variable is a plague dummy that equals 1 for plagued regions after the plague hit interacted with a dummy for second serfdom.

Historical evidence confirms that agricultural productivities evolved differently under serfdom. In particular, the recovery of Russia and the Baltics shows different patterns from Sweden. Broadberry and Korchmina, 2024 note that Russian agricultural output per capita was no different in 1800 than it was in 1700. There is no evidence that strip farming was abandoned in Russia, unlike in Sweden, where this change was made to save labour. Relative labour abundance in serfdom-dominated areas simply may not have necessitated changes to production methods. Crop rotation and new farming machinery also only show up towards the end of the 18th century in Russia, far later than in Sweden. Magnusson, 2007 suggests that this is not driven by the availability of tools through trade, as he shows that all of Sweden could have had access to modern tools but not all areas adopted it. This suggests that for Russia, too, this was a choice. The agricultural history of Prussia, which also had serfdom, paints a similar picture of no change after the plague.

E.3.2 Capital Frictions

Following Antràs and Tubdenov, 2025, APP can be measured as the ratio of inventories over cost of goods sold. Drawing on data for 6-digit NAICS (2017) industries in the US between 2002 and 2016, I construct this ratio from the Annual Survey of Manufacturers (U.S. Census Bureau, 2002–2016). Antràs and Tubdenov, 2025 use firm-level data that I do not have access to. However, they validate their findings with industry-level controls from NBER-CES, for which ASM is the upstream source which, as opposed to NBER-CES, includes inventories, and additionally covers private firms. Antràs and Tubdenov, 2025 finds that Census-based data (via NBER-CES) correlate with 0.8+ with their firm-level measure. To assign early modern goods modern industry-level APP values, I check whether a given 6-digit NAICS industry existed in 1700. 165 of 364 are assessed to have existed, clustered in food and drink, textiles, basic materials, and craft-based goods. All electrical industries, motor vehicles and aircraft, plastics and rubber, most chemicals (except botanical manufacturing and pharmaceutical preparations), petroleum products, pulp-based paper, and most modern precision manufacturing (except clocks) are dropped. I then assign goods into these industries, and find that 155 commodities are assigned into 67 industries, with the rest primary goods. My assumption is not that production

technology was the same in the 1700s, but that APP contains a time-invariant component. This is reflected in the 10 goods with the highest APP, which are wines, distilled alcohols, tobacco, tiles, and some manufactured metal items. The 10 goods with the lowest APP are dominated by animal products (beef, meat, ham, bones), bread, beer, cidre, mead. Thus, APP captures a meaningful dimension of production: for distilled alcohol and wines, more time passes between input costs and sales revenue. Following Böhm-Bawerk, 1889, more financially developed regions have a comparative advantage in such goods.

Table 43 shows that ports with higher pre-war financial development, proxied by their export-weighted average period of production (APP), shift more strongly into capital-intensive exports after the plague, and undergo a stronger export expansion. Table 44 shows that post-plague productivity growth in manufacturing correlates with pre-plague financial development.

Table 43: Pre-war financial development (proxied by export-weighted APP) predicts larger treatment effects

	Sectoral Exports		Exports	
	(1)	(2)	(3)	(4)
Post Plague x APP	-0.00004 (0.00004)	-0.00004 (0.00004)	0.00457** (0.00196)	0.00463** (0.00198)
Post Plague x Capital-Int. x APP	0.00015** (0.00006)	0.00015** (0.00006)		
<i>Fixed Effects:</i>				
– Origin x Sector	✓	✓		
– Sector x Time	✓	✓		
– Area x Year			✓	✓
– Origin x Destination			✓	✓
– Destination x Year			✓	✓
Inventory Measure	Average	End of Year	Average	End of Year
Observations	102,636	102,636	491,277	491,277

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variable in columns 1-2 is the volume of exports by capital intensity. In columns 3-4, it is export volumes. The independent variables are a post plague dummy, the pre-1700 export-weighted APP, their interaction, and annual growing season temperatures. In columns 1-2, I also regress on the interaction of a post plague dummy with a capital-intensity dummy and that interactions interaction with pre-1700 export-weighted APP. Throughout, I also regress on the interactions of the post plague (and post plague x capital-intensity for columns 1 and 2) dummies with pre-1700 export volumes of all individual sectors.

E.4 Human and Social Capital Deepening

I construct measures of goods' skill intensity and contract intensity. Following Berman, Bound, and Griliches, 1994 and Autor, Katz, and Krueger, 1998, I measure skill intensity as the share of non-production workers in total employment from the Annual Survey of Manufactures (U.S. Census Bureau, 2002–2016). I use 2002-2016 data at the NAICS 6-digit level, which covers 155 of the commodities of this paper following the mapping of goods in 1700 to modern-day U.S. industries laid out in Section 5.1. Following Nunn,

Table 44: Impact of plague on sectoral productivity growth, by financial development

	Labour-intensive ag.		Capital-intensive ag.		Labour-intensive man.		Capital-intensive man.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post Plague	0.05484 (0.19104)	0.05631 (0.19116)	0.55398** (0.22410)	0.55388** (0.22425)	-1.68205*** (0.17605)	-1.68540*** (0.17616)	-0.06510 (0.18160)	-0.06578 (0.18171)
Post Plague x APP	-0.00001 (0.00003)	-0.00001 (0.00003)	0.00003 (0.00004)	0.00003 (0.00004)	0.00022*** (0.00003)	0.00022*** (0.00003)	0.00007** (0.00003)	0.00007** (0.00003)
<i>Fixed Effects:</i>								
– Region	✓	✓	✓	✓	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓	✓	✓	✓	✓
Inventory Measure	Average	End of Year	Average	End of Year	Average	End of Year	Average	End of Year
Observations	48,780	48,780	48,790	48,790	48,790	48,790	48,790	48,790

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is log sectoral productivity growth. The independent variable is a plague dummy, the pre-1700 export-weighted APP, and their interaction.

2007 and Rauch, 1999, I proxy social capital intensity with contract intensity, which I measure using the share of differentiated inputs (liberally or conservatively defined as in Rauch, 1999). I use data by Nunn, 2007 at the NAICS 6-digit level. Additionally, I extend the 155 manufacturing goods, matched to NAICS 6-digit industries, by 52 raw and unprocessed goods for which I impute a value of 0 for both the shares of differentiated inputs and the share of non-production workers.

Table 45 finds no shift into export goods of higher human or social capital content after the plague.

E.5 Non-Homotheticity

In this Appendix, I detail the arguments made in Section 6 against non-homothetic demand. First, I show that my finding on productivity growth is robust to permitting non-homothetic demand. Fielser, 2011 provides a trade model with non-homotheticity which allows a neat mapping onto my model under the assumption that $\theta_k = \theta \forall k$, i.e. a fixed intra-industry heterogeneity in the Fréchet distribution. Even under non-homothetic preferences, her model then breaks down to an Eaton and Kortum, 2002 model:

$$X_{ijk} = \frac{\chi_k A_{ik} (w_{ik})^{-\gamma_k \theta} d_{ij}^{-\theta}}{CMA_{jk}} X_{jk}, \quad (35)$$

where I no longer assume $X_{jk} = \alpha_k Y_j$. Note that δ_{jkt} in equation 11 still absorbs the demand side under non-homotheticity. Therefore, my findings on productivity growth are unaffected.

Next, I analyse prices, as non-homothetic demand has tractable implications for relative prices of capital-intensive goods, which should become relatively more expensive. Crucially, the productivity channel advocated for in this paper produces the opposite relative price effect. Forming the ratio of the price indices for capital-intensive and labour-intensive (agricultural) goods in equation 7, I find:

$$\frac{P_{jCA}}{P_{jLA}} = \left(\frac{\chi_{CA} \sum_i A_{iCA} (w_{iA})^{-\gamma_{CA} \theta} d_{ij}^{-\theta}}{\chi_{LA} \sum_i A_{iLA} (w_{iA})^{-\gamma_{LA} \theta} d_{ij}^{-\theta}} \right)^{-\frac{1}{\theta}}, \quad (36)$$

where I impose $w_{iCA} = w_{iLA} = w_{iA}$ due to labour mobility in the agricultural sector and $d_{ijCA} = d_{ijLA} \forall (i, j)$. This ratio takes the same form in a general model of non-

Table 45: The plague leads to a shift into physical capital intensity, not into human or social capital intensity (proxied by skill and contract intensity)

	Only Manufacturing				Imputed for Raw Goods					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post Plague	-0.89901* (0.54362)	-0.91600* (0.55347)	-0.44523 (0.37270)	-0.44282 (0.37414)	-0.44334 (0.37457)	-2.47991** (1.10691)	-2.45044** (1.09028)	-1.56855** (0.78080)	-2.21473* (1.21381)	-2.15805* (1.19053)
Post Plague x Capital-Int.	0.75102* (0.40111)	0.77733* (0.40963)	1.48996** (0.58206)	1.49123** (0.58121)	1.49197** (0.58035)	1.60509*** (0.55170)	1.57660*** (0.53425)	1.74010*** (0.60310)	1.65517*** (0.57170)	1.63603*** (0.55842)
Post Plague x Contract Int.	-0.00022 (0.00016)			-0.00001 (0.00007)		0.00015 (0.00013)		0.00013 (0.00014)		
Post Plague x Contract Int.		-0.00020 (0.00015)			-0.00001 (0.00007)		0.00013 (0.00012)			0.00011 (0.00012)
Post Plague x Skill Int.			-4.91993** (1.96402)	-4.89244** (2.04112)	-4.89804** (2.04315)			-2.08948 (1.61679)	-0.79596 (1.44019)	-0.88498 (1.44818)
<i>Fixed Effects:</i>										
- Origin x Good	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
- Good x Time	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Nunn Measure	Liberal	Conservative		Liberal	Conservative	Liberal	Conservative		Liberal	Conservative
Observations	446,104	446,104	446,104	446,104	446,104	339,253	339,253	339,253	339,253	339,253

Notes: Standard errors clustered at the origin level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. Results from PPML regressions, where the dependent variable is exports in levels. The independent variables are a post plague dummy, the interaction of a post plague dummy with a capital intensity dummy, and the interaction of the post plague dummy with the liberal measure (columns 1, 4, 6 and 9); or the conservative measure (columns 2, 5, 7, and 10) of the share of a good's input that are differentiated according to Rauch, 1999, which in Nunn, 2007 is a measure of contract intensity; and/or the interaction of the post plague dummy with the share of non-production workers averaged across 2002 to 2016, based on 6-digit NAICS industries in the Annual Survey of Manufacturers (columns 3-5 and 8-10). For columns 1-2, 4-7, and 9-10, I use Nunn's data at the 6-digit NAICS level for the 1997 classification. In columns 1-5, only the 155 goods which are covered by NAICS 6-digit industries are included, whereas in columns 6-10, I impute a value of 0 for the share of differentiated inputs and for the share of non-production workers for goods that are raw and unprocessed, bringing the total to 207 commodities.

homothetic demand (Fieler, 2011) as in my model. I will differentiate between three types of prices. First, the price index at the regional level, which incorporates productivities, wages, and trade costs in all exporting regions. Trade costs and productivities in unplagued exporting regions do not change, but wages may as firm market access changes given productivity growth in plagued regions. I assume these changes to be small and scaled down by trade costs to North Sea regions. Productivities in plagued exporting regions change, but their trade with Baltic importers is not observed in my data. Unobserved trade within the Baltic unequivocally lowers the relative price of capital-intensive goods via the productivity and labour cost channels. Price data at the regional level should therefore show a relative decline in the price of capital-intensive goods. Note, however, that this confounds demand- and supply-side factors.

I assemble price data at the regional level from Allen and Unger, 2018.⁶⁴ For plagued Danzig and unplagued Amsterdam, their dataset holds annual prices on 77 products, which I classify according to Appendix Table 7.⁶⁵ I show in Table 46 that labour-intensive agricultural and manufacturing prices rise in Danzig compared to Amsterdam and capital-intensive agricultural prices fall.

Table 46: Product-level prices by sector and year in Danzig and Amsterdam

	Labour-int. Agriculture				Capital-int. Agriculture				Labour-int. Manufacturing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post Plague	10.53*** (3.15e-14)	32.82 (21.90)	114.2* (16.61)	702.0*** (8.00e-12)	-41.24*** (6.44e-14)	-59.72 (10.04)	-16.44** (0.663)	4.615*** (1.99e-14)	0.665*** (3.70e-15)
<i>Fixed Effects:</i>									
– City	✓	✓	✓	✓	✓	✓	✓	✓	✓
– Year		✓	✓			✓	✓		
– Product			✓				✓		
– Product x Year				✓				✓	
Observations	4140	4140	4140	452	5116	5116	5116	248	191

Notes: Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are prices of products by sector in Amsterdam and Danzig. The independent variable is 1 after the plague in Danzig in 1709. The second, third, and fourth specifications cannot be estimated for labour-intensive manufacturing due to lack of observations. No data on capital-intensive manufacturing products are contained in the data set.

Voigtländer and Voth, 2012 make the opposite prediction: a hump-shaped pattern of prices for plagued regions, with prices of capital-intensive goods increasing relative to those of labour-intensive goods. My results reject this prediction. If demand shifted relatively into capital-intensive agriculture, this should have increased its relative price. Instead, the relative price decline of capital-intensive goods supports a supply-driven explanation. These price results also reconcile the evidence that after the plague consumers received a higher share of their calories from meat and dairy (Broadberry et al., 2014) with my assumption of homothetic demand. As capital-intensive agricultural prices drop under constant consumption shares, consumers mechanically demand higher quantities of meat and dairy. Accordingly, non-homotheticity is not required to make sense of this finding.

Owing to the exceptional granularity of the Soundtoll data, I can also analyse export and import prices, which are the other two types of prices analysed. I construct prices at

⁶⁴No data on goods I classify as capital-intensive manufacturing are contained in the data set and few for labour-intensive manufacturing.

⁶⁵While the effects of labour scarcity were passed on to Amsterdam through trade, Amsterdam itself was not affected by the plague and faced higher trade costs to the regions that were plagued.

the bilateral level by unifying historical measures of account and dividing duty amounts by the recorded weight (available for 46% of passages), thus creating a comparable price per kilogram. I will refer to prices of goods sent to ports as import prices and to prices of goods sent from ports as export prices.

Import prices are only indirectly affected: productivities and trade costs in North Sea regions do not change; the indirect effects of firm market access changes on wages are small and weighted down by trade costs. In Table 47, I control for the supply side and show that relative prices of capital-intensive goods are unaffected by the importer's plague status. These importing prices are the relevant metric in my test for non-homotheticity below.

Table 47: Good-level logarithmic import prices

	(1)	(2)	(3)	(4)
Post Plague	0.281*** (0.0597)			
Post Plague x Capital-Int.	-0.0997 (0.116)		0.0875 (0.136)	
0-10 Years Post Plague		0.143 (0.124)		
11-20 Years Post Plague		0.386*** (0.0832)		
21-30 Years Post Plague		0.493*** (0.0982)		
>30 Years Post Plague		0.248*** (0.0703)		
0-10 Years Post Plague x Cap-Int.		0.0200 (0.224)		0.303 (0.195)
11-20 Years Post Plague x Cap-Int.		-0.0243 (0.129)		0.0534 (0.176)
21-30 Years Post Plague x Cap-Int.		-0.122 (0.123)		0.0776 (0.138)
>30 Years Post Plague x Cap-Int.		-0.107 (0.117)		0.0603 (0.152)
<i>Fixed Effects:</i>				
– Area x Sector x Year	✓	✓	✓	✓
– Origin x Destination x Year			✓	✓
– Origin x Destination x Sector	✓	✓	✓	✓
– Origin x Sector x Year	✓	✓	✓	✓
Observations	132411	132411	89334	89334

Notes: Standard errors clustered at the destination level in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01. The dependent variables are prices of goods by sector, origin, destination, and year as recovered from the Soundtoll data. Specifically, the recorded duty amount per good is divided by the weight whenever it is recorded (46% of observations), thus creating comparable prices per kilogram. The independent variables are a post plague dummy that is 1 after the plague and the interaction of this post plague dummy with a dummy for a capital-intensive sector. Columns 2 and 4 split up this dummy and the interaction by time horizon after the plague.

Export prices are directly affected by both the productivity and labour channels. If wages rise in plagued ports i and $\gamma_k < \gamma_{k'}$, the ratio $\frac{P_{jk}}{P_{jk'}}$ falls. Even if $\gamma_k = \gamma_{k'}$, the productivity channel will still produce a relative drop of capital-intensive export prices.

In Table 48, this prediction is confirmed: the (relative) export prices of capital-intensive goods shipped from plagued origins decline significantly. When analysing the time path of this price diversion, I find that the relative drop in prices of capital-intensive goods intensifies over time.

Table 48: Good-level logarithmic export prices

	(1)	(2)	(3)	(4)
Post Plague	0.526*** (0.185)			
Post Plague x Capital-Int.	-0.511** (0.224)		-0.664** (0.298)	
0-10 Years Post Plague		0.253** (0.103)		
11-20 Years Post Plague		0.501*** (0.169)		
21-30 Years Post Plague		0.546*** (0.183)		
>30 Years Post Plague		0.548*** (0.205)		
0-10 Years Post Plague x Cap-Int.		-0.246 (0.153)		-0.375* (0.224)
11-20 Years Post Plague x Cap-Int.		-0.558*** (0.203)		-0.814*** (0.296)
21-30 Years Post Plague x Cap-Int.		-0.639** (0.248)		-0.801** (0.341)
>30 Years Post Plague x Cap-Int.		-0.505** (0.252)		-0.657* (0.340)
<i>Fixed Effects:</i>				
– Area x Sector x Year	✓	✓	✓	✓
– Origin x Destination x Year			✓	✓
– Origin x Destination x Sector	✓	✓	✓	✓
– Destination x Sector x Year	✓	✓	✓	✓
Observations	118108	118108	73091	73091

Notes: Standard errors clustered at the origin level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are prices of goods by sector, origin, destination, and year as recovered from the Soundtoll data. Specifically, the recorded duty amount per good is divided by the weight whenever it is recorded (46% of observations), thus creating comparable prices per kilogram. The independent variables are a post plague dummy that is 1 after the plague and the interaction of this post plague dummy with a dummy for a capital-intensive sector. Columns 2 and 4 split up this dummy and the interaction by time horizon after the plague.

Comparing import prices and export prices allows me to disentangle supply-side from demand-side factors. The null finding for import prices supports my claim that there is no evidence for non-homothetic demand; the significant findings on export prices support my claim of supply-side changes after the plague.

Finally, I directly test for a demand shift in plagued regions. Instead of absorbing non-homothetic demand structures or comparing implied price changes, I test for the presence of non-homothetic demand. I form the ratio of equation 35 between any two sectors k and k' , where I need to assume $w_{ik} = w_{ik'}$ and $\gamma_k = \gamma_{k'}$, and take the ratio of this fraction between a pre-period p , 1700-1710, and any year t after the plague:

$$\frac{X_{ijkt}X_{ijk'p}}{X_{ijk't}X_{ijkp}} = \frac{A_{ikt}A_{ik'p}}{A_{ik't}A_{ikp}} \frac{X_{jkt}X_{jk'p}}{X_{jk't}X_{jkp}} \frac{CMA_{jk't}CMA_{jkp}}{CMA_{jkt}CMA_{jk'p}}. \quad (37)$$

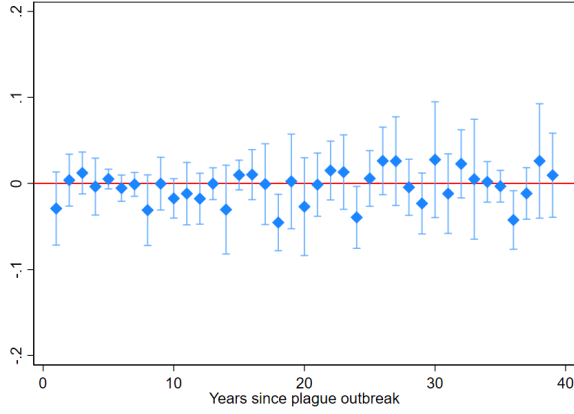
Relative export volumes evolve as a function of relative productivity changes in the exporting region and two demand-side factors: relative demand and relative market access. In my analysis, I form the left-hand side based on trade data and absorb relative productivity changes with exporter-time fixed effects. The argument made above about the behaviour of relative import prices under non-homotheticity simplifies the demand side further. As consumer market access is a price index, the finding that for equal labour shares and wages in a region relative (import) prices do not change carries over to the ratio of consumer market access. This is true for import prices coming from unplugged North Sea ports, which is sufficient here as that is the only trade I observe. Thus, equation 37 simplifies as the final fraction drops out. In this case, I can directly test for the presence of non-homothetic demand while controlling for the supply side.

While the assumption on constant relative prices simplifies the demand side of equation 37, it actually works against me in that any changes in relative demand will be seen as evidence for non-homotheticity, even if they arise from relative market access changes. The only possibility for this ratio to obscure underlying relative demand shifts is a change in relative demand that is offset by a change in relative consumer market access.

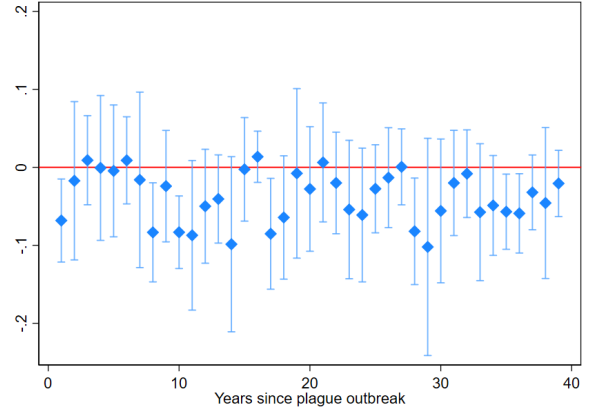
I define the baseline period p as the years 1700 to 1710 and let t be any year after 1710. In essence, the variable I regress on is a double ratio: relative exports to j compared between any post-plague year t and the pre-plague period. To capture changes in the import ratio indicative of non-homotheticity, I present event study results in Figure 44 where I regress on the log of the left-hand side of equation 37. The independent variable is a dummy for a plague outbreak in the importing region.⁶⁶ I absorb the productivity ratio by origin-time fixed effects and the time-invariant component in relative demand by destination fixed effects. I find no evidence of plague-related changes in relative demand.

⁶⁶I also check that there is no evidence for non-homotheticity when using the continuous plague treatment derived in Section C.5. Results available on request.

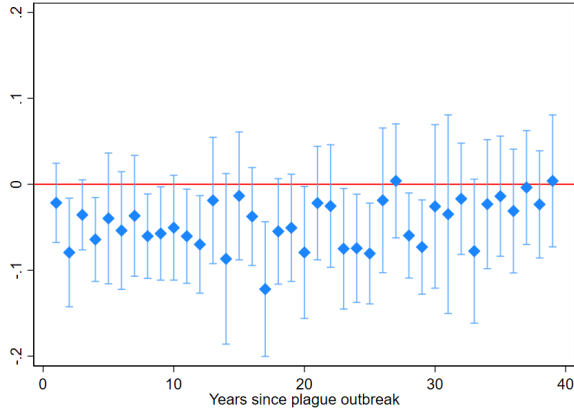
Figure 44: Event studies testing for non-homotheticity



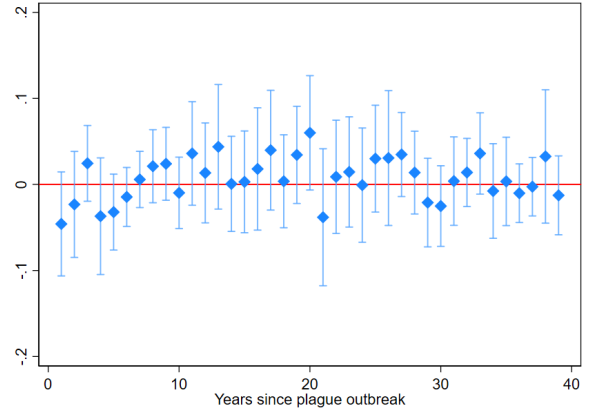
(a) Capital-int. & Labour-int. Manufacturing



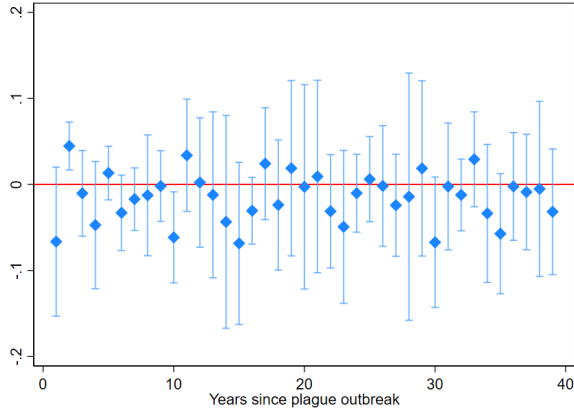
(b) Capital-int. Manufacturing & Labour-int. Agriculture



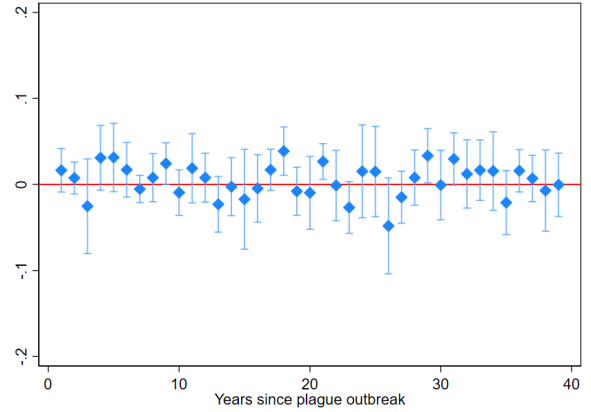
(c) Capital-int. Manufacturing & Agriculture



(d) Capital-int. Agriculture & Labour-int. Manufacturing



(e) Capital-int. Agriculture & Labour-int. Agriculture



(f) Labour-int. Manufacturing & Agriculture

Notes: Event study on the log of the left-hand side of equation 37. The independent variable is a plague dummy on the importer side. Origin-time and destination fixed effects are included and standard errors are clustered at the destination level.

F Counterfactuals

This appendix provides details on the counterfactuals provided in Section 7. Constructing export shares, the outcome of equation 2, according to gravity equation 10, I find that shares can be expressed as:

$$s_{ijk} = \frac{X_{ijk}}{\sum_i X_{ijk}} = \frac{A_{ik}(w_{ik})^{-\gamma_k \theta} d_{ijk}^{-\theta}}{\sum_i A_{ik}(w_{ik})^{-\gamma_k \theta} d_{ijk}^{-\theta}}. \quad (38)$$

When plugging in equation 9 for wages and equations 7 for firm and consumer market access, market shares can be written as a function of productivities, sectoral employment, trade costs, wages, and total income.

To shut down the productivity channel, I assume A_{ik} to be fixed. I also assume fixed trade costs d_{ijk} and destination income Y_j . Only trade between the Baltic and North Seas is observed, and the plague struck almost exclusively on the Baltic Sea, rendering this a reasonable assumption.

In the estimation equation, destination-sector-time fixed effects absorb changes in the denominator. When shutting down productivity growth and ignoring the variation that will be absorbed by fixed effects, one can write $s_{ijkt}^c = \left(\frac{L_{ikt}}{L_{ikPRE}} \right)^{\frac{\gamma_k \theta}{1+\gamma_k \theta}} s_{ijkPRE}$.

I model the population recovery from L_{iPOST} to L_{it} to happen exponentially over 40 years. This is the upper bound, as by 1750 plagued cities had recovered their populations. I assume that after 40 years, regional and not just urban populations had recovered. This yields:

$$s_{ijkt}^c = (1 - m_i)^{\frac{\gamma_k \theta}{1+\gamma_k \theta} (1 - \frac{t}{40})} s_{ijkPRE}.$$

Using these counterfactual destination market shares, one can follow the multiplicative structure of this model's gravity equation to determine the average productivity growth required to bridge from the counterfactual export contraction to the observed export expansion. The average point estimate for counterfactual export shares in the event study is a market share contraction of 1.03 p.p.; the observed average market expansion is 0.98 p.p. Average productivity needs to rise by about 210% to arrive at this export expansion. This - admittedly high - estimate mirrors the magnitudes found in the TFP estimation in Table 4, where the point estimate for labour-(capital-)intensive TFP growth (column 1) implies TFP growth of 105% (205%).